



VCS Quality Services Private Limited

SUPPLY OF ELECTRIC MOTOR DRIVEN
CNG HYDRAULIC BOOSTER
COMPRESSOR PACKAGE OF MIN.
AVERAGE FLOW CAPACITY OF 400 SCM/H
IN GEOGRAPHICAL AREAS OF
RAJASTHAN

TENDER NO:- As per Commercial Volume
DOCUMENT NO:- As per Commercial Volume

TECHNICAL VOLUME

(VOLUME II OF II)

ISSUE DATE : ____/____/____



**HINDUSTAN PETROLEUM CORPORATION
LIMITED**



VCS Quality Services Private Limited

400 SCMH Booster Compressor

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**HINDUSTAN PETROLEUM CORPORATION
LIMITED**



**PROJECT NUMBER:
VCS-HPCL-C221086**



MATERIAL REQUISITION-400 SCMH BOOSTER COMPRESOR

DOCUMENT NO

C221086

RJ

ME

MR

3040

SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC
BOOSTER COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW
CAPACITY OF 400 SCMH IN GEOGRAPHICAL AREAS OF
RAJASTHAN.

C1	05.04.2026	ISSUED FOR CLIENT REVIEW	AB	SA	AB
REV	DATE	DESCRIPTION	PREP	CHK	APPR



**SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC BOOSTER COMPRESSOR
PACKAGE OF MIN. AVERAGE FLOW CAPACITY OF 400 SCMH IN GEOGRAPHICAL
AREAS OF RAJASTHAN**

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SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC BOOSTER COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW CAPACITY OF 400 SCM/H IN GEOGRAPHICAL AREAS OF RAJASTHAN

1. INTRODUCTION

Hindustan Petroleum Corporation Limited (HPCL), a Government of India Undertaking (hereinafter called as the Owner), has been authorized by PNGRB for setting up infrastructure and operation of City Gas Distribution Network for the authorize GA of Rajasthan. Natural Gas will be transported to residential, commercial, industrial (PNG) and automobile consumers (CNG) in the city.

M/s VCS Quality Services Pvt. Ltd (VCS) has been appointed as Engineering Project Management Consultant (hereinafter referred as EPMC), by HPCL for providing Design, Engineering and PMC services for this Project.

2. DEFINITION

Where used in this document, the following terms shall have the meanings indicated below, unless clearly indicated by the context to this order:

PROJECT:	CGD PROJECT AT RAJASTHAN GA
OWNER/COMPANY:	HINDUSTAN PETROLEUM CORP. LTD.
CONSULTANT:	VCS QUALITY SERVICES PVT. LTD.
MANUFACTURER:	THE PARTY WHO MANUFACTURERS & SUPPLIES EQUIPMENT & PROVIDE SERVICES TO OWNER.
MR:	MATERIAL REQUISITION
FLP:	FLAME PROOF
SAT:	SITE ACCEPTANCE TEST
FAT:	FACTORY ACCEPTANCE TEST
CAMC:	COMPREHENSIVE ANNUAL MAINTENANCE CONTRACT
O&M:	OPERATION & MAINTENANCE

3. DOCUMENT PRECEDENCE

It shall be the responsibility of the SUPPLY / VENDOR to inform the PURCHASER of any errors, ambiguities, inconsistencies, discrepancies, or conflict of information that may be found to exist in any document, specification or drawing submitted by the PURCHASER.

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In case of conflict, the order of precedence shall be as follows:

- a) Material Requisition
- b) Data Sheets
- c) Technical Specifications
- d) Basic Documents
- e) Codes and Standards

As a general rule in the event of any discrepancy between technical matter and local laws/ regulations (and documents above listed) the most stringent shall be applied.

MANUFACTURER / VENDOR shall notify PURCHASER of any apparent conflicts between MR, specifications, related datasheets, any code and standards and any other specifications noted herein. (Resolution and/ or interpretation precedence shall be obtained from PURCHASER in writing before proceeding with the design/ manufacturer or completion of services.)

4. SCOPE OF SUPPLY & SERVICES

Design, engineering, supply, inspection, testing at suppliers works, packaging (if any), delivery of Electric motor driven Hydraulic variable suction CNG booster compressor (from supplier plant to HPCL warehouse/Site including loading and unloading with hydra arrangement as per HSE requirements), supply of all Pre-commissioning, Commissioning spares & documentation as per the enclosed engineering standard, specifications and data sheets etc. attached or referred. Supply of compressor shall be at HINDUSTAN PETROLEUM CORP. LTD.

400 SCM/H ELECTRIC MOTOR DRIVEN HYDRAULIC BOOSTER COMPRESSOR PACKAGE FOR RAJASTHAN GA			
A	SUPPLY		
1.0	Design, Engineering, Inspection, Testing, Supply of FLP Motor driven Hydraulic variable suction CNG Booster Compressor with canopy having Avg. flow capacity of 400 SCM/H (at pressure range from 210 to 30 kg/cm ² (g)). The package includes supply of SOV operated all actuator valve along with separate Coriolis Mass flow meter for suction & discharge line and 7 bank priority panel with 3/4" OD Isolation ball valves on each outlet line ,supply of required rating power cables from booster to electrical room , , supply & installation of 3 nos ESD switch with all accessories and auxiliaries, supply of hydraulic oil, flame arrester and other associated works to commission the	Nos.	30

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	compressor as per Technical volume of Tender document complete in all respects including supply of special tools and tackles & Commissioning spares. Followings items to be supplied with 400 SCM/H Booster Compressor package - a) Electric motor driven Air compressor with rain protection shade, Air dryer & Air receiver of Min. 100 WL capacity, motor rating min 1.5 kW & discharge pressure approx. 8 to 12 kg/cm²(g) including FRLS cable, SS tubing, clamps, union joints, fittings & cable tray etc. b) CO₂ flooding system having cylinder capacity (min 45 Kg each cylinder) with 50 % redundancy and both cylinders having Automatic as well as manual operation / control system along with supply of spring balance & panel including their tubing, cables, covering shade, pre-assembly. Simulation testing of the system. c) Design, fabrication and Supply of base frame of canopy to withstand load for 3000WL capacity CNG cascade over canopy structure. d) Supply of FLP type PLC based control panel with HMI, supply of power and control cables from compressor to Electrical room (Approx. length 60 Mtr. for each compressor to be considered) however. cable length shall be vary as per site requirements and associate fixtures required for connections. Comprehensive Monthly Preventive Maintenance and breakdown maintenance of every booster compressor as per SOR and scope of work.		
2.0	LCV unloading post with Coriolis Mass Flow meter, 4" Dial size Pressure gauge (Range from 0 to 400 kg/cm ² (g)) along with block and bleed valve arrangements, SS tubing, Ball valve, SS fittings, electrically conductive Hose 1/2" ID and length of 4.5 Mtr. with breakaway coupling connection & SS retaining springs, Vent tube & 1/2" NPTF Quick connect coupling couplers.	Nos.	30
B	SERVICES		
3.0	Erection & Commissioning		
3.1	Installation & Commissioning at site along with field Performance Acceptance Test	Nos.	30
	Sub-Total for Services (B)		
C	Operation and Comprehensive Annual Maintenance (CAMC) Services Of 400 SCM/H Electric motor driven Hydraulic booster CNG Compressor Including operation, Repair, supply of		



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	consumables, spare parts, Tools & Manpower, etc. a. Deployment of Operators for 24 x 7 days. Operator should maintain the daily log book for operation. b. Preventive maintenance at regular intervals by OEM / authorized contractor as per OEM recommendations. c. Break down maintenance as and when required within CAMC by OEM Note: Comprehensive Annual maintenance (CAMC) during warranty period included in supply of equipment i.e. Item sl. No.-A-1.0 & 2.0		
4.0	Operation of compressor package		
4.1	Operation charges for 1st year i.e. during Warranty period	No. of Shifts	27335
5.0	Comprehensive Maintenance of compressor package		
5.1	Comprehensive Maintenance Charges for 1st year after completion of warranty period	Machine Months	360
5.2	Comprehensive Maintenance charges for 2nd year i.e. after completion of 1st year post warranty	Machine Months	360
5.3	Comprehensive Maintenance charges for 3rd year i.e. after the completion of 2nd year post warranty	Machine Months	360
5.4	Comprehensive Maintenance charges for 4th year i.e. after the completion of 3rd year post warranty	Machine Months	360

NOTES: -

- I. Machine Months calculated on the basis of No. of compressors Multiplying by 12 months.
- II. No. of Shift calculated on the basis of (No. of Package × No. of shift × 365 days)
- III. Duration of one shift should be consider for 8 Hrs. as per statutory guidelines.

5. INSPECTION

Bidder shall appoint TPIA, with reference to documents enclosed separately as per doc. no. **C221086-RJ-ME-TPIA-3040**, for deploying inspection agencies. TPIA charges shall be paid by the Manufacturer/Supplier. Apart from inspection by TPIA, joint inspection may be performed if required by client/consultant.

6. SPECIAL INSTRUCTIONS TO BIDDER

- 6.1 Bidder to note that no correspondence shall be entered into or entertained after the bid submission.
- 6.2 Bidder shall furnish quotation only in case he can supply material strictly as per this Material Requisition and specification / data sheets forming part of Material Requisition.
- 6.3 If the offer contains any technical deviations or clarifications or stipulates any

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technical specifications (even if in line with MR requirements) and does not include complete scope & technical / performance data required to be submitted with the offer, the offer shall be liable for rejection.

- 6.4 The submission of prices by the Bidder shall be construed to mean that he has confirmed compliance with all technical specifications of the corresponding item(s).
- 6.5 Bidder must submit all documents as listed in checklist along with his offer.
- 6.6 The Supplier shall deliver a Certificate confirming to **BS EN 10204 3.2/3.1** stating the quality, the mechanical properties, the chemical analysis, the process of manufacture bought out items and the making of the Booster compressor.
- 6.7 All materials shall be delivered to consortium stores/sites of M/s Hindustan Petroleum Corp. Limited. Detailed addresses are furnished in the technical bid document.

7. INFORMATION/ DOCUMENTS/ DRAWINGS TO BE SUBMITTED BY SUCCESSFUL BIDDER

Successful Bidder shall submit each of the following documents:

- 7.1 Inter-locked Piping & Instrumentation Diagram (P&ID) including Gas, Hydraulics, Cooling system, CO2 flooding system, LCV fill post, Air compression and dryer system & schematic diagram.
- 7.2 General arrangements Drawings, Hookup drawings including hoses/pipes & Tubing's connection & specifications.
- 7.3 General Arrangements Drawings for control panel, Air compressor, Air dryer, complete compressor package & CO2 flooding systems etc.
- 7.4 Inspection & test reports for all mandatory tests as per the applicable code as well as test reports for any supplementary tests, in nicely bound volumes.
- 7.5 Material test certificates (physical property, chemical composition, make, heat treatment report, etc.) as applicable for items in nicely bound volumes.
- 7.6 Statutory test certificates, as applicable.
- 7.7 Foundation drawings & civil works detailed requirements.
- 7.8 OEM Datasheets & catalogs for all Instrumentation to be provided by vendors.
- 7.9 WPS & PQR as required.
- 7.10 Other Drawing & document as specified in vendor data & drawing requirements as with Tender.
- 7.11 Purchase orders/test certificate of origin of material of bought out items soon after placement of order.
- 7.12 Manufacturer's drawings/documents for bought out items, for Purchaser's / Consultant's approval within 4 weeks.
- 7.13 Manufacturer related information for design of civil foundation & other matching items within 6 weeks of FOI / LOI.
- 7.14 All approved drawings / design calculation / maintenance & operating manual documents as well as inspection and test reports for Owner's / Consultants

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reference /record in nicely category-wise bound volumes (in Hard Copy) and in Soft Copy separately.

- 7.15 Filled in data sheet for each instrument tag after sizing, range selection, proper selection of materials etc. shall be bidder's responsibility. Any necessary change required later for meeting the specification shall be done by the vendor without any price or delivery implications.
- 7.16 Bidder shall provide daily checklist, monthly maintenance checklist, breakdown report format, monthly maintenance format and test procedure.
- 7.17 Bidder shall be providing list of special tools and tackles.
- 7.18 Bidder shall provide list of mandatory spares for breakdown and preventive maintenance and same shall be available with vendor during CAMC.
- 7.19 Bidder shall provide PESO document Approval of model for Booster Compressor, & Their supporting Ex-electrical items by CCOE Approval/equivalent lab agency.
- 7.20 Proper tagging, labeling and marking of all the equipment as per the approved drawings.

8. INSTRUCTION TO BIDDER

- 8.0 Bidder to furnish filled up documents / formats as per "Special Instruction to Bidders" of **Volume I OF II** along with the offer.
- 8.1 Booster compressor (including all components) shall be designed and suitable for Natural Gas and shall comply the technical specification of 400 SCM/H Booster compressors, **Doc. No. C221086- RJ-ME-TS-3040.**
- 8.2 Delivery of shall be at M/s. Hindustan Petroleum Corp. Limited designated storage yard/ site and loading and unloading shall be in the Bidder's scope.
- 8.3 The submission of prices by the bidder shall be construed to mean that he has confirmed compliance with all technical specifications of the corresponding item(s).
- 8.4 Purchaser's inspector reserves the right to perform final inspection and witness tests, as indicated in specification of Booster compressor / ITP at manufacture's works prior to shipment.
- 8.5 All drawings, instructions, catalogues, etc. shall be in English language and all dimensions shall be metric units.

9. LIST OF ATTACHMENTS

Sl. No.	Name of Documents	Documents No.
01.	Material Requisition	C221086-RJ-ME-MR-3040
02.	Scope of work – 400 SCM/H Boostercompressor	C221086-RJ-SOW-MR-3040
03.	Technical Specification	C221086-RJ-ME-TS-3040

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04.	Data sheet – a. Mass flow meter b. Booster Compressor c. Intercooler & after cooler d. Pressure Gauge e. Safety relief valves f. Gas Detector g. Flame Detector h. KWH meter i. Discharge filter j. FLP electric motor k. FLP LV electric motor l. Flame arrester m. Air compressor n. Temperature gauge	C221086-RJ-ME-DS-3040 C221086-RJ-ME-DS-3041 C221086-RJ-ME-DS-3042 C221086-RJ-ME-DS-3043 C221086-RJ-ME-DS-3044 C221086-RJ-ME-DS-3045 C221086-RJ-ME-DS-3046 C221086-RJ-ME-DS-3047 C221086-RJ-ME-DS-3048 C221086-RJ-ME-DS-3049 C221086-RJ-ME-DS-3050 C221086-RJ-ME-DS-3051 C221086-RJ-ME-DS-3052 C221086-RJ-ME-DS-3053
05.	Quality Assurance plan- 400 SCM/H Booster compressor	C221086-RJ-ME-QAP-3040
06.	Vendor Drawing & Data Requirement	C221086-RJ-ME-VDDR-3040
07.	List of Recommended TPIA	C221086-RJ-ME-TPIA-3040
08.	List of Recommended vendors for bought out items	VCS-00-00-VL-0001

NOTES: -

- 1.0 Vendor shall handover one set of special tools and tackles to particular GA, if any after the completion of CAMC contract.
- 2.0 Booster compressor & body Earthing cable and GI strips to be supplied by vendor during Final delivery.
- 3.0 Vendor shall also provide maintenance Log Book along with commissioning check sheet during final delivery.



ENERGISING QUALITY

**PROJECT NUMBER:
VCS-HPCL-C221086**



SCOPE OF WORK- 400 SCMh BOOSTER COMPRESSOR

DOCUMENT NO

C221086

RJ

ME

SOW

3040

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BOOSTER COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW
CAPACITY OF 400 SCMh IN GEOGRAPHICAL AREAS OF
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1. INTRODUCTION

Hindustan Petroleum Corporation Limited (HPCL), a Government of India Undertaking (hereinafter called as the Owner), has been authorized by PNGRB for setting up infrastructure and operation of City Gas Distribution Network for the authorize GA of Rajasthan. Natural Gas will be transported to residential, commercial, industrial (PNG) and automobile consumers (CNG) in the city.

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2. DEFINITION

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OWNER/COMPANY: HINDUSTAN PETROLEUM CORP. LTD.

CONSULTANT: VCS QUALITY SERVICES PVT. LTD.

MANUFACTURER: THE PARTY WHO MANUFACTURES & SUPPLIES EQUIPMENT & PROVIDE SERVICES TO OWNER OR TO CONTRACTOR.

SOW: SCOPE OF WORK

3. SCOPE OF WORK ((Refer SOR item no.-A-1.0 & 2.0)

This scope of work for supply of CNG booster compressor package consists of Design/detail engineering, procurement, fabrication, testing, supply and transportation and unloading at designated site. Details of works mentioned below.

- 3.1 Design, Engineering, Supply, assembly, Inspection & testing at supplier's works, field trial runs, and Equipment performance test along with associated electrical, instrumentation etc. as per bid document.
- 3.2 Hydraulic booster with lube oil system and cooling system as required. Hydraulic oil circulation can be considered as lubrication system.
- 3.3 Flame proof Electrical Electric motor as compressor driver.
- 3.4 Single stage filtration at suction & single-stage filtration at discharge so as to limit oil carryover is to be provided.
- 3.5 Differential pressure gauge shall be provided at suction and discharge filter.

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- 3.6 All Instrumentation and HMI shall be displayed from outside of the enclosure.
- 3.7 Instrumentation and control system as specified on data sheets, P&ID including Local panel, Console/Local gauge boards, PLC.
- 3.8 All instruments to be properly tagged as per the approved drawings.
- 3.9 Common structural steel skid for the compressor- Motor combination and for all auxiliary systems.
- 3.10 ESD button (3 Nos.) shall be provided (Control Room, Process Area, one side of compressor).
- 3.11 Aviation lamp shall be provided on outside of enclosure.
- 3.12 Wires mesh type guard for radiator exhaust fan.
- 3.13 PLC based control panel with HMI.
- 3.14 Instrumentation and control system as specified as per tender documents.
- 3.15 Electric equipment / Instruments being required in the Compressor package.
- 3.16 NRV as required for each fluid line to control & minimize the water hammering and pump back pressure effects.
- 3.17 Air/Water-cooled heat exchanger for inter stage and final stage discharge gas.
- 3.18 Close circuit water cooling through air cooled heat exchanger.
- 3.19 Closed circuit cooling water system (console type)/Air cooled as required including heat exchangers, coolant circulation system, fan operated radiator, etc. for inter stage and final cooling of compressed gas, lubrication oil and hydraulic oil.
- 3.20 Priority Panel at Package Discharge as per priority fill system.
- 3.21 Bidder shall provide ¾" or 1" size end of line flame arrester at compressor common vent.
- 3.22 All interconnecting oil, gas, water, air piping, cables within the compressor package.
- 3.23 The provision for overhead mounting of cascade (3000 Water Liter capacity with approximate weight of Approx. 7 tons) should be there & canopy should be designed of enough strength having working space and with ladder arrangement. However, cascade supply in scope of Purchaser and its mounting on the structure shall be in scope of Bidder. Structure Stability compliance Certificate of the unit where cascade will be mounted to be submitted during detail engineering.
- 3.24 All cable glands within the package and for incoming main power cable shall be supplied by vendor. Cable glands shall be of flame proof type & CCOE approved.
- 3.25 Impulse and pneumatic piping/Tubing for all valves, fittings as specified & required for mounting the instruments. Two valve manifold (i.e. Block and bleed valve) /Needle with bleed valves to be provided for Pressure gauges and pressure Transmitters. Flow directions on all tubing/piping to be

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indicated by arrow.

- 3.26 CNG compressor shall have suction line tubing 3/4" and discharge tubing arrangement is 3/4". It is mandatory.
- 3.27 NRV at final suction and discharge line.
- 3.28 CNG compressors shall have the inbuilt gas and fire detectors inside the enclosure.
- 3.29 Structural supports within the compressor package for all piping, instruments etc.
- 3.30 One no. safety relief valve at each stage discharge.
- 3.31 Y-type strainers, valves, sight flow indicators, check valves, auto & manual drain traps etc. as required for various auxiliary systems i.e., frame lube oil, cylinder lubrication system, cooling water systems etc.
- 3.32 Direct Coupling/V-belts/pulleys assembly for driver and pumps assembly.
- 3.33 Acoustic enclosure for Compressor package, with one number L.E.L detectors and one UV detectors in the enclosure with door limit switch.
- 3.34 Common CO2 extinguishing system consisting of two cylinders, piping, valves, cables and both cylinders having automatic as well as manual control systems with online weight loss monitoring & health status of CO2.
- 3.35 Vendor shall provide CO2 flooding system with Min 2 nos. brand new CO2 cylinders of adequate capacity (should not be less than 45 Kg each). However actual size of the cylinder shall be as per compressor enclosure size and necessary calculation shall be submitted and approval shall be taken before finalization / supply. One cylinder will act as main & other as stand by, which shall have identical arrangement and connected to the system.
- 3.36 Inlet and outlet manual and automatic isolating valves for maintenance & emergency. Field Performance test at site.
- 3.37 Supply of all erection & commissioning spares along with Booster compressor package.
- 3.38 One set of spare parts catalogue, as built drawings, and Operation & Maintenance catalogue with each compressor package.
- 3.39 Vendor should provide Training of site engineers /operators at Purchaser's work premises. The travelling, boarding, and lodging of Purchaser's engineers shall be borne by bidder scope.
- 3.40 Each training module shall span for one week and shall cover the equipment constructional features, operational and maintenance procedures, practical hands on experience on assembling, dismantling etc.
- 3.41 The compressor package control system shall be designed for unattended safe operation in automatic mode and shall unload, start, load, stop safely. The compressor shall start in auto in case high bank pressure in dispenser fall below 200 kg/cm²(g) and stop once the pressure in all three banks reaches 255 kg/cm²(g).



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- 3.42 7-line Priority Panel at Package Discharge as per priority fill system along with one bank for suction from stationary bank to compressor.
- 3.43 Selector switch at Booster Compressor panel for valve positioning to automatically switch over the inlet i.e., to take suction from LCV cascade or from Stationary cascade.
- 3.44 Compressor shall be suitable for continuously variable suction pressure from 210 kg/cm²g to 30 kg/cm²g, supplied through LCV mounted CNG storage cascade.
- 3.45 The ingress of oil into CNG adversely affects vehicle emission and storage system. Hence in case of lubricated cylinders, vendor shall supply a proven, maintenance free oil removal system with automatic and manual drain after-cooler to remove oil from removal system to restrict the oil less than 5 PPM in discharge compressed gas.
- 3.46 An oil drain pot outside of the package shall be provided to collect all drains from packing, distance pieces, processes etc.
- 3.47 Electric motor with hydraulic as Compressor Driver.
- 3.48 For metering of natural gas, 3 no. of Coriolis type Mass Flow meter shall be provided 1 No. at the suction, 1 No. at the priority panel of Compressor, and 1 no. at LCV unloading side. (Refer attached GAD for LCV post)
- 3.49 Instrumentation and control system shall provide with 50% redundancy as specified in the tender document. PLC shall be of modular type consisting of rack for mounting cards, CPU, Input output cards, Power supply card, and communication card.
- 3.50 PLC CPU shall be redundant (1 Working+1 Standby) with auto switch over without manual interference. Failure alarm of CPU shall be provided in HMI and same shall be repeated in PLC for remote monitoring. Moreover 20% spare philosophy shall be considered for I / O's for future expansion.
- 3.51 One no. Instrument Air compressor with min. 100 WL capacity air receiver along with desiccant type air dryer and having motor power rating of 1.5 KW.
- 3.52 All solenoid valve shall be operated by air actuated type.
- 3.53 Suitable Priority Fill System with compressor top-up facility inclusive of regulating valves, bypass valve & liquid filled pressure gauges as specified in technical specifications.
- 3.54 Electrical equipment / Instruments used in the Compressor package required for different type of signals like analog, digital signals and alarm.
- 3.55 It is further clarified that Local control panel (With PLC) will be mounted on the package itself and will be directly wired from Instruments to the Local control panel.
- 3.56 Main incoming cable from owners PDB to main control panel of the compressor (as per site requirement) through GI conduit/trenches, Cable from owner's electronic earth pit (EE) for electronic circuit in control panel,





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- cable from owner's main earthing ring to control panel body earth and all interconnecting cables including complete erection accessories like double compression cable gland, cable tags, lugs etc. as required is bidder scope.
- 3.57 Fire and Gas detection system integrated with compressor instrument panel.
- 3.58 Safety relief valve Gas vent and manual vent valve shall be terminated with common header. Separate vent header tubing with proper supporting to be provided.
- 3.59 Operation and Maintenance during the warranty period along with the supply of spares, consumables, lubricants, and manpower, etc., except electricity.
- 3.60 Piping from air compressor and CO2 cylinders up to Compressor enclosures in the scope of bidder.
- 3.61 LCV unloading post with Coriolis Mass Flow meter, 4" Dial size Pressure gauge (Range from 0 to 400 kg/cm² (g)) along with block and bleed valve arrangements, 3/4" SS tubing, Ball valve, SS fittings, Electrically conductive High pressure Hose 1/2" ID and length of 4.5 Mtr. with breakaway coupling connection & SS retaining springs, Vent tube & 1/2" NPTF Quick connect coupling couplers
- 3.62 LCV post to be fabricated by ISMC150 and minimum 3mtr height. Hose will be installed with 3/4" SS Three Way Ball Valve, 3/4" OD end connector for high pressure hose connection. If any item such as clamps, fittings/valves, tube is not listed above i.e. required to complete the assembly of LCV fill post, same shall be supplied by bidder.
- 3.63 A typical drawing is being attached, same shall be used for necessary reference and compliance. Vendor may submit their design for approval during detail engineering.
- 3.64 Complete erection, installation and commissioning of LCV fill post is in the bidder's scope which include Grouting and fixing of Post on the foundation.
- 3.65 Complete Erection, Testing & Commissioning of compressor packages.
- 3.66 Monkey ladder and hand railing shall be provided on canopy for cascade mounting working platform and adequate size to be provided for regular services, operation & maintenance purpose.
- 3.67 Tubing from Air compressor & dryer to compressor enclosure and piping from CO2 flooding system cylinder to the enclosure shall be in the scope of bidder.
- 3.68 Supply of all cables and laying works from purchaser PDB panel to compressor enclosure, CO2 system, Power factor Panel, Air compressor & dryer shall be in the bidder scope (Min. length consider approx. 60 Mtrs, however if required more than estimated quantity, bidder to be arrange the same).





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- 3.69 Bidder shall supply hydraulic oil for initial filling tank and required coolant if any.
- 3.70 During warrantee period if any breakdown occurs in the compressor system that will be rectify and part will be replaced (defective part will be replaced) without any charges by the bidder.
- 3.71 The bidder shall depute adequate numbers of qualified, experienced, and competent persons and supervisors for smooth preventive or breakdown maintenance of the compressors. The maintenance staffs have to be available round the clock daily throughout the year.
- 3.72 All spares, consumables, oil and lubricants required for carrying out the Maintenance of the complete compressor packages including preventive maintenance checkup and any other materials during preventive maintenance of the compressor packages, shall be provided by the bidder.

**3.1 SCOPE OF WORK FOR ERECTION & COMMISSIONING
(Refer SOR item no.-B-3.0)**

- 3.1.1 Bidder's scope of work shall also include packaging, forwarding, insurance, custom clearance if any, handling and unloading at PURCHASER stores/site, and Site Acceptance Test (SAT) of "ELECTRIC MOTOR DRIVEN HYDRAULIC VARIABLE SUCTION CNG BOOSTER COMPRESSOR PACKAGES" as required for dispensing CNG to vehicles at the location described in this tender. Electric type hydraulic CNG Booster Compressor is designed to handle the following capacity to a discharge pressure of 255 kg/cm²:
400 SCM/H @ with variable suction pressure 210 to 30 kg /cm² (g).
- 3.1.2 Bidder's scope of work include erection, pre-commissioning and commissioning, field trial run test. Noise level test, Vibration test and performance test of all packages at sites. The field trial run of the compressor will be for a period of 40 minutes on continuous run basis and minimum of 8 hours on cumulative basis at purchaser site.
- 3.1.3 The field trial run of the Variable Suction Compressor will be from 210 kg/cm² (g) to 30 kg/cm² (g).
- 3.1.4 The variable suction Compressors have to be installed at HINDUSTAN PETROLEUM CORP. LTD. The suction of the compressor will be from Light/Heavy Commercial Vehicle (LCV/HCV) mounted mobile cascades provided by Client and the compressed Natural gas (CNG) will be dispensed in vehicles by the dispensers installed by Client.
- 3.1.5 The package shall be kept under observation for 72 hours after commissioning, for stable operation and no major breakdown. In case of any detect, discrepancies under specified site conditions. Supplier shall first rectify the same and repeat the field trial run. Field test requirements are subject to availability gas based on the location and GA wise.
- 3.1.6 Vendor representative shall mobilize for commissioning at site within 48 hrs.

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After intimation by client/consultant in case site readiness is delay.

- 3.1.7 The vendor shall be responsible for supplying all materials as specified in the Scope of Work, Data Sheet, Schedule of Rates (SOR), and tender documents. Approval of drawings shall not be considered the final criterion for supply. If any items are omitted or not shown in the drawings submitted by the vendor for approval, but are included in the tender scope, the vendor shall still be obligated to supply all such items as per the Scope of Work and tender documents.
- 3.1.8 The vendor's obligation to supply shall not be limited by approved drawings, and any omission in the drawings shall not relieve the vendor of their responsibility to supply all items specified in the tender documents.
- 3.1.9 It will be the endeavor of all the parties to get the performance acceptance test (PAT) at site conducted within a period of **30 days** from the start of commercial operation of a particular package. The bidder has to keep the compressors operational round the clock (i.e., 24X7) as defined by EIC and all the expenditures including spares and consumables, oil etc. to make the compressors operational shall have to be borne by the bidder. The power required to run the compressors will be provided by OWNER/ OMC. The contractor shall maintain the compressors in sound mechanical condition at all times. The contractor shall rectify the defects notified by OWNER immediately and should submit all the history log sheets and spares availability status along with the report in the format mutually agreed between OWNER and the bidder.

3.2 SCOPE OF WORK FOR OPERATION (Refer SOR item no.-C-4.0)

- 3.2.1 The bidder has to carry out operation of the compressors package for **One years** from the date of commissioning.
- 3.2.2 The bidder shall deploy adequate number of operator / supervisors with all required tools, spares, consumables and equipment for smooth operation.
- 3.2.3 Contractor to submit a detailed organogram with key person details before starting operation of the compressor package to Engineer In-charge (EIC).
- 3.2.4 The bidder has to keep the compressors operational round the clock and all the expenditures including manpower, spares and consumables, oil, lubricant etc. to make the compressors operational shall have to be borne by the bidder.
- 3.2.5 The power required to run the compressors will be provided by Client. The contractor shall maintain the compressors in sound mechanical condition at all times. The contractor shall rectify the defects notified by Client immediately and should submit all the history log sheets and spares availability status along with the report in the format mutually agreed between Client and the bidder.
- 3.2.6 Bidder shall provide PPE's to operator/ supervisors/ Engineers at his own

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cost.

- 3.2.7 The contractor is required to carry out all services as mentioned in the Scope of Schedule of Rates on all the 365 days including Sunday and all Holiday & around the clock.
- 3.2.8 Contractor has to ensure the safety of man and machine all the times. Damages of equipment due to negligence will be recovered as per the decision of Engineer-in- Charge, which will be final.
- 3.2.9 Calibration shall be done from government-approved laboratories (NABL/FCRI certified) and shall be carried out at least 15 days prior to the calibration due date.
- 3.2.10 It will be in bidder's scope to carry out calibration of all instruments and equipment as per industrial best practice. In case commissioning is delayed due to any reason and validity is expired, then it will be bidder's responsibility to carry out calibration of all instruments and equipment before commissioning. HPCL shall notify bidder prior to 15 days from date of commissioning.
- 3.2.11 All calibration stickers are stick on every equipment.
- 3.2.12 The contractor's representative/supervisor shall report daily to the Shift-in-Charge for day to day working.
- 3.2.13 The bidder shall keep 1 set of safety relief valves in spare for the purpose of calibration.
- 3.2.14 The bidder shall carry out retesting of pressure vessels periodically i.e., every year or earlier as per Gas Cylinder rules 2016 / Static & Mobile Pressure Vessels Rules.
- 3.2.15 The contractor shall allow weekly rest and daily working hours to his workmen as per the relevant Act/Law and Rule made there under. However, no work shall be left incomplete/unattended on any holiday/weekly rest. Technician/operators provided shall have minimum qualification of ITI. Contract in person or his authorized representative shall provide the services on daily basis to interact with Engineer-in- charge and deployed workman.
- 3.2.16 Regarding work completion, the decision of the Engineer-in-Charge will be final and binding.
- 3.2.17 The contractor shall make his own arrangements to provide all facilities like boarding and transport etc. to his workmen.
- 3.2.18 All personnel of the contractor entering on work premises shall be properly and neatly dressed and shall wear uniform, badges while working on premises of the Owner including work sites.
- 3.2.19 Contractor shall maintain proper record of his working employee's attendance and payment made to them.
- 3.2.20 It will be the responsibility of the contractor to pay as per the minimum wages of the appropriate government applicable under the Central Minimum Wage Act 1948.



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- 3.2.21 The services shall be provided in terms of shift pattern on the round the clock basis. The contractor is responsible to provide effective and efficient services in all shifts and assure that there is no disruption in the services for want of any resources.
- 3.2.22 The successful bidder shall comply with all the rules regarding PF, ESI etc. as stated in the tender document.
- 3.2.23 All the jobs mentioned under scope of services shall be carried out as per sound engineering practices, work procedure documentation, recommendation of the manufacturer/suppliers and as per the guidelines/direction of engineer-in-charge of authorized representative.
- 3.2.24 The contractor shall submit a copy of the daily / weekly / fortnightly / monthly / bimonthly / quarterly and yearly report to the EIC in both soft and hard form. All stationery including the printed material such as compressor parameter logbook, complaint log book, service report, break down summary report etc. shall be in scope of the bidder.
- 3.2.25 The contractor must submit the following documents on monthly basis along with the bill:
- Compressor parameter logbook for the month.
 - Certificate to be given by the bidder stating that they have complied with all the labor regulations and are following the minimum wages act.
 - All consumables, lubricating oil, coolant required for carrying out preventive / any type of maintenance shall be in the scope of supplier during the warranty period. The warranty spares shall be supplied by the vendor during the warranty period as per warranty clause.
 - Any correspondence required to be made with the principal company or OEM or various offices shall be made by the bidder or bidder's agent. All arrangements like phone, computer, Internet etc. required for above correspondences shall be arranged by the bidder at his own cost.

3.3 SCOPE OF WORK FOR CAMC

- 3.3.1 The bidder has to carry out Comprehensive maintenance of the compressors for **five years** from the date of commissioning, however during warranty period CAMC is inclusive of compressor package.
- 3.3.2 During warrantee period if any breakdown occurs in the compressor system that will be rectify and part will be replaced (defective part will be replaced) without any charges by the bidder.
- 3.3.3 The bidder shall depute adequate numbers of qualified, experienced, and competent persons and supervisors for smooth maintenance of the compressors. The maintenance staffs have to be available round the clock

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daily throughout the year.

- 3.3.4 All spares, consumables, oil and lubricants required for carrying out the Operation and Maintenance of the complete compressor packages including periodic breakdown and any other materials required for operation and maintenance of the compressor packages, shall be provided by the bidder.
- 3.3.5 Periodic inspections of Safety Valves, F&G detectors, Transmitters. Pressure vessel gauge and any other equipment as per statutory norms of concerned state Factory Rules.
- 3.3.6 Gas Cylinder Rules 2016 shall have to be carried out by the bidder at his own cost during the period of maintenance by the bidder. The inspections have to be carried out by competent persons as per advice of Engineer-in-Charge and certificates have to be submitted to Client.
- 3.3.7 Regarding work completion, the decision of the Engineer-in-Charge will be final and binding.
- 3.3.8 The contractor shall make his own arrangements to provide all facilities like boarding and transport etc. to his workmen.
- 3.3.9 All personnel of the contractor entering on work premises shall be properly and neatly dressed and shall wear uniform, badges while working on premises of the Owner including work sites.
- 3.3.10 Contractor shall maintain proper record of his working employee's attendance and payment made to them.
- 3.3.11 All the safety rules and regulations prevailing and applicable from time to time at the installations as directed by OWNER will be strictly adhered to by the contractor.
- 3.3.12 The contractor shall establish a central control room to operate 24 hours, seven days a week where complaint regarding non-performance of the Compressors in terms of the contract can be lodged. Further, the contractor shall deploy adequate number of technicians/ supervisors / engineers at various site offices in consultation with Engineer-in-Charge to provide trouble free maintenance of the Compressors.
- 3.3.13 All arrangements for communication from control room to the contract person working on job under the services shall be the responsibility of the contractor, viz. cell phone / walky-talky.
- 3.3.14 The successful bidder shall indemnify the Owner from any claim of the contract labor.
- 3.3.15 Summary of breakdown hour's station wise with analysis shall be submitted to CNG control room on a fortnightly basis both in hard and soft form as per OWNER format.
- 3.3.16 The contractor must submit the following documents on monthly basis along with the bill:

- Preventative maintenance compliance report for that month along with

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the detailed service report.

- Details of the compressor breakdown, summary of break down hours for that month and the cumulative break down hours along with breakdown response time.
- All tools, tackles and fixtures required for carrying out the above maintenance of the compressor shall be in scope of the bidder. The scope will also include handling equipment like crane, forklift, chain pulley block, etc. required during the any maintenances activity.
- The periodic maintenance required to be done as per OEM recommendation shall be taken up promptly. The bidder shall provide the detailed preventative maintenance schedule.

- 3.3.17 Bidder shall confirm Estimated down time required for each type of maintenance schedule.
- 3.3.18 List of spares and their quantities required for each type of maintenance schedule per compressor.
- 3.3.19 The bidder shall plan such maintenances during non-peak hours and in consultancy with the Engineer In-Charge (EIC) of OWNER. Any maintenance that needs to be taken up shall be well planned in advance with due approval of the EIC.
- 3.3.20 The bidder shall use only OEM's certified spares during maintenances. All spares shall be kept in sealed OEM stamped packages. The packages shall be opened in front of OWNER representative during maintenance. In case, the schedule maintenance of the OEM manual recommends checking and replace parts like valve spring, valve plates, piston rings etc. after certain time interval, same shall replace or used further only on approval from the OWNER's representative. However, any unwanted consequences for non-replacement of such parts shall be the responsibility of the bidder and spares, repair required to put back the unit into operation will be to bidder's account.
- 3.3.21 All routine and periodic checks / inspections required to be done as per OEM recommendation shall be done by the bidder. Instruments required for above inspection like Vernier caliper, micrometer screw gauge, fill gauges, bore gauge etc. shall be in scope of the bidder and these instruments shall be calibrated every year.
- 3.3.22 The bidder shall carry out calibration of gas detectors and flame detectors every years or earlier as per requirement or instruction of EIC of OWNER. Also, yearly calibration of all instruments such as pressure gauges, transmitters, switches etc. shall be in the scope of the bidder. In addition to the above all safety relief valves shall also be tested and calibrated every





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year.

- 3.3.23 Mass flow meter shall be calibrated from FCRI lab on every five years or earlier as per requirement or instruction of EIC of OWNER.
- 3.3.24 All routine and periodic checks / inspections required to be done as per OEM recommendation shall be done by the bidder. Instruments required for above inspection like Vernier caliper, micrometer screw gauge, fill gauges, bore gauge, Torque wrench, Db meter & Vibration meter etc. shall be in scope of the bidder and these instruments shall be calibrated every year.

4. EXCLUSIONS

The following are excluded from the scope of the bidder:

- 4.1 All civil works and foundation design, however the bidder shall furnish all the relevant data for design of any pedestal/foundation of all equipment's.
- 4.2 CNG storage Mobile Cascades.
- 4.3 CNG storage Stationary cascade.
- 4.4 All piping beyond battery limits.
- 4.5 CNG dispensers and inter connected SS tubes & fittings. The design, construction, manufacture, supply, testing and other general requirements of the compressor package equipment shall be strictly in accordance with the data sheets, applicable API codes, technical specification, approved drawing/document and shall comply fully with relevant national/international standards, Indian electricity act, Indian electricity rules, regulations of insurance association of India and factories act while carrying out work as per this specification.
- 4.6 Any modification suggested by the statutory bodies either during drawing approval or during inspection, if any, shall be carried out by the bidder without any additional cost and delivery implications.

5. CODES & STANDARDS

Following codes, standards, and regulations to be considered:

- OISD 142:
- IS 5572
- OISD 179, NFPA-52: 2006, NFP-496, NFPA-68, NFPA-70 or equivalent
- NFPA – 37
- NFPA – 12- CO2 Flooding system
- IS: 325/ IEC or International standards. – Standards for electric Motor
- IS: 6382
- Applicable ANSI, ASTM, NEC, NEMA code
- API – 618/API 11 P
- EURO EAN NORM P.E.D., Italian NOR M D.P.R. 47/55
- EURO EAN NORM P.E.D, D.M. 24.5.02 - D.M. 28.6.02
- D.M 24.11.84 parte prima - size. II°, D.M. 24.5.02 - D.M. 28.6.02, DIN 2413, SAE J 514

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- EURO EAN NOR MS, CEI N 60079- 0/CEI EN 60079-14/ CEI, EN 60204-1/ CEI EN 60439-1, ATEX STAN DARD
- API – 661: Specifications for Air cooled exchangers
- ASME Section – VIII Div. – 1/2 Design codes for pressure vessels.
- Gas Cylinder Rules 2016.
- Standard Specifications of Bureau of Indian Standards (BIS).
- Specifications/Recommendations of IEC.
- Indian Electricity Rules.
- Indian Explosives Act.
- Delhi Factory Rules, 1950
- TEMA – C - Water cooled heat exchangers
- ASME / ANSI – B-31.3 Code for Process Piping
- DIN 2413-This standard covers the design of steel bends and bent pipes of circular cross section used in pressure pipelines.
- SAE J 514-Standard for CNG hydraulic tube fittings and O-ring
- CEI EN 60079-10-Classification of area for explosive gas atmosphere
- CEI EN 60079-14-Design, selection and installation of electrical systems for areas with potentially explosive atmosphere.
- CEI, EN 60204-1-Standard for safety of machinery — Electrical equipment of machine
- CEI EN 60439-1-Standard for safety of electrical equipment
- ATEX-Standard for describing electrical equipment and workspace is allowed in an explosive atmosphere.

6. DESCRIPTION OF WORK

- 6.1 Procurement of raw materials etc. from sub-vendors.
- 6.2 Preparation of documentation for design approval by Purchaser/consultant.
- 6.3 Inspection and testing as per Technical specification.
- 6.4 Surface preparation, protective coating and painting as per Technical specification.
- 6.5 Packaging for transportation to site and supply.
- 6.6 Field trial run and performance test at site.
- 6.7 Post Commissioning Operation and Maintenance during Warranty Period with all spares.
- 6.8 Consumables, lubricants and manpower etc., except electricity

7. OTHER CONDITIONS OF WORK

7.1 UTILITIES

- 7.1.1 Cooling water is not available as utility and the package shall be provided with self- sufficient cooling water system for Compressor, as required, with makeup tank. However, cooling water for makeup tank is available. All the electrical equipment in this system shall be suitable for area classification of Hazardous area CLASS-1, DIVISION-1, GROUP-D as per NEC.
- 7.1.2 Bidder to provide Instrument air with an electric motor driven air compressor with min. 100WL receiver & desiccant type air drier system. The

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instrumentation air shall also be used for CNG Dispensers. Hence the system to be designed considering the same.

- 7.1.3 Air Compressor with discharge pressure of 7 kg/cm² suitable for 1.5 KW electric motor rating with dryer shall be supplied by the bidder.
- 7.1.4 Air Compressor to be supplied along with air receiver of min. 100 water liter capacity.
- 7.1.5 Air compressor, drier and air receiver for instrument air, shall be kept off the package in safe area or owner's building or if required then FLP/Non-FLP type motor used for station safe area.
- 7.1.6 Manual drains and automatic moisture trap shall be provided in the system.
- 7.1.7 Air receiver shall be provided with SRV, Pressure switch and pressure gauge with an isolation valve. Air dryer shall be with bypass pass arrangements.
- 7.1.8 Air compressor from air vessel to Instrument connection shall be provided ¼" or 3/8" SS OD tube and MOC shall be SS316/SS304.
- 7.1.9 Two Number Tapping from air receiver and dryer shall be provided as follows:
 - **For Dispenser:** One tapping with ½" isolation valve from air receiver.
 - **For booster Compressor:** One tapping with ½" isolation valve from air receivers.
- 7.1.10 All electrical and instrumentation terminals shall be as specified. Electric power shall be made available by Owner.
- 7.1.11 Purchaser shall provide 415 V, 3Ph, 50Hz, 4 wire electric power for running the compressor and illumination in the electrical room. Supplier shall indicate power/ Feeder (KW/Amp) requirement in the offer, if different. All cabling (supply & laying) from electrical room to booster compressor shall be in bidder's scope. Bidder shall indicate power /Feeder (KW/Amp) requirement in the offer and shall supply power KWH meter for power consumption of the complete package including all accessories to be installed in the control panel.
- 7.1.12 KWH reading shall be available at HMI.
- 7.1.13 Vendor shall ensure and establish their Main motor guaranteed load data including KW, Amp, Voltage, PF and frequency etc. through HMI display via. Multi-function meter or KWH meter in control panel.
- 7.1.14 Vendor
- 7.1.15 For running the compressor and illumination 415 Volt (±10 %) 3-phases 3 Wire, 50 Hz (±5%) shall be provided to starter panel to feed the compressor motor. Bidder shall indicate power/ Feeder (KW/Amp) requirement in the offer.
- 7.1.16 All cabling (supply & laying) from electrical room to booster compressor shall be in bidder's scope. Bidder shall indicate power Feeder (KW/Amp) requirement in the offer. Surge protection devices shall be provided in the control panel.
- 7.1.17 Purchaser shall provide UPS (240 ±1 % V, 50 ±1 % Hz) for LCP. Bidder shall indicate power/ feeder (KW/Amp) requirement in the offer. Bidder to make



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arrangement for conditioning of power supply beyond above limit.

7.2 BATTERY LIMITS

- 7.2.1 All customer interface connections, gas Inlet shall be brought out to the package edge and terminated with ¾ "OD SS-Tube.
- 7.2.2 All customer interface connections (i.e. Gas inlet & gas outlet) shall be brought out to the package edge. Gas inlet shall be terminated in nozzles with isolation valves and gas outlet (priority panel outlet connection) shall be terminated through high pressure ¾" full flow ball valves with 3/4" end connectors.
- 7.2.3 As and where specified on the data sheets all vents (i.e. Relief valve, distance piece and packing) shall be Mani folded and terminated at skid edge outside the enclosure and vented to safe height of 3.0Mtr. At package roof with proper support.
- 7.2.4 All drains from different process equipment, distance piece and packing shall be Mani folded and terminated as single point for customer interface duly flanged with isolation valve.
- 7.2.5 UPS and Non UPS power shall be made available from power distribution board (PDB) in the electrical room. Supply, Erection and termination of all cables and accessories from feeder in electrical room (As per site requirement) shall be in the bidder's scope.
- 7.2.6 Electrical earth pit shall be made available at a distance of about 5mts. from compressor package. Electrical Earthing Cable from this earth pit shall be in the bidder's scope shall be terminated to dedicated earth provided in the panel through proper size of glands. Owner's earthing main ring shall be made available at compressor foundation for equipment earthing. Electrical earthing for motor shall be done through Cable and the body earthing to be done through **GI strip of 25 x 3 mm²** inside the compressor package shall be in the bidder's scope.

8 AS-BUILT DOCUMENTS

On successful completion of hydrostatic testing, the Bidder shall prepare As Built drawings & reports of entire Booster compressor package as specified in scope of work. All "As Built" drawings / reports shall be submitted as below.

Four sets of hard copies of following documents shall be submitted by Bidder. All documents shall be bound together:

- (i) As-built drawing of Compressor package GAD / Fabrication Drawing / P&ID etc.
- (ii) Test Reports/Results/Records



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In addition, the above documents shall also be submitted in electronic media i.e. Pen drive. Software used for the presentation of these documents shall be as follows:

Type of document Software

(iii) Test Reports/Results/Records MS Word/Excel (MS Office 2000)

(iv) Drawings AutoCAD

9 BIDDERS RESPONSIBILITIES

Bidder's responsibilities, besides the scope of work for supply to be performed by him defined earlier, shall also include the following:

- A) Interpretation and verification of data/information furnished by company. Any additional information/data/surveys etc. Required by bidder for detailed engineering shall be obtained by him. Company may assist him in obtaining such information/ data by issuing recommendatory letters etc.
- B) Entire engineering for procurement & fabrication, including drawings, QA/QC procedures, etc. Performed by the bidder for the compressor package system shall be reviewed and approved by company. All works shall be executed based on approved documents only.
- C) Review and approval of bidder's documents by company shall in no way relieve the bidder from his sole responsibility for safe and efficient design, engineering, supply and subsequent smooth operation of the system.
- D) Bidder shall depute independent third-party inspector from list of recommended TPIA (attached with this tender) for carrying out all necessary inspections and review of test results/reports etc.
- E) Pre-commissioning/ commissioning assistance of entire compressor package.
- F) Bidder shall carry out all testing and inspection of materials, equipment etc. Through independent testing institutions, laboratories, if so desired by company.
- G) Any other work not specifically listed above but is required for successful completion of entire system.

10 CHECKLIST: SCOPE OF SUPPLY

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Bidder shall furnish all equipment, drivers, auxiliary systems, instruments, and controls and safety devices as per the enquiry document. Anything required over and above what is specified, for safe and satisfactory operations of the equipment package shall be included by the Bidder in his scope.

Bidder to write 'YES' or 'NO' against each item. Bidder is required to include complete scope, as such 'NO' is not warranted. However, in case for any of the items if Vendor's reply is 'NO', Vendor should give reason for the same.

Bidder's scope of supply shall include but not be limited to the following:

Sl. No	Description	Specified by Purchaser (YES/NO)	Included by Bidder (YES/NO)	Remarks
1	Each compressor package shall be complete with:			
2	Document required for equipment qualification criteria submitted along with User's certificate	Yes		
3	Area Classification: All electrical and electronic components as specified shall be suitable for hazardous area CLASS-1 DIVISION-1 GROUP-D as per NFPA 70 Article 500 or ZONE-1, GROUP IIA/IIB as per EIC/IS	Yes		
4	CCOE/PESO approval certificates of all electrical equipment's of complete package along with Model details. OR CCOE/PESO approval of ex-electrical components application letter if any.	Yes		
5	One no. Modular type DCP fire extinguisher capacity 10 Kg, shall be provided with each Compressor in the separately.	Yes		
6	One LEL, One fire detector (UV) and One FLP Lamp shall be provided in the Enclosure.	Yes		
7	Cooling System at site is not available: It is required to have closed circuit radiators for cooling compressor and auxiliaries if needed which shall be in the Scope of the Bidder.	Yes		



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8	Safety relief valves on each stage discharge of the compressor	Yes		
9	Minimum flow capacity corresponding at suction pressure of 30 Kg/cm ² (g)	Bidder to confirm		
10	Avg. flow capacity corresponding at suction pressure range from 210 to 30 Kg/cm ² (g)	Bidder to confirm		
11	Explosion proof Electrical panel to be installed in the hazardous area.	Yes		
12	Electrical equipment of cooling system to be explosion proof.	Yes		
13	All valves, tubing's, fittings, as specified and required within the compressor package shall be SS-316/SS-316L	Yes		
14	CO ₂ flooding system included as per specifications along with CCOE certificates.	Yes		
15	Instrumentations and control included as per specifications, including three No Mass flow meter (i.e. suction, discharge & LCV post)	Yes		
16	Acoustic enclosures for Motor & compressor for noise attenuation up to 75 Dba @ 1 meter distance.	Yes		
17	Priority Panel with ¾" OD connection at package discharge.	Yes		
18	Cabling with cable trays for all electrical instrumentation cabling within the package	Yes		
19	All couplings and Guards.	Yes		
20	Set of Special tools & tackles and ToolBox (along with list)	Yes		
21	Inspection and testing as specified on the data sheet and Technical Specification.	Yes		
22	Documents: All data and drawings as required as per technical specification.	Yes		
23	Erection, commissioning, and PG test at site of the complete package	Yes		
24	2 Nos. Monkey ladders along with 4 side walkway and there support railing in the scope of the Bidder	Yes		
25	Operation and Maintenance Services as per technical	Yes		



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	specification.			
26	Structural support for mounting the cascade of 3000WLC.	Yes		
27	One no. Instrument Air compressor/dryer package with air storage vessel of 1.5 KW motor rating along with Package.	Yes		
28	LCV unloading post With mass Flowmeter	Yes		

11 GUARANTEED PARAMETERS

SI No	Description	Bidder to indicate
1	Average flow capacity (over full range of suction pressure from 210 kg/cm ² g to 30 kg/cm ² varying on continuous basis) Required min. 400 Sm³/hr.	
2	Minimum flow capacity corresponding to suction Pressure of 210kg/cm ² (g). Bidder to specify	
3	Minimum flow capacity corresponding to suction Pressure of 30 kg/cm ² ; Required min. 200 Sm³/hr.	
4	Sound level of enclosure (required Max. 75+/-3 DBA from 1 meter Distance). Bidder to confirm	
5	Electrical power consumption in KWH with no positive tolerance with overall full range of suction pressure (from 210kg/cm ² (g) to 30 kg/cm ² (g) varying on Continuous basis to compress 400 Std. M3 gas per hour with no negative tolerance without air compressor and Air/water cooled fan. (For Penalty purpose)	
Guaranteed General package Data		
6	Pump shaft power BkW in KW @ RV Set Conditions (No +Ve tolerance)	
7	Net of all auxiliaries/utility/package Air or water cooled Motor & pump loads in KW	
8	Specific power consumption of compressor KWH/KG CNG	
9	Footprint area of compressor package quoted (Length---Mtr. x Width---Mtr. x Height---Mtr.)	

NOTE:

- a) No advantage shall be given in case bidder offers compressor with flows higher than as detailed above for various types.



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- b) Compressor shaft power BkW in KW shall be Pump/Hydraulic power which is always lower than selected motor power.
- c) Bidders offer shall be based on firm and final compressor model on which basis the offer shall be evaluated and no alternate compressor model or change of model, after submission of bid shall be entertained / considered. This is very important, and all bidders shall take full cognizance of this matter before submitting the bid.
- d) Bidder to indicate the capacity and absorbed power of the offered compressors at various suction conditions starting from 210 to 30 Kg/Cm²g (Temperature 35°C max.) and 255 Kg/Cm²g and 52 deg. C (max) discharge condition. Performance curves and tables i.e., **Flow versus suction pressure** and **temperature and power curves** i.e., **absorbed power versus suction pressure** and **temperature** at specified discharge conditions shall be furnished. In addition to above, flow capacity and absorbed power values for suction conditions from 210 to 30 Kg/Cm²g. The graph shall be plotted at various suction pressures ranging from 210 to 30 Kg/Cm²g and at various suction temperatures ranging from 10°C to 35°C. Similarly, the graphs shall be plotted at various discharge pressures ranging from 220 Kg/Cm²g to 255 Kg/Cm² (g), however at 52° C (max) discharge conditions shall be consider.
- e) Bidder to note that the compressor package required shall be suitable for operating at a suction pressure from 210 Kg/Cm²g to 30 Kg/Cm²g at Max. 35°C.
- f) Bidder to note that negative tolerance on the guaranteed capacity will not be acceptable. Also, no advantage shall be given for positive tolerance of the capacity.
- g) Bidder must indicate the guaranteed KWH including all losses such as mechanical transmission power absorbed by compressor auxiliary like cooler fan etc. but excluding air compressor.
- h) Bidders to note that the guaranteed power consumption, as stated in the quoted parameters under Clause 11, sl. no.- 5, must not exceed the quoted value during the actual site performance test.
- i) Bidder shall quote Min. 20 KWH guaranteed Electrical power consumption in KWH with overall full range of suction pressure (from 210kg/cm²(g) to 30 kg/cm²(g) varying on Continuous basis to compress 400 Std. M3 gas per hour without air compressor and Air/water cooled fan.
- j) However, in case guaranteed electrical Power consumption is more than 20 KWH in actual power consumption observed during performance guarantee test, penalty shall be levied for the differential unit. Then penalty shall be calculated based on the formula provided in the tender specification.
- k) No advantage shall be given in case bidder guaranteed electrical Power consumption is lower than 20KWH for this booster compressor package.
- a) Conversion factor for Kg to SCM is 1 kg = 1.337 SCM**

Penalty for Non-Performance during Operation & Maintenance Period

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(a) On normal day (i.e., the day other than the schedule maintenance day):

- (i) The party has to ensure that the equipment's are available for operation for minimum 18 hours per day.
- (ii) If the equipment is down for more than 6 hours on any day. Penalty would be applicable as follows: up to 12 hours: R.S. 20,000/- per day Beyond 12 hours: R.S. 40,000/- per day.

(b) On schedule maintenance day:

- (i) The party would be required to carry out the recommended schedule/preventive maintenance of the equipment for which the party has to indicate the time required for each type of schedule maintenance.
- (ii) If the equipment is down for more than 6 hours & up to 12 hours beyond the time indicated for the agreed schedule maintenance, the party would be penalized R.s. 20,000/- and for more than 12 hours R.s. 40,000/- per day.
- (iii) In any case, the maximum penalty imposed in a month for non-performance of the equipment would be limited to 50% of the Amount of O&M charges to be paid to the party per month per compressor.

(c) Penalty for operator Non-availability & PPE requirements :

In case, the contractor's service personnel are unable to reach the break down site within stipulated time or is unable to complete the maintenance within stipulated time, following penalty will be applicable. This amount will be deducted from the invoice raised by the contractor, at the end of the month.

- 1.1.1 Penalty for 1-hour delay in reaching at all site (i.e., within 3 hours instead of 2 hours) – Rs 500/-
- 1.1.2 Penalty for 2-hour delay in reaching at all site (i.e., within 4 hours instead of 2 hours)- Rs 1000/-.
- 1.1.3 If the shutdown time, which will be calculated from the time the compressor is offered to service person on reaching site, is extended beyond 4 hours, a penalty of Rs 500/- per hour will be applicable. Rs 2500 towards non-availability of mandatory spares at operational sites.
- 1.1.4 Contractor shall not deploy the employee of age less than 18 years in any of the activities. If it is found, then it will be viewed seriously and heavy penalty of Rs. 20000/-per instance and also the termination/blacklist will be done from our approved vendor list.





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- 1.1.5 The contractor shall provide full Personal Protective Equipment (PPE) to each individual employee including, soft hat, eye protection, ear plug, and safety shoes. It is mandatory for all personnel to wear said PPE whilst performing their duties, failing which a penalty @ Rs. 500/- per incidence will be levied in addition to dismissal of the person.

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ENERGISING QUALITY

**PROJECT NUMBER:
VCS-HPCL-C221086**



DATA SHEET OF CNG MASS FLOW METER

DOCUMENT NO

C221086

RJ

ME

DS

3040

SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC
BOOSTER COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW
CAPACITY OF 400 SCM/H IN GEOGRAPHICAL AREAS OF
RAJASTHAN

C1	06.04.2026	ISSUED FOR CLIENT REVIEW	AB	SA	AB
REV	DATE	DESCRIPTION	PREP	CHK	APPR



**SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC BOOSTER
COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW CAPACITY OF 400 SCM/H IN
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MASS FLOW METER (CORIOLIS TYPE) DATA SHEET

General	Sl. No.	Description	Client Specifications	Vendor to Provide
	01.	Tag No.	#	#
	02.	Service	#	#
	03.	Model No.	#	#
	04.	Make	#	#
	05.	Measuring principle	Coriolis type latest edition	-
	06.	Fluid	Natural Gas	-
	07.	Design Pressure	345 bars (g)	-
	08.	Working pressure	250 bars (g)	-
	09.	Nominal Line size (inches)	½" to ¾" (DN15 to DN20)	-
Haz. Area classification & location	10.	Electrical Area classifications	IEC Zone 1 Gr IIA, IIB	-
	11.	Ingress Protection Rating	IP 66/67	-
	12.	Approval & Certification type	ATEX/PESO/IECEX	-
	13.	Hazardous area classifications	Class I, Div. 1, Groups C and D	-
Sensor unit	14.	Sensor unit series	CNG Mass	-
	15.	Type	Coriolis	-
	16.	Function	Mass flow	-
	17.	Connection size / ratings	#	#
	18.	Facings & finish	RF 125AARH	-
	19.	Custody transfer	#	#
	20.	Flow tube Design pressure ratings	#	#
	21.	Combined sensor and process fitting ratings	#	#
	22.	Union to NPT adapter piece ratings	#	#
	23.	Body Material	SS316	-
	24.	Sensor Housing Material leads	SS Hermetically Sealed	-
	25.	Sensor/Wetted Parts Material	SS316	-



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Sensor unit	26.	Enclosure	WP to IP 66/67 Secondary containment	-
	27.	Non-Wetted Parts Material	#	#
	28.	Intrinsically safe	Required	-
	29.	Flow range Max./Min.(kg/min)	#	#
	30.	Batch Accuracy	+/-0.5 %	-
	31.	Repeatability	+/-0.25 %	-
	32.	Stand Uncertainty	#	#
	33.	At 100 % flow (flow rate to be)	#	#
	34.	Jacketing	Required	-
	35.	Zero stability	0.009 kg/min.	-
	36.	Sensor weight (kg)	#	#
	37.	Turn down	10:1	#
	38.	Pressure Relief path	Required	#
Transmitter unit	39.	Enclosure	FLP+WP	-
	40.	Mass & Vol. Flow display	Required	-
	41.	Density	N/a	-
	42.	Display configuration	Local integrated display with sensor module	-
	43.	Power	Both AC & DC supply applicable	-
	44.	Transmitter Output	4-20 mA	-
	45.	10KHz pulse	Required	-
	46.	Modbus communication	Required	-
	47.	HART wireless communication	Required	-
	48.	Transmitter Design Temperature	-40° to 120°C	-
	49.	Transmitter Housing	NEMA 4X(IP66/67)	-
	50.	Electronics interface	#	#
	51.	Load Driving Capability in ohms	600 Ω	-
	52.	Conduit connection	1/2-inch NPTF (Applicable for Brass gland FLP Type)	-
	53.	Diagnosis application software	Pro-link latest version	-
	54.	Power consumption in watt	#	#
	55.	RS485 Baud Rate	#	#



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COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW CAPACITY OF 400 SCMH IN
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	56.	Meter ID	#	#
	57.	Flow rate Measured	Kg/Hr.	-
	58.	Low flow cutoff @	#	-
	59.	High flow cutoff @	#	#
	60.	Transmitter Earthing point	Required	-
	61.	Mass flow cutoff	#	#
	62.	Volume flow cutoff	#	#
	63.	Equipment Ground	Required	-
	64.	Totalizer	shall be non-volatile and non-resettable	-
Fluid data	65.	Fluid State	CNG GAS	-
	66.	Pressure: Operating / Max.	250 Bar(g) 250 Bar(g)	-
	67.	Temperature: Operating / Max.	Refer Technical specifications	-
	68.	Operating density (kg/m3)	170 to 219 kg/m3 @ 15°C	-
	69.	Relative Molecular mass	16.80 kg/k mol.	-
	70.	Operating viscosity	Cp	-
	71.	Maximum allowable pressure drops	Bar(g)	-
Parameter Display	72.	Mass flow rate (kg/Min.)	#	#
	73.	Gas Temperature (°C)	#	#
	74.	Volume flow rate (L/min.)	#	#
	75.	Volume total (Liter)	#	#
	76.	Mass total (kg)	#	#
Certification	77.	AGA11 conformance	Required	#
	78.	Hazardous Area PESO	#	#
	79.	Hazardous Area ATEX	#	#
	80.	Custody Transfer-W&M India	#	#
	81.	Calibration Certificates as per ISO 17025	Required	-
	82.	Material Inspection certificates	Required	-
	83.	Radiographic Test certificates	Required	-



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Inspection & Testing	84.	Pressure Testing	Required	-
	85.	Weld examination	Required	-

NOTE: -

- I. VENDOR TO PROVIDE THE DATA AS MARKED "#".
- II. VENDOR TO PROVIDE INDIVIDUAL DATA SHEET ALONG WITH OPERATING MANUAL & OTHER COMPLIANCE CERTIFICATES DURING DETAIL ENGINEERING.
- III. VENDOR TO PROVIDE PESO CERTIFICATES FOR BOTH TRANSMITTER & SENSOR UNIT ALONG WITH OTHER SUPPLEMENT CERTIFICATES DURING APPROVAL PROCESS.
- IV. VENDOR TO PROVIDE W&M CERTIFICATES FOR MASS FLOW METER DURING DETAIL ENGINEERING.
- V. VENDOR TO PROVIDE TRANSMITTER & SENSOR BODY EARTHING WITH COMPRESSOR MOUNTINGS.
- VI. VENDOR TO PROVIDE LATEST PRO-LINK SOFTWARE TO ONLINE DIAGNOSIS OF METER.
- VII. VENDOR TO PROVIDE CONVERTER CABLES, SOFTWARE & O&M MANUAL FOR FUTURE REQUIREMENTS DURING OPERATION PERIOD.



**PROJECT NUMBER:
VCS-HPCL-C221086**



DATA SHEET OF 400 SCMh CNG BOOSTER COMPRESSOR

DOCUMENT NO

C221086

RJ

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SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC
BOOSTER COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW
CAPACITY OF 400 SCMh IN GEOGRAPHICAL AREAS OF
RAJASTHAN

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**SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC BOOSTER
COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW CAPACITY OF 400 SCM/H IN
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**DATA SHEET CNG BOOSTER
COMPRESSOR**

S. No.	DESCRIPTION	SPECIFICATION	OFFERED
01	No. of unit	As per SOR	
02	Drive	FLP Electrical motor	
03	Duty	Continuous	
04	Location	As per SOR	
05	Feed Gas & Operating Condition		
a.	Suction Pressure	30 ~ 210 kg/cm ² (g)	
b.	Suction Temperature	Max. 35°C	
c.	Discharge Pressure	220 ~ 255 kg/cm ² (g)	
d.	Discharge Temperature	Max. 52°C	
e.	Required Flow rate	400 SCM/H	
f.	Max. flow rate @ 100 % of full flow suction @ 255 kg/cm ²	#	
g.	Service	Natural Gas	
h.	Gas Handled	Compressed natural Gas	
i.	Compressor Type	Non-Lubricating type, Variable suction Pressure Reciprocating	
j.	Compressor Model No.	#	
k.	Capacity control	VTA Automatic type SOV with NO/NC Configuration	
l.	Driver & Driven Type	Electrical Motor Driven Coupled with Oil pump	
m.	Cylinder configuration	#	
n.	No. of stages	02	
o.	Gas composition	Refer tender documents	
p.	Design code	API 618, API-11P 2nd edition, ISO 16923, ISO 12213-2 (2006)	
q.	Driver Motor rating (KW)	#	
r.	Noise level	75 dBA+/-3 @ from 1 meter Distance of compressor	
s.	Oil & moisture filtration level	0.005 µm/mm Max.	
06.	Site Condition		
a.	Site Ambient Temp. (°C)	Min: 2 °C Max: 47.5 °C	
b.	Max. Relative Humidity (%)	90 %	
c.	MSL	205 Meter	
d.	Area of classification	Class1 Div1 Group D or Zone 1 Div1 Group IIA Group IIB	
e.	Std. condition Referred to	Standard Atmospheric pressure (1.033 Kg/cm ² abs) and Temperature of 15.6°C	
07.	Applicable codes & specifications		



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a.	Compressor	As per Tender	
b.	Piping	ASME/ANSI B31.3	
c.	Driver	Refer electrical motor Data sheet	
d.	Air Cooled Exchangers	API 661/OEM Design std.	
e.	Pressure Vessel	ASME Sec. VIII Div.1/2	
f.	Water cooled exchangers	TEMA 'C'	
g.	Control Panel	Refer Tech Specs.	
08.	Manufacturer's Specifications		
a.	Name / Model	#	
b.	No. Off stroke/min.	#	
c.	Type	#	
d.	Service/Stage	CNG / 2	
e.	PESO Approval number	#	
f.	Direction of Rotation from driving end	#	
g.	Compressor weight (kg)	#	
h.	Compressor weight including cascade frame	#	
i.	Differential pressure @ suction	#	
j.	Differential pressure @ Discharge	#	
k.	Differential pressure gauge at suction & discharge	Required	
l.	Foot print area of compressor package (Mtr. x Mtr.)	#	
m.	Frame Lifting arrangement's	Required	
n.	Pressure Relief valve on each stage of Discharge	Required	
09.	Inlet Condition (Suction)		
a.	Suction Pressure kg/cm ²	30 to 220 kg/cm ²	
b.	Suction Temperature °C-Actual	35°C Max.	
c.	Suction Temperature °C-Adiabatic	#	
d.	Molecular Weight(constant) kg/k-mol.	#	
e.	Specific gravity of gas inlet	#	
f.	Mass Density at NTP kg/m ³	#	
g.	Mass Density @ 220 kg/cm ² @30°C kg/m ³	#	
h.	Mass Density @ 30 kg/cm ³ @35°C kg/m ³	#	
i.	Specific heat ratio (Cp/Cv ratio)	#	
j.	Compressibility factor @220 kg/cm ²	#	
k.	Compressibility factor @30 kg/cm ²	#	
l.	Polytrophic Efficiency @ 30 KG/CM ² (%)	#	
m.	Polytrophic Efficiency @ 220 KG/CM ² (%)	#	
n.	Duty (KW)	#	



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o.	Min. specific power consumption /kg (KWH/kg)	#	
p.	Max. specific power consumption/kg (KWH/kg)	#	
q.	Max. power consumption @ suction 30 kg/cm ² KWH	#	
o.	Min. power consumption @ suction 220 kg/cm ² KWH	#	
P.	Driver Rating & Speed (KW/RPM)	#	
q.	Adiabatic power (KW)	#	
r.	Shaft power @ RV Set pressure (KW)	#	
10.	Outlet Condition (Discharge)		
a.	Discharge Pressure kg/cm ²	255	
b.	Discharge Temperature-Actual	52°C Max.	
c.	Discharge Temperature-Adiabatic	#	
d.	Molecular Weight	#	
e.	Specific gravity of gas outlet	#	
f.	Mass Density @ 250 kg/cm ²	#	
g.	Specific heat ratio (Cp/Cv ratio)	#	
h.	Compressibility factor	#	
i.	Polytropic Efficiency @ P ₂ /P ₁ =8.3	#	
j.	Duty (KW)	#	
k.	Isentropic Power Required	#	
l.	Actual power Required	#	
m.	Volumetric efficiency %	#	
11.	Process End line specifications		
a.	Suction Line (Size)-inch.	3/4"	
b.	Discharge Line (Size)-inch.	3/4"	
c.	All gas piping/ tubing, valves, fittings material	SS316/SS316L	
d.	Pressure Ratings	5000 PSIG. (315 kg/cm ²)	
12.	Vibrations		
f.	Max. Cylinder Vibration mm/sec.	10 mm/sec. unfiltered peak velocity	
g.	Max. Vibration @ compressor base mm/sec.	5 mm/sec. unfiltered peak velocity	
13.	Compressor controls		
a.	Automatic start / stop on storage pressure level and manual start stop	Pneumatic operated Actuators (Automatic)	
b.	Actuator No. of acting type	#	
c.	Automatic drain of separators	manual	
d.	Air vent protector	Required	
e.	Fail safe Mode	#	
f.	Actuator driving degree of shaft	#	



**DATA SHEET OF 400 SCM/H CNG BOOSTER
COMPRESSOR**

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g.	Actuator output torque	#	
h.	Std. operating pressure	#	
i.	Pressure supply range	#	
j.	Actuator air supply requirements / min.	#	
k.	Actuator Air vent (consumption)/min.	#	
	Compressor to start automatically after power Interruption with 10 second's delay.	#	
	Automatic recovery of gas form gas recovery vessel.	#	
	Automatic closing of suction isolating valve on Compressor trip.	#	
	Intermediate devices (to be included in supply)	#	
	-Solenoid valves	#	
	-Manual -mounted in a local panel.	#	
	-Controller (For auto control)	#	
	-Pressure switches (For auto control)	#	
l.	Compressor shall unload on Power failure/ stoppage through (Vendor to indicate) ☐ Automatically ☐ Manually	#	
m.	Compressor shall load on start through (Vendor to indicate) ☐ Automatically ☐ Manually	#	
14.	Cylinder Packaging Data	1st stage	2nd stage
a.	Cylinder bore & Stroke length (mm)	#	#
b.	Single / Double Acting	#	#
c.	Liner (yes/ No.)	#	#
d.	Type of Valves	#	#
e.	Piston Displacement (M3/Hr)	#	#
f.	Volumetric Efficiency (%) Min/ Nor/max	#	#
g.	Mean Piston Speed(m/sec) [<3.5 m/sec]	#	#
h.	Suction / discharge valve gas velocity (m/sec)	#	#
i.	Relief valve setting (kg/cm ² g)	#	#
j.	Max. allowable cylinder pressure (kg/cm ² g)	#	#
k.	Max. allowable cylinder temperature (°C)	#	#
l.	Pneumatic Test By (kg/cm ² g)	#	#
m.	Hydrostatic test (kg/cm ² g)	#	#
n.	Hydrostatic test water jacket (kg/cm ² g)	#	#
o.	Suction / discharge nozzle orientation	#	#
p.	No. of suction / discharge valves	#	#
q.	Piston Rod Dia. (mm)	#	#
r.	Piston Load	#	#
	-Max. Rod load gas compression, Kg	#	#



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	-Max. Rod load gas tension, Kg	#	#	
	- Rod load (Inertia alone) compression, Kg	#	#	
	- Rod load (Inertia alone) tension, Kg	#	#	
	-Rod load rated Pr. (gas + inertia) compression, Kg	#	#	
	-Rod load rated Pr. (gas + inertia) tension, Kg	#	#	
	-Rod load at R.V set Pr. (gas + inertia) compression, Kg	#	#	
	-Rod load at R.V. set Pr. (gas + inertia) tension, Kg	#	#	
s.	Max. Permissible Piston Rod loads (kg)	#	#	
t.	Rod packing cooling by liquid (yes/No.)	#	#	
u.	Rod packing lubrication (Yes/No)	#	#	
v.	Rod packing vent to	#	#	
w.	distance piece type	#	#	
x.	Cylinder jacket cooling by	#	#	
y.	Cylinder wall thickness	#	#	
z.	Oil & gas seal durability Hrs.	#	#	
15.	Lubrication system			
a.	Splash/ Force feed Lubricated/ Mini Lubricated/Non lubricated	#		
b.	Type of bearing on Main end / Big end	#		
c.	Bearing No. if any	#		
d.	Grade / viscosity of lube oil	#		
e.	Oil sump/Tank capacity	#		
f.	Change lube oil after (Hrs.)	#		
g.	Air Breather	Required		
h.	Oil Return line filter	Required		
-	Filtration level in microns	#		
-	Return line filter differential pressure (kg/cm ²)	#		
-	Grade / viscosity of lube oil	#		
i.	Oil level indicator	#		
j.	Oil flow rate @ 100 kg/cm ² RV set point	#		
k.	Oil flow rate @ 150 kg/cm ² RV set point	#		
l.	Force required to push upward stroke	#		
j.	Force required to push downward stroke	#		
k.	Max. current achieved at upward stroke	#		
l.	Max. current required at downward stroke	#		
m.	Oil pump Type (vane/gear/Piston)	#		
n.	Oil pump Volumetric capacity LPM @ max.	#		



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	pressure				
o.	Oil pump Volumetric capacity LPM @ min. pressure	#			
p.	Oil pump Volumetric capacity LPM @ Nor. Pressure	#			
q.	Oil cooler type (tube & shell Type, fin type, Plate type)	#			
r.	Gas Cooling Methods/Media	#			
s.	Water cooling methods /Media	#			
t.	Oil cooling methods / Media	#			
16.	COOLERS	Oil Cooler (Required)	After cooler	Inter coolers	
a.	Cooler Type				
b.	Tube Material				
c.	Tube sheet material shell Material				
d.	Shell Material				
e.	ASME / IBR CODE STAMP/TPIA	Yes	Yes	Yes	
f.	Suction Strainer	Required	Required	Required	
17.	Instrumentation				
a.	Pressure gauge	Required (Refer technical Vol. II)			
b.	Temperature gauge	Required (Refer technical Vol. II)			
c.	Pressure Transmitter	Required (Refer technical Vol. II)			
d.	Temperature transmitter	Required (Refer technical Vol. II)			
e.	Differential pressure gauge	Required (Refer technical Vol. II)			
f.	Mass flow meter	Required (Refer technical Vol. II)			
g.	Gas detector	Required (Refer technical Vol. II)			
h.	Flame detector	Required (Refer technical Vol. II)			
j.	Pressure switch	Required (Refer technical Vol. II)			
k.	Temperature switch	Required (Refer technical Vol. II)			
l.	level indicator	Required (Refer technical Vol. II)			
j.	Flow indicator	Required (Refer technical Vol. II)			
k.	Safety relief valve	Required (Refer technical Vol. II)			
l.	Flame arresters	Required (Refer technical Vol. II)			
j.	Level transmitter	Required (Refer technical Vol. II)			
18.	Other Instrumentation				
a.	Junction Box with interconnecting wiring	Required			
b.	Pressure Relief Valve at discharge each stage	Required			
c.	Pressure Relief Valve at suction to compressor	NA			
d.	Instruments for closed circuit cooling water system	Required			
e.	Hour meter	Required			



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f.	Gas detection system	Required		
g.	Flame detection system	Required		
h.	Forced feed lubrication failure to stop comp.	Required		
i.	Priority fill panel	Required		
j.	Emergency shut down system	Required		
19.	MATERIAL OF CONSTRUCTION & GRADES	1 st stage	2 nd stage	
a.	Cylinder & Head	#	#	
b.	Liner	#	#	
c.	Piston	#	#	
d.	Piston Ring	#	#	
e.	Piston Rod	#	#	
f.	Oil scraper Ring	#	#	
g.	Gas sealing's	#	#	
h.	Non return Valve	#	#	
i.	Cylinder Packing Ring	#	#	
j.	Tie rod	#	#	
k.	Piston rod seal ring	#	#	
l.	Piston rod F.O.S(factor of safety)	#		
l.	Piston Rod Yield strength (SMYS)	#		
l.	Piston Rod Hardness	#		
k.	Cylinder wall surface finish (Rz & Ra)	#		
20.	General Electrical Parameters			
a.	Supply Voltage	415+ 10% volt, 3 phases,50+ 5%Hz		
b.	Protection	Flame proof & weather proof enclosure		
c.	Insulation	Class F with class B temperature rise		
d.	Mounting	Horizontal Foot Mounting		
e.	Earthing System	As per IS 3043		
21.	Radiator cooler Parameters			
a.	Service	#		
b.	Overall fins area	#		
c.	Size Type (Induced Force)	#		
d.	Surface per Unit-Finned Tube (m2)	#		
e.	Heat Exchanged (KW)	#		
f.	Craft/No. of Bays	#		
g.	Bare Tube (m2)	#		
h.	MTD. Eff °C	#		
i.	Transfer Rate * Finned Tube: Bare Tube, Service Clean W/m2 °C	#		



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j.	Water temperature inlet (warm)	#								
l.	Water temperature out (cold)	#								
m.	Ambient air temperature inlet(cold)	#								
n.	Ambient air temperature outlet(warm)	#								
o.	Water mass flow at max. pressure	#								
p.	Water mass flow at min. pressure	#								
q.	Water mass flow at normal pressure	#								
r.	Max. Air flow rate @ full load of compressor	#								
s.	Min. Air flow rate @ when one blower works other standby mode	#								
22.	Performance Data (Bidder to calculate)	Gas		Gas		Gas				
		In	Out	In	Out	In	Out			
Gas Cooling	Inter Cooler 1st Stage	Flow Kg/s								
		Temp °C								
		Pr. Kg/cm ²								
		Total Heat KJ/Kg								
		Heat Transfer rate j/sec.								
	Inter Cooler 2nd Stage	Flow Kg/s								
		Temp °C								
		Pr. Kg/cm ²								
		Total Heat KJ/Kg								
		Heat Transfer rate j/sec.								
	After Cooler Stage	Flow Kg/s								
		Temp °C								
		Pr. Kg/cm ²								
		Total Heat KJ/Kg								
		Heat Transfer rate j/sec.								
Oil cooler	Flow Kg/s									
	Temp °C									
	Pr. Kg/cm ²									
	Total Heat KJ/Kg									
	Heat Transfer rate j/sec.									
Water / Air cooler stage	Flow Kg/s									
	Temp °C									
	Pr. Kg/cm ²									
	Total Heat KJ/Kg									
	Heat Transfer rate j/sec.									



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Total Heat all stream in KJ/Kg	Gas	Oil	Water	Air

NOTE: -

1. Given temperature is booster compressor package outlet temperature. However, compressor discharge temperature will be provided by vendor in their compressor datasheet.
2. Compressor vendor to confirm the compressor ratio.
3. Compressor vendor shall ensure the suitability of the material of construction for the Booster compressor for 35 °C suction temperatures & corresponding discharge temperature(s).
4. The discharge pressure provided is the discharge pressure at the end of final stages of compression.
5. The discharge pressure at each stage must be provided by vendor.
6. The discharge temperature at each stage must be provided by vendor.
7. Polytrophic efficiency will be provided by compressor vendor.
8. The given duty is the Absorbed power of the compressor.
9. Vendor to consider 10% margin on the flow rate to design compressor.
10. Design Life of the compressors shall be minimum 25 years.
11. Ambient Temperature: 2 °C MIN / 47.5 °C MAX.
12. The given suction and discharge properties is based on Design Operating case. Also, vendor to design compressor suitable for Normal gas composition range as specified above.
13. **# VTC - Vendor to Confirm**



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DATA SHEET OF INTERCOOLER & AFTER COOLER

DOCUMENT NO

C221086

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3042

SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC
BOOSTER COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW
CAPACITY OF 400 SCMH IN GEOGRAPHICAL AREAS OF
RAJASTHAN

C1	07.04.2026	ISSUED FOR CLIENT REVIEW	AB	SA	AB
REV	DATE	DESCRIPTION	PREP	CHK	APPR



**SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC BOOSTER
COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW CAPACITY OF 400 SCM/H IN
GEOGRAPHICAL AREAS OF RAJASTHAN**

1	GENERAL			
2	PROJECT: SUPPLY OF 400 SCM/H CNG BOOSTER COMPRESSOR			
3	OWNER: M/S HINDUSTAN PETROLEUM CORP. LTD. SITE:			
4	Item No.:	Service: Intercooler / After cooler for Compressor Package		
5	NOTE: ■ SCOPE OPTION / INFORMATION SPECIFIED BY PURCHASER □ INFORMATION REQUIRED FROM VENDOR.			
6	□ Manufacturer:	Type: □ Forced Draft □ Induced Draft		
7	□ Bundle Size: m x m x m	Bundles/Section	Number of Units:	
8	Bundles/Unit:	In Parallel / Series	Section Size:	
9	Surface/Bundle: m ²	Bare Tube: m ²	Section/Unit:	
10	Surface/Unit: m ²	Bare Tube: m ²	Plot Area/Unit:	
11	PERFORMANCE (Of One Unit)			
12	Heat Exchanged: kcal/hr.	MTD (Corrected): °C		
13	Transfer Rate: kcal/hr. m ² °C	(Finned Surface)	(Bare Surface)	
14	TUBE SIDE			
15	Fluid Circulated	WATER	Gravity: Liquid API SG @ 15.4EC	
16	Total Entering fluid kg/hr.	Enthalpy / Latent Heat kcal/kg		
17	Operating Temperature	In:	Out:	Fouling Resistance hr. m ² °C/kcal
18	Operating Pressure Passes / Bundle kg/cm ²			
19	AIR SIDE			
20	Temperature	In: 47.5	Out:	Altitude (Mtr.)
21	Total Flow/Unit kg/hr.	Static Pressure kg/cm ²		
22	Quantity/Fan kg/hr.	Driving Power/Fan kW:		
23	Face Velocity m/sec.	Fan protection class:		
24	CONSTRUCTION (Each Bundle)			
25	Design Pressure: kg/cm ² g	Test Pressure: kg/cm ² g		
26	Design Temperature: °C			
27	Code Requirements:			
28	Type of Tubing:	Tube Material:	Fin Material: Al	
29	Tube Bare Tubes (no's):	No. of rows:	O.D.:	BWG/Thk
30			Length:	
31	Fins: Spacing /inch. O.D.	Root Dia.	Thickness:	
32	Header Type: Plug / Cover	No. of Splits:	Material:	
33	Plugs/Gaskets	Side Frame: C.S. Inside Zinc Protected		
34	Nozzles	In:	Out:	
35	Couplings	Vent:	Drain:	
36	CONSTRUCTION (Each section)			
37	Structure	CS	Sec/Gr. No.	Design Wind Load: kgf /m
38	Plenum Chamber	CS inside Zinc Protected		Type:
39	Fans	No.	Dia.	RPM
40	Blades	Material:	No./Fan	Pitch Angle (Design):
41	Hubs	Material:	Pitch: Auto variable / Adjustable (No.)	
42	Louvers	Material:	Type:	Mfr.
43	Weights kg Each Section (Dry):		Full of Water:	
44	Each Bundle (Dry):		Full of Water:	
45	Blower Dia.:	Projected width:	No. of blades:	
46	Max. Vol. of water flow (LPM):		Max. Vol. of Gas flow (LPM):	
47	Max. Vol. of Air flow (CFM):		Max. Vol. of Oil flow (LPM):	



**SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC BOOSTER
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43	APPLICABLE SPECIFICATIONS API Standard 661
44	REMARKS 1. Air coolers shall be designed for 20% excess capacity than required normally.
45	Exchanger shall be designed with air side temperature of 47 °C.
46	Separate data sheet shall be filled by the bidder for each service i.e., Inter cooler and After cooler
47	Vendor to calculate Details of (Q) Heat to be dissipated by water @ max. water temperature 60°C , (V) water flow rate (L/min.) in side cooling coil , Inlet temperature of water (T water) , inlet Ambient temperature (T ambient) & actual cooling capacity of cooler in KW.
48	Vendor to provide Radiator cooler performance curve for KW/K Vs. flow rate & Pressure loss. Vs. flow rate of water inside Radiator cooler during detail engineering.
49	During cooling load calculation vendor shall consider Max. Ambient temperature as per detail mentioned in technical tender documents.

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DATA SHEET OF GLYCERINE FILLED PRESSURE GAUGE

DOCUMENT NO

C221086

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BOOSTER COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW
CAPACITY OF 400 SCMH IN GEOGRAPHICAL AREAS OF
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COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW CAPACITY OF 400 SCMH IN
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PRESSURE GAUGE DATASHEET			
S. No.	DESCRIPTION	SPECIFICATION	OFFERED
1.	Type of service	CNG	
2.	Pressure Gauge Type	Bourdon sensing	
3.	Mounting	Local	
4.	Model	#	
5.	Case Filling	Liquid Filled Case	
6.	Dial Size of Pressure Gauge	100/63 mm	
7.	Color	White with black inscriptions	
8.	Case Material	SS316 with bayonet bezel Phenol with screwed bezel	
9.	Bezel Ring	Bayonet type SS316	
10.	Window	Shatterproof glass	
11.	Enclosure	WP to IP 65 as per IEC 60529 / IS 2147	
12.	Pressure Elements	Bourdon type	
13.	Elements/Socket Material	SS 316	
14.	Accuracy	+/-0.5% FSD	
15.	Zero Adjustments	Micro pointer	
16.	Connection	#	
17.	Connection location	Bottom or Back	
18.	Movements	SS316	
19.	Range	0 to 400 kg/cm ²	
20.	Least Count	10 kg/cm ²	
21.	Ref. Standards	EN 837 Latest version	
22.	Protection	WP to IP68 (IS:13947 part I / IEC:60529)	
23.	Bourdon	SS316	
24.	Connection	½" or ¼" NPT(M)	
25.	Over Range	As per EN 837	
26.	Zero Adjustment	#	
27.	Socket	22 mm SQ. in SS316	
28.	Blow out Disc	Required	
29.	Temperature Suitability	-20°C to 60°C	
30.	Temperature Effects	Within +/-0.4 % FSD/10°C	
31.	Diaphragm seal	#	
32.	Wetted parts materials	#	
33.	Process connection size & ratings	#	



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34.	Flushing & filling connection with	#	
35.	Over range protection	130 % of FSD	
36.	Temperature error	#	
37.	Approvals	#	
38.	Make		
39.	CE conformity	Pressure equipment directive 97/23/EC, PS > 200 bar, module A, pressure accessory	
39.	Approvals	#	
40.	Certificates	2.2 Test report per EN 10204 (e.g., state-of-the-art manufacturing, material proof, indication accuracy) 3.1 inspection certificate per EN 10204 (e.g., indication accuracy)	
41.	Shipping weight(kg)	#	

42	Notes.
a.	(#) Marked data to be furnished by Vendor/Manufacturer
b.	At the time of approval of DS, Supplier shall furnish DS with Catalogues.
c.	Vendor to provide all Pressure Tag List with Aluminum embossed plate (where mention with Tag no./Date of calibration/Due date of Calibration /Items serial no etc.)
d.	Vendor to provide Pressure gauge spare items like Glycerin Liquid & Rubber filling cap with mandatory spares.
e.	Vendor to provide Operating and Instruction manual before dispatch.
f.	Vendor to provide Leak rate of pressure gauge on High pressure application of natural gas.
g.	Gauge shall withstand steady pressure, over pressure & cyclic pressure.
h.	Gauge shall withstand mechanical shock & Vibration within accuracy range.
i.	Gauge shall be equipped with Manifold valve or bleed valve and venting system in case of maintenance.
j.	Gauge shall have designed to satisfy requirements to operate in aggressive Environment.



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DATA SHEET OF SAFETY RELIEF VALVES

DOCUMENT NO

C221086

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BOOSTER COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW
CAPACITY OF 400 SCMH IN GEOGRAPHICAL AREAS OF
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**SAFETY-RELIEF VALVE
DATASHEET**

S. No.	DESCRIPTION	SPECIFICATION	OFFERED
1.	Type of Service	CNG	
2.	Tag No.	#	
3.	Model No. / valve series	#	
4.	Make/Model	Refer approved vendor list	
5.	Orifice	#	
6.	Inlet specification	#	
7.	Outlet specification	#	
8.	Set pressure	Refer Technical specifications	
9.	CDS pressure	#	
10.	Relieving temperature	45°C	
11.	Back pressure	#	
12.	Min. Seat Leakage pressure	#	
13.	Max. Seat Leakage pressure	#	
14.	Allowable Leakage rate @ BPM (bubble per min.) at Min. seat leakage	#	
15.	Allowable Leakage rate @ BPM (bubble per min.) at Max. seat leakage	#	
16.	Design code	ASME sec. VIII Div. 1&2	
17.	Operating Temperature Range	-10°C to 60°C	
18.	Valve Design temperature Range	#	
19.	Min. Set pressure point	#	
20.	Max. Set pressure point	#	
21.	Max. outlet pressure	#	
22.	Seal Type	Metal to Metal seat /O-ring soft seal	
23.	Material of Construction		
a.	Base	SS316	
b.	Cylinder	Carbon steel SA-216 gr. WCB	
c.	Disc Insert	SS316	
d.	Disc Holder	SS316	
e.	O-Ring	#	
f.	Guide	SS316	
g.	Spindle	416 SS	



DATA SHEET OF SAFETY RELIEF VALVE

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h.	Spring	17-7 PH SS	
i.	Spring washers	416 SS	
j.	Adjusting Bolt	SS 316	
K.	Adjusting Bolt Nut	SS 316	
l.	Threaded Cap	#	
L.	Metal to Metal Seat material	#	
24.	Effective area in Sq. mm @ Min. set pressure	#	
25.	Effective area in Sq. mm @ Max. set pressure	#	
26.	Service Condition		
a.	Fluid	Natural gas	
b.	State	Pure compressible gas (Natural Gas)	
c.	Molecular weight (	16.04	
d.	Specific heat Ratio (Cp/Cv) @ 15.5°C, 1 atm.	1.31	
e.	Relieving temperature	#	
f.	Overpressure %	10 %	
g.	Max. Fixed Blow down %	≤ 20 %	
h.	Blow down %	#	
i.	Back pressure	#	
j.	Critical flow nozzle pressure	#	
k.	Effective coefficient of discharge (Kd)	#	
l.	Combination correction factor (Kc)	#	
m.	Compressibility factor(z)	0.9	
n.	Required effective discharge area	#	
o.	Recommended std. effective orifice area	#	
27.	Seat Leakage Test Pressure	#	
28.	Shell Pneumatic Test pressure & Media	#	
29.	Cylinder Hydro test pressure & Media	#	
30.	Base Hydro Test Pressure	#	



**SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC BOOSTER
COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW CAPACITY OF 400 SCMH IN
GEOGRAPHICAL AREAS OF RAJASTHAN**

31.	Notes.
a.	(#) Marked data to be furnished by Vendor/Manufacturer
b.	At the time of approval of DS, Supplier shall furnish DS with Catalogues.
c.	Vendor to provide all pressure safety valve Tag List with Aluminum embossed plate (where mention with Tag no. /Date of calibration/Due date of Calibration /Items serial no etc.)
d.	Vendor to provide Operating and Instruction manual before dispatch.
e.	Process data shall be as per tender specification mentioned elsewhere.
f.	At the time of approval, vendor shall furnish sizing calculation and catalogues of SRV.
g.	Vendor to confirm SRV cold differential test pressure (CDTP).
h.	This Relief requires valves OPEN when system pressure reaches the set pressure and CLOSE when system pressure falls below the set pressure.
i.	Vendor to provide Inlet pressure of relief valve Vs Flow rate curve during detail engineering.
j.	Vendor to provide Maximum & Min. set pressure range of safety relief valve during detail engineering.
l.	Differential pressure (ΔP) equals inlet pressure (set pressure plus overpressure) at flowing conditions minus back pressure.
m.	Vendor to share Back pressure flow correction factor curves.



ENERGISING QUALITY

**PROJECT NUMBER:
VCS-HPCL-C221086**



DATA SHEET OF GAS DETECTOR

DOCUMENT NO

C221086

RJ

ME

DS

3045

SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC
BOOSTER COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW
CAPACITY OF 400 SCMH IN GEOGRAPHICAL AREAS OF
RAJASTHAN

C1	07.04.2026	ISSUED FOR CLIENT REVIEW	AB	SA	AB
REV	DATE	DESCRIPTION	PREP	CHK	APPR



**SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC BOOSTER
COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW CAPACITY OF 400 SCMH IN
GEOGRAPHICAL AREAS OF RAJASTHAN**

Type	Infrared sensor with Transmitter, with Local Display or Indicator,
Measurement Range	0-100% LEL
Operating Medium	Natural Gas.
Operational and Certified Temperature Range	65 °C Maximum
Power Supply	18 to 32Vdc (24Vdc nominal)
Power Consumption	< 4.5W max
Current Demand	< 235 mA at 18V
	< 190 mA at 24V
	< 155 mA at 32V
Output Signals :-	mA, Relay, Hart
(all 3 below mentioned outputs must be available in separate connectors/ ports)	
1. Analog (mA):	4-20mA, Non-Isolated, Current Source or Sink. (Default Configuration is Current Source).
Maximum 4-20mA Loop Resistance	600 Ohms
Measuring Range (0-100% FSD)	4-20 mA
Inhibit	1 to 3Ma (Default 2mA)
Warning	0 to 6mA (Default 3mA)
Fault	0mA
Over-Range	20 to 21.5mA (Default 21 mA)
2.Contact output:	Relay output: (High alarm, Low alarm, Trip, Fault) Contact rating: 2 A, 230 VAC/24 VDC
3. HART	
protocol	HART protocol
	External HART Port connector (without opening of Housing cover of Gas Detector) is required in transmitter for calibration and parameter accessing.
Status Indicator:	Status indication: LED indication OR Local Display. Separate indication for No gas leak, Low alarm, High alarm, optics failure and calibration in progress shall be available
Vibration	As the detector shall be vibrations arising mounted inside CNG compressor canopy, the same shall be capable of withstanding from reciprocating compressor.
Accuracy	± 5 % LEL through-out the range
Response Time	Less than 5 seconds
ACCESSORIES	Housing: Anodized aluminum with powder coated finish or equivalent (Explosion proof enclosure) Mounting: Roof mounted or wall mounted. For threaded type detectors suitable termination box (approved for use inside hazardous area) shall be provided. The detector shall be supplied with suitable Sunshade/ Deluge protection, Strom baffle, Weather protection, Dust barrier, Gassing Cap, etc. as applicable protection against dust particles.
	2 No's Cable Glands (3/4" Double compression)



**SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC BOOSTER
COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW CAPACITY OF 400 SCMH IN
GEOGRAPHICAL AREAS OF RAJASTHAN**

WARRANTY	4 year warranty against sensor and electronics for the supplied product against CAMC.
	Onsite Service to Be Provided During Warranty Period
Safety Approval /Certificate	For both sensor, Transmitter and Terminal box / Junction Box
ATEX/UT/UT/CSA /FM/CCOE	ATEX: BAS992259X II 2GD EEx d IIC T100°C (T _{Ambient} -40°C to +55°C) T135°C (T _{Ambient} -40°C to +65°C)
	UL / CSA Class 1, Div. 1, groups B, C, and D (-40°C to +65°C) GOST and SAA
	CE, IEC, EEC, EMC, CENELEC AND Exd.
	IP66/67, NEMA 4
CCOE	CCOE approval Must
Performance Approval	EN61779 EXAM, BVS 03 ATEX G 016 X, CSA, FM C22.2 152
EMC COMPLIANCE SOFTWARE	EN 50270; EN 50271
Manuals (soft and Hard Form)	Operation, Maintenance, Calibration and Troubleshooting
Software (soft and Hard Form)	Software or additional device, cables required for calibration or troubleshooting of instrument shall be supplied free of cost with instrument.
GAS DETECTOR CALIBRATION	Must be inbuilt or else calibration device must be supplied along with GD at free of cost.
CALIBRATION PROCEDURE	(BOTH ZERO AND SPAN) FOR THE MODEL QUOTED TO BE PROVIDED IN DETAIL



ENERGISING QUALITY

**PROJECT NUMBER:
VCS-HPCL-C221086**



DATA SHEET OF FLAME DETECTOR

DOCUMENT NO

C221086

RJ

ME

DS

3046

SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC
BOOSTER COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW
CAPACITY OF 400 SCMH IN GEOGRAPHICAL AREAS OF
RAJASTHAN

C1	07.04.2026	ISSUED FOR CLIENT REVIEW	AB	SA	AB
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**SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC BOOSTER
COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW CAPACITY OF 400 SCMH IN
GEOGRAPHICAL AREAS OF RAJASTHAN**

Flame detector Type:	UV and IR Detector
Wave Length	UV radiation over the range of 185 to 260 nanometers (1850 to 2600 angstroms)
	IR radiation in the 4.35 micron range
Field of View	120° cone of vision (or) Better.
Sensitivity	Approved Performance Specification-50 feet (15.2m) distance for a 1 sq. ft. (0.092m ²) heptane fire.
Response Time:	Less than 5 seconds
Status Indicator:	LEDs indicate status of Healthy, fault, and alarm conditions
Operating Medium	Natural Gas.
Nominal supply voltage:	24 VDC
Supply voltage range:	20 to 36 VDC
Maximum supply current:	400 mA during COPM only
Typical current:	80 to 150 mA Depend on relays ETC
Maximum output signal load:	600 Ohms
Electrical Specifications [OUTPUT: mA, Relay, Hart]	
(all 3 below mentioned outputs must be available in separate connectors/ ports)	
1. Analog (mA):	
Output signal range:	0 to 20 mA *
FAULT signal:	0 to 0.2 mA *
COPM fault signal:	2.0 ± 0.2 mA *
Ready signal:	4.0± 0.2 Ma
IR only signal:	8.0 ± 0.2 Ma
UV only signal:	12.0 ±0.2 mA
WARN signal:	16.0 ±0.2 Ma
ALARM signal:	20.0 ± 0.2 mA
2. Relay Contacts: (Alarm, Fault and Auxiliary)	Fire relay (alarm)- Both Contact output NO and NC required. Fault relay - Contact output NO and NC required. Contact rating 30 VDC, 2 A or 125 VAC, 1 A
3. HART	
Protocol	HART protocol
	External HART Port connector (without opening of Housing cover of Flame Detector) is required in transmitter for calibration and parameter accessing.
Vibration	As the detector shall be vibrations arising mounted inside CNG compressor canopy, the same shall be capable of withstanding from reciprocating compressor.
Approvals (sensor, Transmitter and Terminal box / Junction Box)	CCOE, CSA, FM, ATEX, HART Registered, and AMS Aware



DATA SHEET OF FLAME DETECTOR

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**SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC BOOSTER
COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW CAPACITY OF 400 SCMH IN
GEOGRAPHICAL AREAS OF RAJASTHAN**

	Class 1, Div. 1, groups B, C, and D (-40°C to +65°C), Type 4X, Exd iic, T5
CCOE	CCOE approval Must
	IP54
Reliability	IEC 61508
AMS Aware:	Certified by AMS
RFI/EMI Protection:	Complies with EN 50130-4, EN 61000-6-4
WARRANTY	Four year warranty against sensor and electronics for the supplied product against CAMC.
	ONSITE SERVICE TO BE PROVIDED DURING WARRANTY PERIOD
ACCESSORIES	Housing: Anodized aluminum with powder coated finish or equivalent (Explosion proof enclosure) Mounting: Roof mounted or wall mounted. For threaded type detectors suitable termination box (approved for use inside hazardous area) shall be provided. The detector shall be supplied with suitable Sunshade/ Deluge protection, Strom baffle, Weather protection, Dust barrier, Gassing Cap, etc. as applicable protection against dust particles. .
	2 No's Cable Glands (3/4" Double compression)
FLAME DETECTOR BUILT-IN TEST	AUTOMATIC AND MANUAL TEST
FLAME DETECTOR IMMUNITY	False alarm sources like Arc Welding, X-Ray's or hot vibrating object, lightening.
Manuals (soft and Hard Form)	Operation, Maintenance, Calibration and Troubleshooting
Software (soft and Hard Form)	Software or additional device, cables required for calibration or troubleshooting of instrument shall be supplied free of cost with instrument.
FLAME DETECTOR CALIBRATION	Must be inbuilt or else calibration device must be supplied along with FD at free of cost.
CALIBRATION PROCEDURE	(BOTH ZERO AND SPAN) FOR THE MODEL QUOTED TO BE PROVIDED IN DETAIL



DATA SHEET OF FLAME DETECTOR

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ENERGISING QUALITY

**PROJECT NUMBER:
VCS-HPCL-C221086**



DATA SHEET OF KWH METER

DOCUMENT NO

C221086

RJ

ME

DS

3047

SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC
BOOSTER COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW
CAPACITY OF 400 SCMH IN GEOGRAPHICAL AREAS OF
RAJASTHAN

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**SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC BOOSTER
COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW CAPACITY OF 400 SCMH IN
GEOGRAPHICAL AREAS OF RAJASTHAN**

SPECIFICATION FOR KWH METER

1	Meter type	3 Phase 4 wire Static Energy meter
2	Accuracy class	0.5 S as per IS14697
3	Connection	Transformer Operated
4	Rated Voltage	240V(P-N), 415V(P-P) $\pm$ 10%
5	Rated Basic current	5A
6	Rated maximum current	10A
7	Rated Frequency	50 HZ
8	Power Factor range	Zero lag- Unity- Zero lead
9	Standards compliance	IS 14697, IEC 62052-11, IEC 62053-21, IS 15959
10	Calibration	Meter shall be calibrated at factory and no modifications should be permissible
11	Energy Audit Data	a. Meter serial number b. Date and time c. Cumulative forwarded active energy d. Cumulative reactive energy - Lag e. Cumulative reactive energy - Lead f. Cumulative forwarded apparent energy g. Cumulative Maximum Demand in kW and kVA with date and time
12	Metering philosophy	Metering should be 2 quadrant lag only and programmed accordingly
13	Auto/Manual Scroll mode	a. LCD test b. Date c. Time d. Cumulative Active Energy e. Cumulative Apparent Energy f. Cumulative Reactive Energy – Lag g. Cumulative Reactive Energy – Lead h. Active Maximum demand with date and time i. Apparent Maximum demand with date and time j. Active load k. Reactive load l. Apparent load m. Phase wise power factor n. Average power factor o. R phase voltage p. Y phase voltage q. B phase voltage r. R phase current (line) s. Y phase current (line) t. B phase current (line) u. Instantaneous average power factor with sign for lag/lead v. Frequency reading
14	Maximum demand Registration	Meter should store and display maximum demand in kW/kVA with date and time. Demand integration period should be 30 minutes. It is preferred that MD is computed using separate counter rather than by Difference of initial and final energy counter.
15	Auto Reset of Maximum Demand	Meter should reset to zero and date should be customizable date



SPECIFICATION OF DISCHARGE FILTER

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ENERGISING QUALITY

**PROJECT NUMBER:
VCS-HPCL-C221086**



SPECIFICATIONS OF DISCHARGE FILTER

DOCUMENT NO

C221086

RJ

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3048

SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC
BOOSTER COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW
CAPACITY OF 400 SCM/H IN GEOGRAPHICAL AREAS OF
RAJASTHAN

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**SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC BOOSTER
COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW CAPACITY OF 400 SCMH IN
GEOGRAPHICAL AREAS OF RAJASTHAN**

SUPER FINE FILTER (COALESCING FILTER)

Super fine filter for removal of liquid (e.g. water & oil) and solid particles down to 0.01 microns out of compressed natural gas

Residual Oil Contents less than 0.01

mg/m³. Filter Designed:

Paint Compatible

CE mark in accordance with European Directive for Pressure Equipment, PED (97 / 23 / EC)

Are designed to meet the ATEX European Directive for Explosion Protection, (94 / 9 / EC)

All natural gas filters in accordance to CE Eex 2GD IIB T6.

Standard equipment:

Complete filter including manual drain.

Specification Filter Elements.

Filter Fabric: Borosilicate Microfiber Fabric coated with polypropylene photopolymer support - fabric.

Drainage Layer: Parafil - Fibre fabric incorporated in the filter fabric. (Without Foam Sock)

Rib Mesh : Stainless Steel VA

1.4306 Temperature :
+ 1 ° C to + 80 ° C

Direction of Flow: From Inside to Outside.

NOTE:

-Bidder to get the inspection done by TPIA accordance with European Directive for Pressure Equipment, PED (97 / 23 / EC) if CE marks are not available.

-Bidder to provide Filter sizing calculation during detail engineering.

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ENERGISING QUALITY

**PROJECT NUMBER:
VCS-HPCL-C221086**



DATA SHEET FOR ELECTRIC MOTOR

DOCUMENT NO

C221086

RJ

ME

DS

3049

SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC
BOOSTER COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW
CAPACITY OF 400 SCM³/H IN GEOGRAPHICAL AREAS OF
RAJASTHAN

C1	07.04.2026	ISSUED FOR CLIENT REVIEW	AB	SA	AB
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**SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC BOOSTER
COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW CAPACITY OF 400 SCMH IN
GEOGRAPHICAL AREAS OF RAJASTHAN**

ITEM NO.:			As per requirement
QUANTITY:			
DESCRIPTION:			
A.	APPLICABLE SPECIFICATION AND STANDARDS		IS:325 / IEC / EQUIVALENT INTERNATIONAL STANDARDS
B.	SERVICE CONDITIONS:		
		Max. Ambient Temp. (Deg. C)	
		Min. Ambient Temp. (Deg. C)	
		Design Ambient Temp. (Deg. C)	
		Altitude Above MSL (MTS)	
		Relative Humidity (Max.) (%)	
		Environment	
		Location (Indoor / Outdoor)	
		Area (Safe / Hazardous)	
C.	SYSTEM CHARACTERISTICS:		
		Systems Voltage with ± %	415 V ±10%
		Number of Phases	3
		Rated Frequency with ± %	50 Hz ±5%
		Combined Variation	±10%
		Fault Level	25 KA
		Space Heater Supply	If Required
		Low Voltage Stator Winding Heating Supply	NA
D.	Motor Rating / Details:		
		Rated Output	As per Vendor
		Rotor Type	Squirrel Cage
		Syn. Speed (RPM)	As per pump and fan vendor
		Direction of Rotation	Bi-Directional
		Insulation Class	'F', Temperature Rise Limited to 'B'
		Duty	S1, Continuous
		Winding Treatment	Moisture Protection Varnish
		Insulation Process	Anti-Corrosive Treatment
		Starting Method	As per Standard Practice
		Starting Current	Vendor to Furnish
		Minimum Voltage Start at Terminal	80% of 415V
		Starting Torque	Min. 200% of FLT
		Pull Out Torque	
		No. of Hot Starts	2 Nos.
		No. of Cold Starts	3 Nos.
		Shaft Extension	Required
		Type of Coupling	
		Earth Terminals	2 Nos. on Body & 2 Nos. on T. Box
		Greasing Arrangement	Yes
		Name Plate	Yes, as per IS:325
		Starter Connection	Vendor to Furnish
		Efficiency at	
		100% load	
		75% load	
		50% load	
E.	ENCLOSURE:		
		Degree of Protection	EExd IP55



DATA SHEET FOR LOW VOLTAGE ELECTRIC MOTOR

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**SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC BOOSTER
COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW CAPACITY OF 400 SCM/H IN
GEOGRAPHICAL AREAS OF RAJASTHAN**

		Mounting Arrangement	As per requirement.
		Type of Cooling	TEFC
F.		TERMINAL BOX	
i)		Terminal Box (Main)	1 No.
		Type	
		Fault Withstand	
		No. of Terminals	6 Nos.
		Side of Terminal Box seen from the Driven End	RHS
ii)		Auxiliary Terminal Box	--
	1.	Separate Terminal Box for	
		Space Heaters	If Required
		Thermistors	--
G.		TESTS TO BE WITNESSED	
		Type Tests	CMRS test certificate to be furnished
		Routine Tests	As per IS:325
H.		ACCESSORIES	
		Anti-condensation Heaters	Yes
		PTC Thermistors	NA
		Voltage Rating of Space Heaters	230V, If applicable
		Foundation Bolt	Yes
		Cable Glands	Required
		Earthling Terminals	Body & T. Box
		Motor peak Amplitude Vibration at no Load at Bearing should not exceed	40 Microns for 1500 RPM 15 Microns for 3000 RPM
		Max. Motor Noise Level Measured at a Distance of 1. Mts. from Motor	75 dB
I.		CABLES	
	1.	Type & Size	
		- Motor	Vender to furnish
		- Space Heater	Vender to furnish, if Required
		- Thermistor	NA
	2.	Cable Lugs	
		- For Motor	Copper
		- Space Heater	Copper, if Required
		- Control Cables	NA
	3.	Cable Glands - Type Material	
		- Motor	FLS-Double Compression, Ni-Cd Plated
		- Space Heater	FLS-Double Compression, Ni-Cd Plated, If required
		- Control Cables	FLS - Double Compression Ni-Cd Plated
		J. PAINTING	
		TYPE	Epoxy
		SHADE (AS PER IS:5)	692 (Smoke Grey)

NOTE: - Certificate from compressor Block/Hydraulic Cylinder manufacturer towards guaranteed shaft power calculation at 400 SCM/H per hour compression of gas on given parameters & from motor manufacturer towards their KWh consumption (Kwh on guaranteed parameters) on above guaranteed shaft power to be calculated along with bid documents.

LIST OF MOTORS

	DATA SHEET FOR LOW VOLTAGE ELECTRIC MOTOR	Documents No. C221086-RJ-ME-DS-3049	
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**SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC BOOSTER
COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW CAPACITY OF 400 SCM/H IN
GEOGRAPHICAL AREAS OF RAJASTHAN**

Sl. No.	Description	KW	Duty	Quantity

NOTE:- All motor are in Hazardous area classifications Zone-1 & Zone-2 therefore motor shall flame proof vendor shall furnish the list of motor(s).



ENERGISING QUALITY

**PROJECT NUMBER:
VCS-HPCL-C221086**



DATA SHEET FOR LOW VOLTAGE (LV) SQUIRREL CAGE INDUCTION MOTOR

DOCUMENT NO

C221086

RJ

ME

DS

3050

SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC
BOOSTER COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW
CAPACITY OF 400 SCMH IN GEOGRAPHICAL AREAS OF
RAJASTHAN

C1	07.04.2026	ISSUED FOR CLIENT REVIEW	AB	SA	AB
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**SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC BOOSTER
COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW CAPACITY OF 400 SCMH IN
GEOGRAPHICAL AREAS OF RAJASTHAN**

1.	Motor tag no.			
2.	Voltage (V)	415V ± 10%	Phase 3	Frequency (Hz.) 50 Hz+3%
3.	Fault level (KA)	NOT APPLICABLE		
4.	Method of starting	As per Standard Practice		
5.	Phase	THREE	Connection – STAR-DELTA	No. of terminal – 6
6.	Design Ambient temp (°C)	50 °C (IN ACOUSTIC ENCLOSURE)		Temp. rise (°C) 70 °C (Maximum)
7.	Cable size (mm ²)	AS PER SPEC ATTACHED		Type CU. COND. PVC INS.
8.	Enclosure type	IP 55, Ex-d, IIA, IIB, T3		Cooling TEFC
9.	Insulation class	CLASS-F with temperature rise of CLASS B		
10.	Haz. Area classification/ Gas Group	ZONE-1, GROUP-IIA, IIB, Temp Class T3 as per IS/IEC		
11.	Type of explosion protection: Ex (d)		Applicable standards: IS/IEC	
Technical particulars from Driven equipment manufacturer				
12.	Suggested Motor Rating in KW/ Manufacturer		# / #	
13.	Shaft kw/kw at end of curve		# / #	
14.	Speed/ rotation of equipment from Coupling End		# / #	
15.	Starting/ max. Torque required (m-kg)		# / #	
16.	WK ² of equipment including/ excluding ly wheel (kg-m ²)		# / #	
17.	Thrust up/ down (kg)		# / #	
18.	Equipment/ coupling type		# / #	
19.	Starting Condition-On no load/ Under loaded condition		#	
Technical particulars from motor manufacturer				
20.	Manufacturer	*		
21.	KW Rating	*	No. of poles	*
22.	Frame designation	*	Mounting (Horizontal)	*
23.	Full load speed (Max. 1500 rpm)	*	Full load Torque (m-kg)	*
24.	Starting torque as % of full load torque	*		
25.	Full load current (A)	*		
26.	Starting current at 100% Voltage (A)	*		
27.	Breakdown or pull out torque %	*		



**DATA SHEET FOR LOW VOLTAGE (LV)
SQUIRREL CAGE INDUCTION MOTOR**

**Documents No.
C221086-RJ-ME-DS-3050**

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**SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC BOOSTER
COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW CAPACITY OF 400 SCMH IN
GEOGRAPHICAL AREAS OF RAJASTHAN**

28.	Rotation viewed from coupling end	*		
29.	Starting time at 75%V	* (sec.)	Starting time at 100%V	* (sec)
30.	Time (Te) for increased safety motors at 100% Voltage (secs.)		NOT APPLICABLE	
31.	Locked rotor with stand time cold/ hot at 75% V(sec)	*	At 100% V(sec)	*
32.	WK ² of motor (kg m ²)	*		
33.	Power factor at 100% load	*	Power Factor at 75% load	*
34.	Efficiency at 100% load	*	Efficiency at 75% load	*
35.	Space heater watts/ volts	*/ 240V AC		
36.	Bearing type/ no. DE	*/*	Bearing type/ no. NDE	*/*
37.	Type of Lubrication	*		
38.	Weight of motor (kg)	*		
39.	Canopy required/ Not required	NOT REQUIRED		

NOTE:-

- 1) # To be filled by Bidder based on package design
- 2) * To be filled by motor Manufacturer
- 3) Bidder shall quote only one make of motor. Bidder filled the above data sheet along with OEM data sheet to be submitted during detail engineering
- 4) Motor terminal box shall be suitable as mentioned detail in cable size indicated on data sheets below.



**SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC BOOSTER
COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW CAPACITY OF 400 SCMH IN
GEOGRAPHICAL AREAS OF RAJASTHAN**

POWER CABLE SIZE FOR 415V MOTOR

Sl. No.	Motor Rating KW	CABLE SIZE mm ²		No. of Cores	CABLE DIAMETER - mm (APPROX)		
		CU	AL		Overall	Under Armor	Over Armor
	Below 3.7 KW	4		3			
	3.7	6		3			
	5.5	10		3			
	7.5	10		3			
	11.0	16		3			
	15.0	16		3			
	18.5	16		3			
	22.0	16		3			
	30.0	25		3			
	37.0	25		3			
	45.0	35		3			
	55.0	50		3			
	75.0	95		3			
	90.0	120		3			
	110.0	2x95		3			
	125.0	2x120		3			
	132.0	2x120		3			

NOTE:-

- 1) Above table is valid for 2/4/6 pole motors. For low speed motors cable sizes shall be defined at the time of detail engineering.
- 2) Cable size for motor space heater (if required) shall be 3x2.5 mm² with Cu Conductor.
- 3) Cable will be 650/1100V copper conductor, FRLS-XLPE insulated, FRLS-PVC extruded inner sheath armored with overall FRLS-PVC sheath.



**DATA SHEET FOR LOW VOLTAGE (LV)
SQUIRREL CAGE INDUCTION MOTOR**

**Documents No.
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ENERGISING QUALITY

**PROJECT NUMBER:
VCS-HPCL-C221086**



DATA SHEET FOR FLAME ARRESTER

DOCUMENT NO

C221086

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3051

SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC
BOOSTER COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW
CAPACITY OF 400 SCMH IN GEOGRAPHICAL AREAS OF
RAJASTHAN

C1	07.04.2026	ISSUED FOR CLIENT REVIEW	AB	SA	AB
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**SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC BOOSTER
COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW CAPACITY OF 400 SCMH IN
GEOGRAPHICAL AREAS OF RAJASTHAN**

GENERAL DESIGN SPECIFICATIONS

SR. NO.	DESCRIPTION	UNITS	DATA
01	FLAME ARRESTER - TYPE	~	In-line/End of line (Flame Arresters with Vent tube rain protective type if Inline type flame arrester used)
02	TAG NO.	~	#
03	SERVICE FLUID	~	Natural Gas (Dry)
04	LINE EQUIPED WITH	~	Cartridge filter
05	LINE SIZE & QTY.	Inch	#
06	MAKE / MODEL OF FLAME ARRESTER	~	#
07	MOLECULAR WEIGHT OF GAS	Kg/K-mol.	17.96 ~ 19
08	GAS DENSITY @ P,T	Kg/m3	219
09	LINE PRESSURE (MIN. / MAX / DESIGN)	kg/cm2	200 / 240 / 280
10	DESIGN GAS FLOW RATE	SCMH	400
11	OPERATING TEMPERATURE (MIN. / MAX.)	°C	1 ~ 52 °C
12	DESIGN TEMPERATURE (MIN. / MAX.)	°C	-29 ~ 65 °C
13	FLAME ARRESTER - CATEGORY	~	Inline Detonation Flame Arrester Type (#)
14	MIN. VENT HEIGHT FROM UNPROTECTED SIDE	Mtr.	Min. Height of 3 Meter above
15	RATED INLET PRESSURE	kg/cm2	255
16	FLAME ARRESTER INSTALLATION ORIENTATION	~	VERTICAL

MATERIAL OF CONSTRUCTION

SR. NO.	DESCRIPTION		MATERIAL GRADE
17	MAIN BODY	~	High Temperature carbon Steel (HTCS) #
18	STUD & NUT	~	Hot dip Galvanized
19	DISC	~	#
20	GASKET	~	#
21	ELEMENTS	~	SS316 (CRIMPED-RIBBON ELEMENTS)
22	BANK ASSEMBLY	~	#
22	CORE	~	#



**SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC BOOSTER
COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW CAPACITY OF 400 SCMH IN
GEOGRAPHICAL AREAS OF RAJASTHAN**

DIMENSION & WEIGHT			
SR. NO.	DESCRIPTION		DATA
23	OVERALL LENGTH OF FLAME ARRESTER	mm	#
24	WIDTH OF FLAME ARRESTER	mm	#
25	HEIGHT OF FLAME ARRESTER	mm	#
26	SHIPPING WEIGHT	Kg	#
SPARE & ACCESSORIES			
SR. NO.	DESCRIPTION		DATA
27	GASKETS	~	Spare required
28	LIFTING TYPE	~	Handle Required
29	ARRESTER ELEMENTS BANK	~	Spare required
30	DESIGN CODE	~	ISO16852 (As per Latest Edition)
31	FLANGE RATING	Pounds	#
32	FLANGE DESIGN STD.	~	ASME B16.5
33	MAX. EXPERIMENTAL SAFE GAP(MESG)	mm	#
34	HYDROTEST PRESSURE BODY	kg/cm2	1.5 X design press.
35	FLANGE DRILLING	~	#
36	FLANGE FACE FINISH	AARH	AS PER ASME B16.5
37	PRESSURE DROP	Psi.	SHALL NOT EXCEED 1 psi
38	HAZARDOUS AREA CLASSIFICATION	~	CLASS I, DIVISION 1, ZONE 0,1 (GAS GROUP I, IIA, IIA1)
39	MAXIMUM OPERATING PRESSURE	kg/cm2	#
40	MAXIMUM OPERATING TEMPERATURE	°C	#
41	SHORT TIME BURN	~	#
42	LEAK TEST & MEDIA	~	1.1 times of Max. Operating Pressure (Test Media- Air)
43	LEAK TEST HOLDING TIME	Min.	Min. 5 Minutes
44	UNPROTECTED PIPE LENGTH	Mtr.	Min. 3mtrs.
45	CERTIFYING AGENCY / APPROVALS	~	#

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GEOGRAPHICAL AREAS OF RAJASTHAN**

46	FLAME VELOCITY	Mtr/sec.	#
47	NO. OF ELEMENTS BANK	Nos.	#
48	ATMOSPHERIC TEMPERATURE CONDITIONS	°C	(-)20°C to (+)60°C
49	ATMOSPHERIC PRESSURE CONDITIONS	Atm.	0.78 Atm. to 1.08 Atm.
INSPECTION & TESTING			
50	FLAME TRANSMISSION TEST	~	Required
51	BURNING TEST (IF REQUIRED)	~	Required
52	FLOW TEST	~	Required
53	FLAME SPEED VS RUNUP DISTANCE CURVE	~	Required
54	ARRESTER FINISH	~	Red colour (#)
55	PNEUMATIC TEST PROCEDURE	~	#
56	HYDRO TEST PROCEDURE	~	#
57	HYDROSTATIC TEST	~	#
58	OPERATION & MAINTENANCE PROCEDURE	~	#

NOTE: -

- 1) VENDOR TO PROVIDE THE DATA AS MARKED "#".
- 2) FLANGES AND GASKETS SHALL CONFORM TO ASME B16.5 RESPECTIVELY.
- 3) VENDOR TO PROVIDE THE BURN BACK VELOCITY CONSIDERED FOR FLAME ARRESTOR DESIGN. THIS BURN BACK VELOCITY SHALL BE DIFFERENT FOR DIFFERENT GAS COMPOSITION
- 4) VENDOR TO PROVIDE THE FLAME ARRESTER PROCESS CALCULATION.
- 5) VENDOR TO MENTION THE MAXIMUM INITIAL OPERATING PRESSURE (IOP) WHICH THE FLAME ARRESTERS CAN HANDLE FOR VARIOUS FLAMABLE GAS MIXTURES.
- 6) FLAME ARRESTER MUST BE DESIGNED FOR MAXIMUM HEAT RELEASED BY FLAME, AND HENCE VENDOR TO PROVIDE THE HEAT TRANSFER CAPACITY FOR WHICH THE FLAME ARRESTER IS DESIGNED FOR. THIS REQUIRED KNOWING HOW MUCH HEAT ENERGY WAS ABSORBED BY THE ARRESTER PER UNIT VOLUME OF GAS IN ORDER TO LOWER THE FLAME TEMPERATURE.
- 7) FLAME ARRESTER SHALL BE CCOE/PESO CERTIFIED VENDOR TO PROVIDE PESO CERTIFICATES.
- 8) FLAME ARRESTER BODY-COLOR SHALL BE RED & RAL GRADE VENDOR SHALL CONFIRM ALSO PAINT ARE FLAME & HEAT RESISTANT.

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DATA SHEET FOR AIR COMPRESSOR

DOCUMENT NO

C221086

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3052

SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC
BOOSTER COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW
CAPACITY OF 400 SCM³/H IN GEOGRAPHICAL AREAS OF
RAJASTHAN

C1	07.04.2026	ISSUED FOR CLIENT REVIEW	AB	SA	AB
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**SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC BOOSTER
COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW CAPACITY OF 400 SCMH IN
GEOGRAPHICAL AREAS OF RAJASTHAN**

1.0	NO. OF UNITS	:	As per SOR	DRIVE: ELECTRIC MOTOR	
1.1	DUTY	:	Intermediate	LOCATION: HPCL GA	
1.2	Model No./Make	:			
2.0	OPERATING CONDITIONS				
2.1	service / stage	:	Air / #		
2.2	Gas handled	:	Air		
2.3	Composition	:	Air (Refer air composition)		
2.4	Corrosive due to:		NA		
2.5	Molecular Weight at Intake (Avg.)		Refer Air composition		
2.6	Cp/Cv at intake / compressibility Factor		Refer Air composition		
2.7	Relative Humidity		90 %		
2.8	Suction Temperature (°C)	:	Max. Ambient 47°C		
2.9	Suction Pressure (Kg/cm ² , Gauge)	:	ATM		
2.10	Discharge Temperature (°C)	:	#		
2.11	Discharge Pressure (Kg/cm ² , G)	:	10 Kg/cm ² g		
2.12	Required Capacity (CFM.)	:	# (Refer Technical specification)	Electric Motor Rating: # KW	
2.13	Drive arrangement Direct Coupling/V-belt		#	#	
2.14	standard Conditions referred to	:	Standard Atmospheric pressure (1.033 Kg/cm ² abs) and Temperature of 15.6°C		
2.0	SITE CONDITIONS				
2.1	Place			Installation: Outdoor	
2.2	Site Ambient Temp. (°C)		Min: 2 °C Max :47 °C	Max. Relative Humidity (%) : 90%	
2.3	MSL – 267 m		Max. Altitude Derating correction	#	
2.4	Area Classification	:	Class1 Div1 Group D or Zone 1 Group IIA/IIB (if Air Compressor placed inside package)		
3.0	APPLICABLE CODES & SPECIFICATIONS				
3.1	Compressor	:	#	Piping Code	#
3.2	Driver	:	Electric motor	Control Panel: Refer Tech Specs	
3.3	Air Cooled Exchangers	:	#	Testing Code	#
3.4	Pressure Vessel	:	ASME Sec. VIII Div.1	Material Code:	#
3.5	Fabrication Code	:	#	Design code	#
4.0	MANUFACTURER SPECIFICATIONS				
4.1	Vessel Capacity	:	#	Hydro test Pressure(kg/cm ²)	#
4.2	Shell Thickness	:	#	Duration of test	#
4.3	Disc Thickness	:	#	Testing Media	Water
4.4	Working Pressure(kg/cm ²)	:	10 kg/cm ² (g)	Test position	#
4.5	Motor Rating HP	:	#	Motor Load (FLA)	#



**SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC BOOSTER
COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW CAPACITY OF 400 SCMH IN
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4.6	Piston Displacement (CFM)	:	#	Non-time delay Fuse	#
4.7	Electrical supply Voltage	:	#	Time Delay Fuse	#
4.8	Drain Point	:	Required	Inverse time circuit breaker	#
4.9	Pressure Gauge	:	Required	Pressure switch	Required
q2 4.10	Safety Valve	:	Required	Air End fan	Required
4.11	Phase Preventer	:	Required	Check valve	Required
4.12	Suction Filter & breather	:	Required	Compressor Motor overload Relay	Required
4.13	Crank case Oil pump	:	Required	Noise level	#
4.14	Air Inlet tube OD	:	#	Air outlet Tube OD	#
4.15	Compressor grouting	:	Required	Air compressor Shipping Weight	#
4.16	Dimension(L×W×H)	:	#	Compressor Auto start/Stop pressure	#
4.17	Compressor Suction flow rate (CFM)	:	#	Compressor Discharge flow rate (CFM)	#
4.18	No. of compression Stages	;	#	Mainline filter model	#
4.19	Nominal Filtration Range	:	0.3µm	Filtration efficiency	99.9 %
4.20	Oil mist density @ outlet Max.	:	#	Filter Element model	#
4.21	Rated capacity at Full load operating pressure	:	ACFM (#)	Max. full flow operating pressure	#
4.22	Drive motor nominal efficiency	:	#	Total package input power @ zero flow	#
4.23	Total Package Input Power at Rated Capacity and Full Load Operating Pressured	:	#	Specific Package Input Power at Rated Capacity and Full Load Operating Pressure (KW/CFM)	#
4.29	Isentropic efficiency	:	#	Unloaded valve	Required
4.30	On off light indicator	:	Required	Main Circuit Breaker	Required
4.31	Lubricating low oil pressure Indicator	:	Required	Free Air Delivery	#



**PROJECT NUMBER:
VCS-HPCL-C221086**



DATA SHEET FOR TEMPERATURE GAUGE

DOCUMENT NO

C221086

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3053

SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC
BOOSTER COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW
CAPACITY OF 400 SCM³/H IN GEOGRAPHICAL AREAS OF
RAJASTHAN

C1	07.04.2026	ISSUED FOR CLIENT REVIEW	AB	SA	AB
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**SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC BOOSTER
COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW CAPACITY OF 400 SCMH IN
GEOGRAPHICAL AREAS OF RAJASTHAN**

TEMPERATURE GAUGE			
S. No.	DESCRIPTION	SPECIFICATION	OFFERED
1	Type of Service	CNG	
2	Type	Refer technical specification	
3	Well	Required	
4	Mounting	Local	
5	Dial Size	100 mm	
6	Color	White with black inscription	
7	Case material	SS316	
8	Window material	Shatterproof glass	
9	Connection location	Bottom	
10	Accuracy	+/- 1% FSD	
11	Enclosure	WP to IP 65 as per IEC 60529/IS 13947	
12	Zero adjustment screw	Required	
13	Stem	Bimetal	
a.	Type	#	
b.	Material	#	
c.	size	#	
d.	Dia.	#	
e.	Filled system	#	
f.	Compensation	Case adjustable union	
14.	Bulb type	#	
a.	Bulb Material	SS316	
b.	Bulb Union Thread	½" BSP(M)	
15.	Extension Type	Rigid	
16.	Bulb dia.	To be suitable with thermos-well connection	
17.	SAMA Class	#	
18.	Capillary Material		
a.	Armor flexible	#	
b.	Armor Material	#	
c.	Armor length	#	
19.	Over range protection	Yes	
20.	Thermo-well		
a.	Material	SS316	
b.	Design standards	As per ASME PTC 19.3TW	
c.	Wake frequency calculation	#	
d.	Root diameter	#	



DATA SHEET FOR TEMPERATURE GAUGE

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e.	Bore diameter	#	
f.	Tip diameter	#	
g.	Tip Thickness	#	
h.	Head Length	#	
i.	Immersion Length	#	
j.	Process connection	#	
k.	Stem style	#	
l.	Instrument connection	½" BSP(F)	
19	Temperature gauge Make	Refer AVL	
20	Thermo-well make/Model	Refer AVL	
21	Thermo-well Ordering code	#	
22	Over Range Protection	130 %	

23.	Notes.
1.	(#) Marked data to be furnished by Vendor/Manufacturer
2.	At the time of approval of DS, Supplier shall furnish DS with Catalogues.
3.	Vendor to provide all Temperature gauge Tag List with Aluminum embossed plate (where mention with Tag no./Date of calibration/Due date of Calibration /Items serial no etc.)
4.	Vendor to provide Operating and Instruction manual before dispatch of cascade.
5.	Vendor to provide Thermo-well operating manual with Thermo-well GA drawings for detail engineering.
6.	U-length shall be selected in such a way that the thermos-well tip shall be preferably at the Centre of pipe to sense the temperature accurately.
7.	Vendor to provide wake frequency calculation of thermos-well as per ASME PTC 19.3 TW including:
a.	Frequency Limit
b.	Dynamic stress Limit
c.	Static stress Limit
d.	Hydrostatic Pressure Limit
8.	All Thermo-well are tested as per ASME PTC 19.3 TW and final test report to be submitted for review during inspection.



ENERGISING QUALITY

**PROJECT NUMBER:
VCS-HPCL-C221086**



TECHNICAL SPECIFICATIONS-400 SCMh BOOSTER COMPRESSOR

DOCUMENT NO

C221086

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3040

SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC
BOOSTER COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW
CAPACITY OF 400 SCMh IN GEOGRAPHICAL AREAS OF
RAJASTHAN.

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**SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC BOOSTER
COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW CAPACITY OF 400 SCMH IN
GEOGRAPHICAL AREAS OF RAJASTHAN**

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**SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC BOOSTER
COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW CAPACITY OF 400 SCMH IN
GEOGRAPHICAL AREAS OF RAJASTHAN**

1. INTRODUCTION

Hindustan Petroleum Corporation Limited (HPCL), a Government of India Undertaking (hereinafter called as the Owner), has been authorized by PNGRB for setting up infrastructure and operation of City Gas Distribution Network for the authorize GA of Rajasthan. Natural Gas will be transported to residential, commercial, industrial (PNG) and automobile consumers (CNG) in the city.

M/s VCS Quality Services Pvt. Ltd (VCS) has been appointed as Engineering Project Management Consultant (hereinafter referred as EPMC), by HPCL for providing Design, Engineering and PMC services for this Project.

2. DEFINITION

Where used in this document, the following terms shall have the meanings indicated below, unless clearly indicated by the context to this order:

PROJECT:	CGD PROJECT AT RAJASTHAN GA
OWNER/COMPANY:	HINDUSTAN PETROLEUM CORP. LTD.
CONSULTANT:	VCS QUALITY SERVICES PVT. LTD.
MANUFACTURER:	THE PARTY WHO MANUFACTURERS & SUPPLIES EQUIPMENT & PROVIDE SERVICES TO OWNER.
TS:	TECHNICAL SPECIFICATIONS

3. SCOPE

This specification along with applicable codes as referred, describe the minimum requirements for design, engineering, supply, inspection, testing, supply including packaging, forwarding, insurance, custom clearance if any, handling and unloading as well as at purchaser stores/site, package performance test(FAT),site acceptance test(SAT) of Motor driven Hydraulic Booster compressor with variable suction pressure range from 210 kg/cm²(g) to 30 kg/cm²(g) up to discharge pressure of 255 kg/cm² & Hydraulic type CNG Booster Compressor is designed to handle a flowrate of 400 SCMH from a variable suction pressure of 210 — 30 kg/cm²(g) to a discharge pressure up to 255 kg/cm²(g).

Additional features other than those indicated herein which call for a better design, increase in efficiency, enhance reliability, optimization may be accepted subjected to Client's approval. The Compressor package shall be shipped in completely assembled condition. Gas supply line and delivery connection shall be made at site. The vendor shall bid their main offer with items according to the

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technical specifications mentioned below.

Offered package shall be complete with compressor, electric motor, hydraulic pump and piping, cooling system, suction and discharge filters, control panel safety and control devices and other accessories required for automatic and safe operation of the system.

4. GENERAL DESCRIPTION

A. GENERAL DATA

1.1	Compressor Type	
1.1.1	Oil lubricated	
1.2	Type of Cooling	Gas cooling and cylinder cooling should be as per manufacturer design.
1.3	No. of compression stage	02
1.4	Cylinder	As per Manufacturer design
1.5	Intake Temperature	Max. 35°C
1.7	Max. motor power	To be indicated with 10 % margin over BKW as per Guaranteed parameters

B. COMPRESSOR PERFORMANCE DATA

2.1	Gas pressure at compressor inlet	210 kg/cm2(g) to 30 kg/cm2(g)
2.2	Compressor Discharge pressure	255 kg/cm2 (g) @ 52°C Max.
2.3	Compressor Discharge Temperature	Compressor Discharge temperature 52 °C (After cooler) with max. Ambient air temperature of 47.5°C and gas inlet temperature of 35°C (max.).
2.4	Compressor speed	To be indicated by bidder
2.5	Ambient conditions	Max. 47.5 °C
2.5.1	Ambient Temperature	2°C to 47.5 °C.
2.5.2	Max. Relative humidity	90 %
2.6	Required guaranteed capacities of electric motor driven compressor packages at rated suction pressure and discharge pressure as mentioned below:	

C. GAS COMPOSITION

Natural Gas composition		
Components/Elements	Normal (Mol. %)	Design (Mol. %)
Methane	82.43 – 99.10	97.98771
Ethane	7.27 – 0.90	1.82832
Propane	3.47 – 0.00	0.04419
i-Butane	0.65 – 0.00	0.00187





**SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC BOOSTER
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n-Butane	0.78 – 0.00	0.00171
i-Pentane	0.17 – 0.00	0.00023
n-Pentane	0.13 – 0.00	0.00007
Hexane	0.10 - 0.0	0.00013
Carbon dioxide (Co2)	4.93-0.0	0.00
Nitrogen	0.06-0.0	0.13567
H2S	0.01	0.00
Total	100	100
Net Cal. Value (Kcal/SCM)	8150 – 8950	8236.314
Gross Cal. Value (Kcal/SCM)		9143.349

D. CNG SPECIFICATIONS

The CNG specification should meet the **ISO 15403:2000 (E)** natural gas quality designation for use as a compressed fuel for vehicles.

The proposed specification of the CNG is as follows:

Gas temperature: -10 °C to 65° C

Oil content: 10 PPM

Particulate: < 5 micron

Odorant Matter: S10 mg/sm3 (Ethyl Mercaptan)

E. DAUGHTER BOOSTER STATION SPECIFICATIONS

Suction Pressure: 210 to 30 kg/cm2 (g)

Outlet Pressure: 220 to 255 kg/cm2 (g)

Design Pressure: 280 kg/cm2 (g)

Design temperature: 65°C

Compressor Flow Capacity: 400 SCMH

Stationary Cascade Capacity: 3000 WL

Note: - Hydraulic type Motor driven reciprocating type booster compressor is designed for variable suction from LCV mounted mobile cascade from pressure 210 kg/cm2(g) to 30 kg/cm2(g).

Performance curves and tables i.e., **Flow versus suction pressure and temperature and power curves** i.e., **absorbed power versus suction pressure and temperature at specified discharge conditions** shall be furnished. In addition to above, flow capacity and absorbed power values for suction conditions from 210 to 30 Kg/Cm2g in steps of 0.5 Kg/Cm2 shall also

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be given in tabular form. The graph shall be plotted at various suction pressures ranging from 210 to 30 Kg/Cm²g and at various suction temperatures ranging from 20°C to 35° C. Similarly, the graphs shall be plotted at various discharge pressures ranging from 220 Kg/Cm²g to 255 Kg/Cm² (g), however at 52°C (max) discharge conditions.

F. CLIMATE CONDITIONS

- Ambient Temp. Min/Max °C: 2 °C / 47.5 °C
- Design wet bulb temp (WBT), 0°C: 27 °C
- Design relative humidity %: 90
- Altitude above MSL, M: 205
- Wind velocities km/hr. (max): 160
- Air Cooler Design 0°C: 47.5°C DBT, 27°C WBT & 90% RH

5. BASIC DESIGN CRITERIA OF COMPRESSOR

- 5.1 Following specification is intended to give the bidder the technical and operating conditions the compressor must fulfil. Compressor shall be hydraulic booster type suitable for variable suction pressure.
- 5.2 The bidder shall meet all applicable statutory codes, national law and local regulation for safety and environment protection.
- 5.3 The design shall conform to API 618 or other relevant reputed international standards but approved by CCOE (bidders to indicate).
- 5.4 Offered package shall be complete with compressor, electric motor, hydraulic pump and piping, cooling system, suction and discharge filters, controls panel safety and control devices and other accessories required for automatic and safe operation the system.
- 5.5 The supply shall include all interconnecting piping/tubing/cables.
- 5.6 Cooling system shall be of closed-circuit type. Ultimate cooling shall be by air.
- 5.7 NRV as required for liquid line in pump discharge to control & minimize the water hammering and pump back pressure effects.
- 5.8 Flame arrester shall be provided at common vent & excessive pressure relief exhaust line to minimize the risk of fire and other possible impingement hazards during pressure relief.
- 5.9 To prevent the discharge of gas into the enclosure, all safety relief valves within to be connected to a manifold. From this connection a single pipe passes through the enclosure roof to a vent stack fitted with spark arrestor/flame arrestor to allow satisfactory dispersion of gas at a safe height from working platform of booster compressor

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- 5.10 Pressure relief valve shall installed & sizing calculation in accordance with **ISO 4126-1** & manufacturer instruction.
- 5.11 The compressor package control system shall be designed for unattended safe operation in automatic mode and shall **unload, start, load, stop safely.**
- 5.12 The compressor shall start in auto in case high bank pressure in dispenser falls below 200 kg/cm² and stop once the pressure in all three banks reaches to 255 kg/cm².
- 5.13 Compressor shall be suitable for continuously variable suction pressure from 210 kg/cm² (g) to 30 kg/cm² (g), supplied through LCV mounted CNG storage cascade.
- 5.14 Compressor should also stop when suction pressure falls below 30 kg/cm² (g).
- 5.15 Compressor shall be suitable for discharge pressure from 250 kg/cm² to 220 kg/cm², corresponding to suction of 210 kg/cm²(g) to 30 kg/cm²(g).
- 5.16 Compressor shall be designed to ensure flow capacity as indicated in data sheet.
- 5.17 Due to space constraint, in some places installation of the stationary cascade (**capacity 3000 WL) of (Approx 7.0 tons)** on the top of the canopy of compressor may be required following PESO norms. The bidder will therefore design the canopy to take the load of a stationary cascade of 7.0 tons. The bidder shall also provide 2 nos. stair case/monkey ladders (in line with the gas cylinder rules) for safe climbing on the top of the canopy along with hand railing on the top for ease maintenance and operation. Provision should be made such that, the cascade can be placed anywhere above the canopy. the bidder shall ensure that adequate space (minimum 2 feet width) walk way in front side of cascades (i.e., valve mounting side) is available for carrying out routine checking/ Maintenance.
- 5.18 Vendor shall provide priority fill system with total 7 line banking Cascade (Low/Med./High), Dispenser (Low/Med/High) & For gas Recompression Bank from Medium & Low bank cascade to Priority fill provision when LCV not available at station.
- 5.19 Vendor shall provide PLC Controlled Priority fill system with compressor top-up facility inclusive of regulating valves by pass valve & liquid filled pressure gauges all mounted in a stainless steel structural.
- 5.20 The priority fill system is to be installed to ensure that vehicle filling takes priority over cascade filling and direct CNG TO THREE STORAGE BANKS IN CORRECT SEQUENCE.

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- 5.21 The compressor shall shut down once all three-cascade storage banks are filled to 255 Kg/cm² (g).
- 5.22 Compressor shall start on pressing of manual start push button & automatically when the cascade storage high bank pressure of compressor falls to 200 Kg/cm² (g) and shutdown automatically when all 3 banks of stationery cascade are filled to a pressure of 255 Kg/cm²(g).
- 5.23 Full bore ball valves shall be provided so that compressor can take suction either from LCV cascade or stationary cascade.
- 5.24 All fittings and tubes used in priority system shall be of stainless steel of suitable pressure rating (5000 PSIG).
- 5.25 End connections shall be 3\4" size pipe OD.
- 5.26 Complete erection, installation and commissioning of LCV fill post is in the bidder's scope which include Grouting and fixing of Post on the foundation. (Foundation will be prepared by client based on drawing provided by vendor).
- 5.27 Complete Erection, Testing & Commissioning of compressor packages.
- 5.28 **CASE-I (Valves positioned to take suction from LCV cascade.)**
- 5.28.1** If the LCV cascade pressure is more than 200 kg/cm², the gas dispensing should take place directly from LCV to dispenser bypassing booster compressor.
- a. Compressor shall start on pressing of manual push button or auto start when the LCV Cascade and Stationary cascade at all banks pressure falls below 200 kg/cm². The priority of filling s hall be as follows;
- **First priority:** Priority panel shall first fill the vehicle through dispenser
 - **Second priority:** If no vehicle is to be fueled, priority panel shall fill the stationary cascade. The compressor shall shutdown automatically when either all stages of stationary cascade are filled to a pressure of 255 kg/cm² or pressure in mobile cascade is less than 30 kg/cm².
- 5.29 **CASE-II (Valve positioned to take suction from Stationary cascade)**
- b. Dispensing shall be done through stationary cascade without compressor running, if stationary cascade pressure is more than 200 kg/cm².
- c. Compressor shall start on pressing of manual/auto start push button if stationary cascade pressure is less than 200 kg/cm². Dispensing into the vehicle should take place as usual. Compressor shall trip if either there is no vehicle for fueling or pressure in stationary cascade is less than 30 kg/cm².

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Note: - Automatic switch over shall be provided at Booster compressor panel to switch valve positioning as per above Case I and Case II.

- 5.30 The compressor package control system shall be designed for unattended safe operation in automatic mode and shall unload, start, load, stop safely. The compressor shall start in auto in case high bank storage pressure falls below 200 Kg/cm² (g) and stop once the pressure in all three banks of storage cascade reaches 255 Kg/cm²(g).
- 5.31 Noise level shall not exceed 75+/-3 DBA at 1m from the compressor package enclosure from any direction.
- 5.32 The compressor package control system shall be so designed that the first item to go into alarm condition shall "Lock out" to indicate the cause of the trip though the cause of the trip may have disappeared. The lock out condition shall be manually reset.
- 5.33 An automatic restart shall be provided on restoration of power with a 10-second delay after temporary interruption. Existing alarm condition shall remain indicated.

5.34 Prime mover (Electric Motor)

The motor shall be flame proof/ explosion proof and confirm to IS: 2148 & IEC 60079-1 suitable for zone 1 group II area as per IS/IEC. The Motor shall be of standard frame size as per IS/IEC and rated for continuous duty with high efficiency and shall be designed for star-delta starting. The Motor shall be provided with class 'F' insulation; however, temperature rise shall be limited to the temperature specified for class 'B' insulation as per IS and shall be suitable for voltage variation of 415V+ 10%. The bidder shall indicate the guaranteed total power requirement in KW. The motor rating shall be 110% of the greatest BkW required by the compressor.

Electric Motor Specification

- a) Type of drive **Totally Enclose Fan Cooled (TEFC) high efficiency as per IEEMA** standard-19-2000
- b) Protection Flame proof & weather proof enclosure
- c) Insulation Class F with class B temperature rise
- d) Mounting Horizontal Foot Mounting
- e) Specification standard By Bidder
- f) Supply Voltage(assumed) 415+ 10% volt, 3 phases, 50+ 5%Hz

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- | | | |
|----|--|-----------|
| g) | Synchronous speed | By bidder |
| h) | Motor rating | By bidder |
| i) | Motor Efficiency | By Bidder |
| j) | Power factor | By Bidder |
| k) | Speed of motor | By Bidder |
| l) | Nos. of hot starts of motor 2 hot and 3 cold starts per hour | |
| m) | Coupling Type | By Bidder |
| n) | Torque speed curve | By Bidder |
| o) | starting torque, speed, thermal withstand curve load, current speed curve, Efficiency power factor vs load curve by Bidder | |
| p) | Motor Accessories | |
| | i) Compressor grooved flywheel (if any) | |
| | ii) Motor grooved drive pulley (if any) | |
| | iii) Drive VEE belts (if any) | |
| | iv) Flexible coupling for direct drive | |
| | v) Drive guard | |
| | vi) Adjustable motor slide rails for belts tensioning to be used (if any) | |
| q) | Electrical Supply parameter's | |

All electrical shall be suitable for the following supply conditions.

Electrical operating voltage: AC, 3 phase, 415 V, 50 Hz

Electrical control voltage: 240 VAC, 50 Hz (under supplier's scope)

HPCL shall provide 415 +/-10% volts, 3 phase and 50 + 3% Hz electrical connection at CNG station electrical PDB panel only.

5.35 Cooling System

Each compressor package shall be complete with its own cooling system. The cooler shall be air-cooled heat exchanger. The gas temperature after after-cooler **shall not exceed 52-degree C**. Special attention to be given while designing the gas cooler considering the local conditions. Bidders shall ensure that final delivered gas temperature is less than 52-degree C. Cooler design

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shall be on the basis of 10% extra load corresponding to maximum severe operating conditions based on thermal duty. Gas cooler shall be design as per manufacturer standard. For cooling of the heat exchanger, a cooling fan to be provided.

Cooling system to be in a single enclosure. Bidder to submit cooling sizing calculation for review. Direction of flow should be marked on the pipe line and nomenclature of all vessels, Manifolds, Actuators (e.g., 1st stage suction & Discharge, 2nd stage suction & discharge etc.), 1st stg. Intercooler & 2nd stg. Intercooler should be written on them. Cross head inspection windows if applicable should be transparent for ease of inspection during running. Set values should be prominently marked on the gauges.

Packages design should be such that its vent should not go upward (package vent in vertical direction not required) i.e., opening of package vent should be in horizontal directional with duct arrangement.

5.36 Oil Filter

The ingress of oil into CNG adversely effects vehicle emission and storage system. Contractor to supply a proven, maintenance free oil removal system after after-cooler to remove oil from compressed gas. The offered oil mist removal system shall restrict the oil to less than 5 PPM in discharge of compressor.

5.37 Piping & Appurtenances

The materials for gas piping shall be ASTM A269 Hardness Rb80 or less and suitable for bending and flaring O.D tolerance shall not exceed +/- 0.005 ". The piping / tubing at the outlet of the compressor and of priority fill system shall be of seamless stainless steel of proper pressure rating and specifications as under:

SS TUBING SPECIFICATIONS:

- Seamless SS Tubing
- Material of construction Stainless Steel 316L
- Tube hardness shall be less than 80 RB
- Sizes: Metric system (inch)
- Max Working Pressure: 350 Kg/cm² g

SS FITTINGS & VALVES SPECIFICATIONS:

- Material of construction SS-316/SS-316L
- Sizes: Metric/SI
- Standard: ASTM/ ASME/ DIM
- End connections: Single or double ferrule Compression type / NPT
- Max Working Pressure: 350 Kg/cm² g
- PSV Vent Line to be extended above the package to safe height.
- Priority system should be designed so that the gas flow from mobile cascade to dispenser is possible even with the Compressor shutdown & de-energized
- Dedicated Air compressor of adequate capacity

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5.38 Electrical System

- All electrical equipment of compressor package shall be installed in accordance with NFPA 70, NEC for Class 1, Division 1, Group D, and IS 5571 and shall have approval of a recognized certifying authority.
- OWNER shall provide 415+ 10% volts, 3 phase and 50+ 3% Hz electrical connection at CNG station electrical panel only. Vendor shall distribute electrical power to all equipment and control system by providing cables and suitable switch-gear distribution panel.
- The electrical power supply distribution panel, switch gear panel and starter shall be in flame proof construction. Certificate from recognized agency to the effect that equipment supplied and or installed conform to above area classification. All devices shall meet the requirement for the specified area classification in which they are installed, including instrumentation leads.
- Heavy duty on-load phase changeover should be provided for H.E motor.
- Semiconductor fuses to be provided, where applicable.
- All illumination fittings should be single phase AC supply based and LED type only.
- All wire/ cable to be used in compressor and panel shall be of copper conductor and FRLS type through proper cable tray conduit etc.
- Sufficient space to be provided for Motor JB for cable glanding work.
- Necessary arrangement to be provided in package for easy extraction of motor from package.
- Chain pulley to be provided in package for easy extraction of motor from package.
- Individual Multifunction meter to be provided for metering of package total energy and for Main motor for monitoring parameters (viz; KVAH, KWH, Voltage, and current, PF, Frequency, MDI (KVA), and MD (KWH).
- Multifunction meter also required for fan motor electrical parameters monitoring.
- Bidder to ensure that spares and service support of all switchgears, instruments, or meter etc. used in package/ panel, shall be available in Indian market.
- The power factor (PF) of the whole electrical system should not be below 0.90. Motor feeder shall be provided with energy meter, heavy duty switch, HRC link type with single phase presenter fuses, contractors (AC-3 Duty), bi-metal relay switch fuse unit, voltmeter, push buttons, earth leakage relays, indication lamps

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for start/stop/trip/Co2 system etc. Ammeters shall be provided for all motors above 3.7. KW rating.

- For maintaining / achieving Power Factor 0.90, Bidder shall install additional equipment/capacitor bank which are in Bidder's scope.
- Stop push buttons shall be lockable and have stay put except in case of critical devices such as lube oil pumps etc.

5.39 Earthing System

- The design & installation of earthing system shall be as per **IS 3043 or equivalent** international specification. One or more no of earth plates with provision of inter connection to main earth grid shall be provided. All hardware used for earthing system shall be hot dip galvanized or zinc passivated. All cables shall be terminated at equipment by means of double compression type compression glands and shall be flame proof cable glands if located in hazardous area.
- Phase sequence preventer (current based) shall be provided.

5.40 Vibration

- Compressor maximum vibration of cylinders shall not exceed 10 mm/sec unfiltered peak velocity. Maximum vibration level of installed compressor frame shall not exceed an unfiltered peak velocity of 5mm/sec or as per IEC Code unfiltered peak-to-peak vibration whichever is less. The bidder shall provide for all structural support within the package so that these levels can be achieved.

6. HYDRAULICS SYSTEM & CONTROLS

- 6.1 The use of variable-speed electric drives to power hydraulic pumps shall provide with the development of high-speed controls, quick-response three phase induction directly coupled with FLP motors.
- 6.2 The improved star delta converter of Variable frequency or speed drive system with motor-pump combinations shall coupling by splines & key connection to drive the shaft similarly Motor & pump alignment , level to be maintained by digital level gauge for excess flexural shaft rigidity & pump failure.
- 6.3 The power, Voltage, current & responsiveness to match systems controlled by electrohydraulic valves or a variable-displacement piston/vane pump, while also reducing energy consumption by adjusting flow & pressure of pumps.
- 6.4 The complete package shall provide with hydraulic drive system with Hydraulic piston/vane pump, direction valve, pressure regulator, flow regulator, Oil temperature transmitter, Reciprocating valve, Air breather, Oil glass level indicator with temperature indicator, Level transmitter etc.

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- 6.5 Vendor shall calculate LPM required to operate Hydraulic system drive and pump shaft power.
- 6.6 NRV shall provide on pump discharge for flow smoothness & pressure compensation by pump at initial effects.

7. INSTRUMENTATION & CONTROLS

- 7.1 All the Instruments and Control Shall Be Suitable for Area Class I, Group D, Division1
- 7.2 All package mounted transmitters & temperature elements; flow transmitter shall be intrinsic safe as per IEC 79-11 and solenoid valves, switches and related junction boxes shall be flame proof 'd' as per IEC 79-1. Other special equipment / instrument, where intrinsic safety is not feasible or available, shall be flame proof/ explosion proof as per IEC 79-1.
- 7.3 All solenoid coils, power contactors etc. shall have operating voltage of 240VAC, and 50 Hz.

PG shall be Direct-mounting type having element of bourdon tube and dial size of minimum 63/100 mm. It shall have shatterproof glass. Connection shall be ½" NPT (M) or 1/4" NPTM from bottom. Enclosure shall be weatherproof to IP65. Protection shall be 130% over range. Accuracy shall be $\pm 1\%$ of FSD. PG shall have blown out disc facility. For higher pressure applications (above 60 Bar), it shall be solid front type. Pressure gauge dial shall be white, non-rusting plastic with black figures. The dial face shall be marked with pressure element material. Pointers shall have micrometer adjustment. Pressure gauge sensing element shall be of SS 316 and moving elements of SS 304, as a minimum.

- 7.4 The temperature gauge shall be generally gas field in steel field type /bimetallic type. Capillary tubing shall be min. SS304 with SS flexible armoring. The gauge shall have an accuracy of $\pm 1\%$ FSD and 63/100mm dial size. The range shall be 1.5 times of operating temperature. Skin type temperature gauges shall not be used.
- 7.5 Pressure Transmitter and Temperature Transmitters shall be used for CNG Gas application with 4-20 mA output signals to PLC. Ex proof enclosure. The units of measurement for pressure shall be Kg/cm² (g)/Bar (g) and temperature shall be degree C. Pressure and temperature switches are not acceptable.
- 7.6 RTDs are 4 wire type and element shall be Pt100 as per DIN 43760 & accuracy class A and thermos-well's immersion length shall be suitable for the line size. All RTDs shall have duplex elements. RTDs sheath OD shall be 8 mm and material SS 316. Cable entry shall be ½" NPT (F). Enclosure shall be WP to IP-65. All RTD shall be supplied with flange or socket welded type

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thermowell. Thermowell shall be SS316, drilled bar stock type. 50% insertion inside the pipe from top shall be considered for calculating wake frequency calculation. This calculation shall be submitted along with data sheet for approval.

- 7.7 Vendor shall provide thermowell wake frequency calculation and they follow the latest **ASME PTC 19.3 TW-2016** standard for vibrations, which is a critical wake frequency for thermowell design.
- 7.8 Oil Level switch/oil level transmitter to be provided for both High & low oil level indication.
- 7.9 Coriolis type mass flow meter with element, transmitter and integrated display shall be provided at inlet and outlet of compressor. All the electronic shall be Ex proof or intrinsically safe. Accuracy shall be $\pm 0.5\%$. Refer data sheet enclosed with tender document.
- 7.10 All package mounted transmitters & temperature elements shall be intrinsic safe as per IEC 79-11 and solenoid Valves, switches and related junction boxes shall be flame proof 'd' as per IEC 79-1. Other special equipment / instrument, where intrinsic safety is not feasible or available, shall be flame proof/ explosion proof as per IEC
- 7.11 Panel shall be complete with start and stop push buttons, hours run meter, power on and fault indication lamps, fault reset button. All necessary timers and intrinsically safe relays to control the system on an automatic starting and stopping basis shall be provided. The compressor package control system shall be designed for unattended operation in automatic mode and in case of any fault it will go in a safe mode.
- 7.12 Calibration certificates required for all instruments such as Mass Flow Meter, Pressure transmitters, Pressure gauges, Temperature gauges, Temperature transmitters, Gas detectors, Flame detectors, Pressure safety valve, vibration switch etc.
- 7.13 Refer data sheet of gas detectors and flame detectors enclosed with this document. Gas detectors and flame detectors should be mounted with the canopy.

7.2 PLC CABINET & PANELS

- 7.21 PLC cabinet shall be ex-proof, Zone 1 & 2, gas group **IIA & IIB**, weatherproof min IP 65. Dia cast aluminum alloy LM 6 MOC with Inside & outside epoxy powder coating, with shade RAL 7032, gasket neoprene rubber with single door, glass arrangement for HMI view. Panel shall be provided is Ex-proof cable gland. 10% spare gland entry of each size with SS plug.
- 7.22 HMI shall be provided with text/graphic display and operating system software for interlocking, monitoring and control. All operational buttons shall

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be on display except the Emergency stop button. Display system shall be weather proof to IP65. This should be provided in the flame proof panel with HMI mounted on the door of the panel. The HMI screen shall be back side of the toughened glass. During running of the compressor the HMI should be assessable through the external push button provided on the panel. The PLC shall be interfaced with SCADA in future. All the parameters on the PLC shall be available on the HMI. Bidder shall provide application program for PLC, HMI on LCP (licensed one set) along with all interfacing adaptors and cables. Bidder shall also provide one set of source & object code s for PLC, HMI on LCP (in both forms, hard & soft).

- 7.23 Panel shall be complete with start, stop push buttons, alarm acknowledge, alarm rest & test button for checking healthiness of annunciation system, HMI, power on, fault indication lamps, fault reset button. All necessary timers and intrinsically safe relays to control the system on an automatic starting and stopping basis shall be provided. The compressor package control system shall be designed for unattended operation in automatic mode and in case of any fault it will go in a safe mode.
- 7.24 Redundant 230 V AC to 24 V DC power supply unit (PSU) shall be provided with diode circuit for auto switch over. For compressor healthiness status shall be provided with lamp in front of the control panel for healthiness of both PSU.
- 7.25 In addition to in build in PLC, surge protection device for 24 V DC, 230 V AC UPS power supply, 230 V AC non-UPS power supply at inlet shall be provided in cabinet.
- 7.26 Each cable shall be neatly tagged & dressed for each instrument.
- 7.27 Compressor package shall be provided with a PLC based local control with HMI for local operation & monitoring, which shall be mounted on the package enclosure. All the interlock, monitoring and controlling of the CNG compressor package shall be done through PLC based control system.
- 7.28 PLC shall be configured as a remote terminal unit of supervisory computer and data acquisition system complete with Ethernet Port RS 485 (MODBUS TCP/IP) shall be readily configurable for communication over MODBUS TCP protocol through Leased Line/MPLS/VSAT/RF. PLC shall be capable of carrying out on line routines for at least ten separate loops without affecting the scan, cycle & up dating time etc.

7.3 PLC SPECIFICATIONS

- 7.30 PLC hardware shall be in accordance with IEC-61131. PLC shall be 32 bit microprocessor, min 8 MB RAM, and support to protocols DNP 3.0 over TCP/IP, DNP 3.0 over Modbus, Modbus (TCP/IP). Etc. as min.
- 7.31 PLC shall be of modular in construction, rack for mounting cards, CPU, Input output cards, Power supply card, communication card. PLC CPU shall be redundant (1 W+1S) with auto switch over without manual intervene. Failure

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alarm of CPU shall be provided in HMI and same shall be repeated in PLC for remote monitoring.

- 7.32 PLC shall be of modular in construction with EEPROM, redundant power supply for CPU and HMI, redundant power supply for load, non-redundant I/O's, communication cards for connecting mass flow meter, communication card /port for future SCADA connectivity.
- 7.33 The selected I/O shall have 20 % spare of total capacity (20% in each used card) for future use. 10% of used IO or Minimum two of each card out of 20% shall be extended up to terminal block with necessary isolator/ barriers.
- 7.34 Mounting of PLC components such as CPU, HMI, I/O's in one JB and power supply relay barriers/isolators, fuses, MCB, electrical earthing bus bar in other JB. PLC components / system shall be tropicalized, adopted with complete wiring and necessary terminals. Wiring to be color-coded with cross printed ferruling in position
- 7.35 The units of measurement for flow shall be Kg/hr., for pressure shall be Kg/cm² (g) or and for temperature shall be degree C.
- 7.36 PLC shall be suitable for controlling of compressor parameters as indicated in instrumentation and all other parameters that are recommended by the compressor manufacturer for 24x7 operation. PLC shall be suitable interfacing port for Laptop for programming, configuration and diagnosis. Type of communication with laptop shall support up to 30 meters distance from compressor to safe area. The cable for laptop communication through RJ-45/RS-485 Ethernet port to be supplied by bidder.
- 7.37 A dedicated Modbus (RS 485) slave communication port shall be provided for Remote terminal unit (RTU) interface. A separate Modbus TCP/IP communication ports shall be provided for GPRS modem for communication with SCADA system.
- 7.38 All the parameters shall be time stamped in PLC. It shall be possible to synchronize PLC clock with GPS system.
- 7.39 PLC shall be 20% spare each type of input output channels for future use. All these IOs shall be wired up to the spare terminal block inside the PLC cabinet. Diagnosis feature shall be available in CPU and I /O used in PLC. The PLC/RTU shall comprise the following subsystems:

- Redundant Central processor with system software.
- Power supply unit
- Analogue input
- Contact (digital) input
- Contact (digital) output
- Modbus Serial ports configurable (RS 232/485)
- Ethernet port

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- Diagnostic port

- 7.40 The Digital Output shall be configured for pulse duration. No separate program or logic will be acceptable at PLC end. On restart/ power failure, PLC shall not reset the output circuit, shall not generate false control signal and shall necessarily cancel all pending control signal.
- 7.41 Configurator software licenses shall be preferred in software (software key) form instead of hardware (dongle). Multiuser software licenses shall be provided for common GA.
- 7.42 Suitable bypass for interlocks shall be provided for start-up.
- 7.43 The compressor package control system shall be so designed that the first item to go into alarm condition shall lock out to indicate the cause of the trip though the cause of the trip may have disappeared. The lock out condition shall be manually reset. A change over set of contacts shall be provided for Owner's use to give a remote indication of alarm and trip.
- 7.44 In case of fault, a warning hooter shall operate, the sound of which should be audible at distance of at least 15 meters. Further the fault alarm and emergency stop PB shall be duplicated in the CNG station control room. Acknowledgement/resetting of fault shall be possible only from compressor panel. Emergency stop PBs shall be mushroom head turn lockable type. Once the fault is acknowledged or compressor is under normal maintenance, the valves of priority panel shall take the position so that gas available in the stationary CNG storage cascade can be dispensed.
- 7.45 Configuration and diagnostic tool should be able to connect to remotely over TCP/IP.
- 7.46 Successful bidder to include in scope live demonstration of remote monitoring of all PLC logged parameters in one machine at his works. OWNER may ask for the same. However, this may be required to be demonstrated at site.
- 7.47 The PLC System offered shall be supplied with monitor and memory card for Processing of live data and stored data. PLC shall be capable of
- a) All the process parameters as per P&ID
 - b) Compressor Control & Emergency Shut down
 - c) Fire and gas detection and monitoring
 - d) Graphics, Data acquisition, monitoring & logging, viewing, modifying set point and range of all process parameters for which transmitters are provided.
 - e) Record the last 20 Alarms of abnormal operations on separate page.
 - f) PLC shall be capable for display of flow meter data for flow rate and flow totalizer (i.e., Gas Suction, Gas Discharge) and power consumed by the motors, compressor running hour:
- 7.48 Compressor package shall be provided with the following indicators:

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- a) Pressure transmitter and pressure gauge at suction, 1st & 2nd stg. discharge, cascade Low, Med, and High bank. Distribution header.
- b) Pressure data at final discharge: Bidder shall provide Pressure Transmitter with suitable interlock and Pressure Gauge for manual visibility at final discharge. In addition Safety Relief Valve shall also be provided for the safety of Package.
- c) Pressure transmitter at Dispenser Emergency line.
- d) Pressure gauge & temperature gauge at each stage of compressor.
- e) Temperature transmitter shall be provided at 1st discharge after compression and final discharge.
- f) MFM at suction & discharge.
- g) Oil pressure indicator on each pressure lubrication system.
- h) Oil levels indicator
- i) Oil temperature indicator
- j) Oil level transmitter
- k) Water level indicator
- l) Water temperature transmitter
- m) Water pressure indicator
- n) Water pressure transmitter
- o) Each compressor non resettable running hour's calculation in PLC. Alarm shall be generated after preset point.
- p) Compressor jacket water coolant temperature indicator on local gauge panel.
- q) Hydraulic oil cooler outlet temperature gauge.
- r) Hydraulic oil discharge line and return line shall be provided with pressure gauge.
- s) Air compressor discharge pressure transmitter.
- t) Co2 system healthy indication with online weight loss indication.

7.49 The Compressor package shall be provided with the following trip devices:

- a) Low & High oil level protection devices
- b) High & Low oil temperature devices
- c) Low & High Oil pressure or Flow.
- d) High discharge gas Temperature on each stage.
- e) High discharge gas Pressure on each stage.
- f) Low & High suction pressure protection devices
- g) High & low water temperature reached
- h) High & Low water pressure reached
- i) Coolant flow low devices
- j) Low signal / unhealthy signal of mass flow meter.
- k) Flame detection
- l) Gas detection
- m) Emergency stop devices
- n) Fail safe/ wire break alarm for safe operation
- o) Interlocking provision in PLC program for tripping of machine
- p) Electrical Faults such as over voltage, under voltage, earth leakage, single phase, Phase Reversal, over current.
- q) When earth current fault exceed the set points.

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- r) Each package shall be furnished with an Audible and visual alarm system for annunciation Equipped on compressor trip

7.50 Compressor package shall be furnished with the following trip logic that shall stop the compressor and suction of compressor shall be isolated:

- a) On high oil temperature
- b) On low suction gas pressure
- c) On high discharge pressure
- d) On high discharge gas temperature
- e) On coolant flow low
- f) On fire detection
- g) On gas detection
- h) On pressing manual stop button at local control panel
- i) On pressing emergency stop devices
- j) On High oil Level

7.51 Compressor package shall be furnished with following tripping circuit (the motor shall stop and suction of compressor shall be isolated)

- a) On actuation of gas detector alarm.
- b) On actuation of flame detection alarm.
- c) On pressing of manual stop button at compressor package
- d) On pushing of emergency stop device

7.6 EMERGENCY SHUT DOWN DEVICES

ESD button (**3 Nos.**) shall be provided (Control Room, Process Area, one side of compressor). A separate hooter for customer interface room shall be provided with annunciation window alarm of individual protection device. The emergency shut down (ESD) system is also in scope of vendor. A fail-safe system shall be designed and incorporated to isolate cascades storage from dispensers, stop compressor isolate the compressor suction storage line and cut off power supply on activation of ESD switch. This ESD switch shall have to be manually reset to restart the compressor package again. Red ESD button. ESD should be installed with rain protection canopy.

7.7 CABLE SPECIFICATIONS

Cables shall be Flame retardant low smoke (FRLS) type shall be used for all field instruments except ESD device. Voltage Grade of cables shall be 650/1100 V.

Fire resistance type for Fire detection fire detection system, ESD service, instruments used for ESD purpose, and ESD push buttons.

1T x 1.5mm² shall be used for gas detection system. Quad cable shall be used for RTD. 1P/2P x 1.5 mm² shall be used from field instruments and 2/3 core x 1.5 mm² /2,5mm² for power to cabinet or SOV.





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Multi pair cable individual and over all shielded shall be used for all the along signals and Multi pair over all shielded cable shall be used for all the digital input /output signals.

Single Pair and quad Shielded Cable

Each core shall be 1.5 mm² made of 7 stranded annealed electrolytic copper conductors. Each strand shall be 0.53 mm dia. For Intrinsically safe application, primary insulation shall be low density poly ethylene (LDPE). For non-intrinsically safe application, primary insulation shall be 85° C polyvinyl chloride (PVC) as per IS5831 Type C. Thickness shall be 0.5 mm minimum.

A pair shall have twisted cores and number of twists shall be not less than 10 per meter. Color of core insulation shall be black blue in pair and black, blue and brown in triad.

Individual pair shall be shielded. Shield shall be Aluminum backed by Mylar/polyester tape with the metallic side down helically applied with either side 25% overlap and 100% coverage. Minimum shield thickness shall be 0.05 mm. Drain wire shall be 0.5 mm² multi-strand bare tinned annealed copper conductor. The drain wire shall be in continuous contact with Aluminum side of the shield.

Inner and outer jacket shall be made of extruded flame retardant 90° C PVC to IS 5831-Type ST2. Oxygen index of PVC shall be over 30%. Temperature Index shall be over 250° C. The thickness of the jacket shall be as per IS-1554 Part 1.

Inner jacket color shall be black. Outer jacket color shall be black except, for cables to be used in intrinsically safe systems it shall be light blue. A rip cord shall be provided for Inner jacket.

Armor over Inner jacket shall be galvanized steel wire as per IS-1554 Part.

Multi-pair Cable with Individual Pair Shield and Overall Shield:

Generally, the cable shall be same as single pair shielded cable conductor sizes shall be 1.5 mm² made of 7 strands of annealed electrolytic copper conductor. Each strand shall be of 0.53 mm dia.

Overall shield shall be of Aluminum backed up by Mylar/polyester tape helically applied with the metallic side down with either side 25% overlap and 100% coverage. Minimum shield thickness shall be 0.05 mm. Drain wire shall be similar to individual pair drain wire and shall be of the overall shield. Pair identification shall be with numbers at interval of not more than 250 mm as per vendor's standard.

Multi-pair/core Cable with only Overall Shield

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These cables shall be same as above except that the individual pair shall not have shielding.

Fire resistance cable

Fire resistance cable will be multi core copper conductor, Mica tape with overlap as fire resistance layer Extruded XLPE insulated per IEC 60092-351, Braid of copper wires designed to act as collective screen, sheath as per 60092-359. Fire resistance cable shall be tested for fire resistant properties 750 Deg. C for 3 hours as per IEC 60331-31.

Control Cable

Single pair control cables shall be used between field mounted solenoid valves and junction boxes/local control panels. Multi pair control cables shall be used between junction boxes/local control panel and control room mounted devices in general. These cables shall have only overall shielding. These control cables shall have 1.5 sq mm conductor size with 7 stranded conductors of annealed electrolytic grade copper, with each strand of 0.53 mm diameter.

Power Supply Cables

All power supply cables shall be as per IS-1554 Part I and shall have copper /aluminum conductors depending on conductor size. Power cables shall be 3-core, minimum 2.5 mm² copper conductors with PVC insulation, galvanized steel armoring and overall PVC sheathing.

Earthing Cables

Earthing cable shall be 1C x 6 mm², outer sheet Yellow with Green strip, voltage grade 1100V. Annealed Bare Copper conductor, PVC insulated Un-armored.

7.8 PRIORITY FILL SYSTEM

Bidder shall supply 7-line priority fill (3-bank) system with compressor top-up facility inclusive of regulating valves, check, bypass valves & liquid filled pressure gauges all mounted in a stainless-steel panel. All fittings and tubes used in priority system shall be of stainless steel of suitable pressure rating. The priority fill system is to be installed to ensure that vehicle filling takes priority over cascade filling and direct CNG to three storage banks in correct sequence. The compressor shall shut down once all three-cascade storage banks are filled to 255 Kg/cm² (g). Compressor shall start on pressing of manual start push button & automatically when the cascade storage high bank pressure of compressor falls to 200 Kg/cm² (g) and shutdown automatically when all 3 stages of stationery cascade are filled to a pressure of 255 Kg/cm² (g).

The priority fill system (In Bidder' s scope) shall ensure the filling of vehicle, storage cascade in correct sequence. Control system shall be designed such

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that in case of any fault, discrepancy or abnormality, it will go in safe mode. All controls shall be made in fail-safe mode failure of any control shall not lead to operation of equipment in unsafe condition.

8. SAFETY

- a. All controls shall operate in a fail-safe mode i.e. failure of any control shall not lead to running of equipment in unsafe mode. Fail safe control shall be available through hardware for all trips & also in software. The Compressor Package shall trip if any of the enclosure is opened while the machine is running.
- b. The hazardous area classification Class-I, Division I, Group D as per NEC or Zone I, Group II A/ II B as per IS/ IEC. Certificate from recognized agency to the effect that equipment supplied and/or installed conform to above area classification. All Devices shall meet the requirement for the specified area classification in which they are installed, including instrumentation leads.
- c. Bidder needs to submit copy of valid type approval for proposed compressor packages (Model) from PESO along with the bid.
- d. All exposed rotating parts shall be provided with adequate guards of non-sparking (Metallic) type.
- e. Driver belt if used shall be of anti-static and fire-resistant type.
- f. Piping shall be arranged in a manner so as to provide clear headroom and accessibility within the package. Adequate clearances shall be provided for all the engineered components.
- g. All material used in the package shall be flame & Fire retardant.
- h. Relief valves shall be provided at each stage of discharge of the Compressor with setting as per cl.7.20.4 of API-618 with R.V. venting as per cl. 7.20.4 of API-618 or OEM std. practice. All vented to common relief valve header.
- i. Provide all the required warning notices, barriers, safety boards, padlocks etc. for safe commissioning of the equipment's.
- j. Complete enclosure shall body shall be pasted with Fire and flame retardant based PU foam for any sudden flame exposure creates due to any possible causes.

9. CARBON DIOXIDE (CO₂) FLOODING SYSTEM

- a) CO₂ flooding system should be installed for the protection of CNG compressor by automatic actuation system. The package should be protected by automatic operated CO₂ flooding system designed as per NFPA-12/ IS 15528: 2004 and OISD 142.

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- b) Gas Detection by installation of hydrocarbon gas detector (IR type) with self-check function and transmitter with adjustable alarm levels (0-100%) with preset of 10%, 20% and 40%. Each enclosure should have at least 1 nos. gas detector.
- c) Installation of flame detector (UV-IR type) with self-check function & transmitter and Multi-sensor detectors, alarm on detection shall be provided. Self-check function to generate fault alarm and trip alarm in case of flame detection Indication lamp shall be provided for fault and trip signal in PLC. Each enclosure should have 1 nos. flame detector.
- d) CO2 flooding system will consist of Min 2 nos. brand new CO2 cylinders of adequate capacity (should not be less than 45 Kg each). However actual size of the cylinder shall be as per compressor enclosure size and necessary calculation shall be submitted and approval shall be taken before finalization / supply. One cylinder will act as main & other as stand by, which shall have identical arrangement and connected to the system.
- e) The cylinders should be placed in a shed raised above ground level to protect from weather and direct sunrays as per Gas Cylinder Rules, 2016.
- f) Cylinders shall be fitted with automatic actuated Valves, Solenoid valves for automatic actuation No extra utility as air, and inert Gas shall be made available by OWNER /used by the supplier to operate the system other than the UPS.
- g) The CO2 system shall be designed to operate on UPS in the event of power failure to handle emergency situations
- h) The system shall be designed to operate on 24 V DC supply. FRLS (Fire resistant low smoke) cables shall be used for the wiring of the system.
- i) Compressor shall trip on detection of flame at preset level and automatic discharge of CO2 gas shall take place from the main cylinder simultaneously.
- j) Cylinder should be ISI marked as per IS: 7285(latest edition) and CCOE approved.
- k) Control philosophy shall be such that in case detection of fire by fire detector cylinder shall discharge CO2 automatically.
- l) Online weight measurement system for cylinders shall be provided.
- m) The System shall be designed to operate on 24 V DC supply.
- n) FRLS (Fire resistant low smoke) cables shall be used for the wiring of the system.
- a) **Interlock of CO2 Flooding system with compressor as per following sequence:**

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- Compressor shall trip on detection of gas at preset level.
 - Compressor shall trip on detection of flame at preset level and automatic discharge of CO₂ gas shall take place from the main cylinder simultaneously. Compressor shall not start if the CO₂ Flooding System is faulty, not working, SWITCHED OFF etc.
 - Compressor shall not start if the CO₂ flooding system is faulty, not working, SWITCHED OFF etc. The compressor shall be able to start only when the CO₂ Flooding System is in healthy working condition.
 - Maintenance Override Switch shall be provided to keep the system off during maintenance.
 - Selector switch shall be provided to put Main/Stand by Cylinder in line at the turn of a switch as per requirement.
- b) Alarm panel for CO₂ Flooding System shall be integral with the main compressor panel. Necessary displays as system ON, OFF, FAULT, RESET, Gas/ Flame indication, Remote actuation of solenoid valve, distinguished hooter etc., shall be provided for CO₂ flooding system.
- c) CO₂ Cylinders shall be provided outside the package at a safe place, where it is not exposed to fire in case of fire in the compressor. Facility shall be made to operate the system both manually and also from remote with the help of a switch/ call point and with help of pull down lever on cylinders.
- d) Suitable online weight (CO₂) loss monitoring/ indication device to be provided to ascertain the health of the CO₂ flooding system.
- e) All installation shall be compatible for hazardous area **Class 1, Division 1, Group-D** for Methane Gas.
- f) One Blinking aviation lamp / Hooter as per requirement shall be provided at the top of compressor canopy suitable for hazardous area for fire indication.
- g) The system designed by the supplier shall be duly approved by Owner/ Owners representative.
- h) Technical specifications, Operation and Maintenance Manual, CCOE Certificate, Approval/Manufacturing certificates for cylinders and cylinder valves, gas detectors, flame detectors, solenoid valves etc. shall be furnished by the supplier along with system.

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- i) Software and hardware, calibration procedure shall be provided by the supplier along with the supply sufficient enough to handle the system independently.
- j) Necessary tools (1 set) shall be provided with the system.
- k) System shall be offered for testing to OWNER by the supplier after commissioning at site by creating actual Gas leak and Gas fire situations and actual discharge of CO₂ Gas from the Cylinders. This shall form a part of performance test and thereby acceptance of the package. The cylinders have to be refilled by the vendor at no extra cost to OWNER after performance test. If the system fails during testing, subsequent testing and refilling would be at vendors cost.
- l) Warning and Operating instructions to be displayed at equipment as per the statutory/ safety regulations.
- m)** Piping of CO₂ flooding system shall be seamless high pressure pipe of **Schedule 40 of 50 mm dia.** Of appropriate length with a minimum safe distance of 4 Meter from CNG Compressor, the fittings like elbows, Tees, Union, sockets should be of same schedule and capacity for installation in a high pressure system as per **NFPA-12.**
- n) All CO₂ flange connection shall be ANSI class 300 # as per NFPA-12.
- o) Flameproof online weighing system, complete frame with shed and all accessories should be of good quality, weighing scale should be of reputed make.
- p) Specifications:
Non Return Valve for CO₂ High Pressure Hose:
As per BIS specifications
Operating Media: CO₂
Body Material: Brass, BIS: 319
Ball: SS 316
Pin: SS 316
Seal: Teflon (PTFE)
Working Pr.: 60 Kg/cm²
Test Pressure: 90 Kg/cm² for 1 min
Weight: 70gm
Outlet Size: ¾" BSP at manifold end
Inlet Size: 1/2" BSP at CO₂ Discharge Hose end
Temp. Range: -29° C to 65°C
Hose Adaptor:
As per BIS specifications
Operating Media: CO₂
Body Material: Mainly Brass
Test Pressure: 250 Kg/cm²
Max. Working Load: 150 Kg/cm²
Temp. Range: -29°C to 65° C



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Discharge Nozzle:

As per BIS specifications

Operating Media: CO₂

Body Material: Lead Tin Bronze as per BIS: 318:1981

Design Nozzle Pr.: Not less than 20.6 kg-f/cm² at 27° C

Test Pressure: 140 kg-f/cm²

Marking for Code No. (On the basis of equivalent single orifice dia.): As per BIS: 6382:1982

Temp. Range: -29° C to 66° C

High Pressure Hoses:

As per BIS 7285:1974

Operating Media: CO₂

Hose Type: Double wire braided (perforated) rubber covered

Min. Bursting Pr.: 420 kg-f/ cm² at 54° C

Length: 40 cm

Cross-section: 1/2"

End Connection: 1/2" BSP (F) x W21.614 TPI

End Fittings: Brass

Temp. Range: -29°C to 66° C

Following warning and caution signage shall be marked on the housing/package:

Caution notice **"This Machine may automatically start at any time"**

"Flammable Gas"

"CNG Do's & don't instructions"

"No open Flames"

"No Smoking"

10. DOCUMENTS

Vendor shall submit documents during engineering stage for review & approval to client/consultant.

- 10.1 Specification /data sheet with statutory approval certificate, W&M certificate, PESO certificate of all the instruments as per P&ID,
- 10.2 Instrument index
- 10.3 Input output list
- 10.4 Power consumption calculation,
- 10.5 Cause & effect
- 10.6 Cables specification
- 10.7 Cable schedule with termination details
- 10.8 Operation & control philosophy
- 10.9 PLC specification & architecture.
- 10.10 Operation and Maintenance Manual (In English) - 02 Copies
- 10.11 Calibration certificates of all instruments & devices
- 10.12 P&ID with tag numbers for each item, line size, valve size, type of connection, instrument symbol
- 10.13 Connectivity with PLC, interlock number.
- 10.14 Bill of Material with Tag No & Technical Specifications



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- 10.15 Wiring Diagram of Electrical & PLC Panel
- 10.16 Specifications of Electric Motor & Characteristic Curves
- 10.17 Foundation Drawings
- 10.18 Capacity vs. Suction Pressure curve
- 10.19 Capacity vs. Energy Consumption curve
- 10.20 List of spares for operation and maintenance as per MR. The list of spares should include ordering information
- 10.21 Specification and manufacturer's catalogues. Bidders to keep inventory of spares as mentioned in the list in Centralized stores at one or two GA's location as finalized by the owner. The availability of the spares will be checked by the owner on regular basis.
- 10.22 List of special tools & tackles to be provided along with the bid.

11. SKID & ENCLOSURE

- 1. The maximum allowed temperature within the enclosure shall be 5° C above ambient temperature. Adequate ventilation fans shall be provided to meet the above and also to account for heat dissipation of the coolers.
- 2. Interlock shall be provided to start the exhaust fan to vent out any entrapped gases in the enclosure before starting the main compressor. In case heat exchanger fan is compressor shaft driven, the same can't be utilized as ventilation fan.
- 3. The compressor package shall consist of single enclosure for Compressor and Electric Motor. The equipment shall be mounted on one common skid. The Enclosure to restrict maximum noise level to 75+/-3 dB (A) at 1 meter from the enclosure in any direction.
- 4. Material used in the enclosures shall be fire retardant. Rain water should not enter into the enclosure and forced ventilation system shall be provided.
- 5. The enclosure shall have doors for normal access and removable wall panels for ease of maintenance. All the pressure, temperature, oil level, lube oil pressure, coolant temperature, coolant level indicators shall be visible from outside of enclosures and shall be mounted on gauge panel shall be visible from outside without opening the package door.
- 6. Enclosures shall have internal flame roof lighting arrangement. For handling all heavy arts for maintenance purpose lifting arrangement i.e., beam fitted with necessary arrangement shall be provided in enclosure.
- 7. The Compressor shall be located inside an acoustic enclosure. All Coolers, Cooling System, lubrication system along with interconnecting piping shall be inside an enclosure. Enough headroom shall be made available for easy access and maintenance of all equipment. The piping layout with respect to the compressor, intercoolers, and auxiliaries' location shall be subject to Purchaser's approval during detailed engineering Stage.



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8. Components such as pressure gauges, temperature, pressure switches, filter automatic ball valves, safety valves etc., which require in-situ adjustment, maintenance and reading, shall be easily accessible.
9. Conduits and tubing shall be arranged in orderly and systematic manner and shall be routed neatly to enter the back of display or monitoring panels
10. Routing service item such as, but not limited to, oil filters, inter stage gas filters, inlet and outlets gas filters and drive belt shall be located to facilitate easy one-man servicing.
11. One person should be able to access oil inlet and drains to allow addition or drainage of oil without removing panels or adjacent components and without the need of the pump.
12. Items which must be operated & monitored during operation shall be readily accessible without opening the door Suitable gradients shall be provided on the enclosure roof for rain drainage and to avoid water pockets.

11.1 Painting & protection

Packing shall be sufficiently robust to withstand rough handling during ocean shipment & inland journey. Sling points shall be clearly indicated on crates.

Painting of Internal process piping should be as per international color coding standard, i.e. Gas Line-Yellow, Water line- green, Airline-Blue, Fire suppressing system - Red etc. The paint shall be chosen, primed and applied to have a service life of ten years the exterior of equipment and enclosure is required to be corrosion Free for Ten years.

Painting/power coating (Primer & Finish Coat) Following primer and finish coats to be applied on the canopy and all structural parts as a minimum:-

A. Primer: Two component epoxy zinc phosphate primer with minimum volume solids of 59% an initial cure of 75 minutes at 25 deg. C and a weight of around 2.52 Kg / Liter.

No. of Coats: 1

DFT: 75 (micron) each

B. Primer: Two component intermediate coat with epoxy high build MIO (micaceous iron oxide) of minimum volume solids of 80% an initial cure of 60 minutes at 25 deg. C a weight of around 2.1 kg/ liter.

No. of Coates: 1

DFT: 100 micron

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C. Finish Coat: Acrylic Polyurethane paint

No. of Coats: 50 (micron) each coat

Total DFT: 100

Total DFT after application of primer and paint shall be 275 (micron) minimum.

12. INSPECTION & TESTING

- a) Inspection shall be carried out as per Quality Control Table/ Quality Assurance Plan approved by Owner or its representative. Supplier shall submit the Quality control table (complying with tender QCT/ QAP) for approval.
- b) The bidder has to provide 15 days advance notice prior to said inspection & test.
- c) Bidder shall keep following data available for at least 5 years for examination by purchaser.
- d) All necessary certification of materials, such as mill test reports.
- e) Purchaser specification for all items on bills of materials.
- f) Test data to verify that requirement of the specification have been met.
- g) Result of quality control test.
- h) Pressure retaining parts including auxiliaries shall be hydrostatically tested with water at following minimum test pressure for a minimum period of 1 hour:
 - Cylinder: 1-1/2 times maximum allowable working pressure.

12.1 Mechanical String Test /Factory Acceptance Test

Mechanical String Test for 4 hrs. Is a mandatory requirement to be performed at packager's shop before dispatch in presence of Owner's representatives (or a third party as arranged by OWNER). This test can be clubbed up with the Mechanical Run Test of compressor as specified above, provided the job driver & lube Oil system is used for the test. String test shall be on natural gas preferably. Air/ N2 can be used for string test purpose if natural gas is not available in the shop.

12.2 Erection, Testing & commissioning at Site

The bidder shall be responsible for erection, testing, commissioning & performance test and noise level test of all packages at site. Commissioning of various equipment and systems shall be carried out by the bidder as per the accepted procedures and as per the instruction of the manufactures of the equipment. The units will be considered commissioned only after the successful site tests are carried out by the bidder. The bidder shall ensure integrity of compressor package and safety of electrical supply system available at back end while testing package, at site.

12.3 Field Trial Run

Bidder shall conduct a field trial run of each compressor package for 8 hrs. (can be in multiple runs) at CNG Station as per QAP in which satisfactory

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operation of complete package together with all accessories/auxiliaries' controls shall be established for specified operating conditions without any major breakdown prior to the start of operation and maintenance period as defined in the contract. During the field trial run the bidder will be allowed a maximum of THREE attempts to complete the above specified test i.e., only for reasons not attributable to the vendor. The Equipment shall be considered commissioned after the successful completion of Field Trial Run. All punch points raised by OWNER should be completed before performing the FTR. The bidder shall record data of field trial run

12.4 Noise Level Test

During the field trial run, noise level test shall be carried out and bidder shall demonstrate /achieve the granted noise level. All necessary instruments /accessories required for fields trial run and noise level test shall be arranged by the bidder.

12.5 Performance Acceptance Test (PAT)

Before conducting performance acceptance test at site, Bidder is required to clear all punch points (if any) raised by OWNER / OWNER' s authorized representative.

Compressor Package Performance test at sites shall be carried out as per ASME PTC9/Approved test procedure. All necessary instruments/accessories required for this test at site shall be arranged by the bidder and repatriated after successful performance test by the bidder. All such instrument shall be pre-calibrated. Bidder to quote for performance test per compressor package separately at site inclusive of boarding, lodging office space, local transport for bidder personnel and hiring of local contractor, crane etc. bidder shall be liable to pay all local taxes, levies applicable and strictly comply with rules, laws prevailing in India. Performance test shall be conducted at site for minimum 8 hours on Cumulative basis at guaranteed parameters as quoted by the bidder (**PAT procedure shall be in accordance with ASME-PTC-09/ OEM Testing procedure**).

However, if load is not available at site intermittent running for 8 hours shall be permitted with maintaining minimum continuous operation of $\frac{1}{2}$ hour. Bidder to submit PG test procedure for review / approval. Complete package shall be performance tested as a module whereby along with motor & compressor performance bidder shall demonstrate all controls, shutdown, trips/alarms etc. The test shall be the basis of, acceptance/rejection of the package thereon. Bidder shall submit the detail test procedure for the same, which shall be approved by OWNER. The test for the package shall be witnessed by OWNER/ OWNER' s representatives.

13. GUARANTEE, LOADING & PENALTY CRITERIA

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This section described the guarantee parameters, which the booster compressor package must fulfil, the penalty for shortfall in guaranteed parameters and rejection of compressor package by the purchaser.

The guaranteed parameter shall be adjusted to account for variation in gas composition and prevailing ambient condition during testing. Necessary calculations correction curves shall have to be furnished by bidder along with bid, which shall be

Final & no deviation shall be permitted afterwards.

In case of any inconsistency in manufacture and /or operation of supplied compressor package, Bidder shall at his own risk and cost, eliminate the defects to the satisfaction of owner. For loading and compensation purpose, power consumption with suction pressure of 210 to 30 kg/cm² (g) and discharge pressure equal to dispensing pressure may be considered. Dispensing pressure will depend on empty vehicle pressure to be fueled and compressor discharge pressure may not be 250 kg/cm² (g) continuously.

For power consumption purpose discharge pressure may be taken 230 kg/cm². The bidder shall furnish the guaranteed value for the following:

Compressor Capacity: Compressor shall guarantee the capacity as mentioned in Guaranteed Parameters.

Compressor BkW: Bidder shall indicate guaranteed BkW including all losses such as mechanical, transmission etc.

Motor Power Output of the prime mover (KW)

Total power required for the package including power consumed by accessories.

13.1 Compressor Capacity

Bidder shall guarantee average capacity of 400 SCM/H from suction pressure 210 to 30 kg/cm² and discharge pressure of 250 Kg/cm² at 52° degrees centigrade (Max).

For calculation purpose **1kg of CNG = 1.337 SCM/H**

If bidder quotes less than **20 KWH for 400 SCM/H** capacity (i.e. excluding Auxiliaries power like Cooling fan, pump & air compressors etc.), No advantage will be given for quoting .The same shall be used to establish the capacity at site during package performance test, however if bidder quote more than 20 KWH for 400 SCM/H penalty shall be imposed at site.

Penalty towards Excess Energy Consumption:

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At the start of O&M period or even at any point of time during the O&M period, cost towards excess power consumption beyond quoted figure shall be deducted from O&M bills.

Following calculations shall be used for deduction towards excess power consumption.

$$F = (G-Q) \times H \times N$$

Where

F = Monthly Penalty in Rs

G = Monthly Actual power consumption

Q = Guaranteed consumption rate quoted by supplier for every 300 Kg of CNG x CNG produced during the month

H = Cost of power Rs 10/kwh

N = No. of operating units.

Penalty towards Package Efficiency Loss

This penalty shall be imposed on compressor blocks not capable of delivering rated capacity of 400 SCM/H

Following calculations shall be used for penalty towards package efficiency loss:

$$F = 2 \times \{(400 \times H \times RD \times AD) - M\}$$

Where,

F = Penalty Amount in Rupees

H = Hours clocked in a month

RD = Average RD for the month using GC Data by clients.

AD = Air Density = 1.22541

M = Discharge mass flow during the month in Kgs

To calculate above mentioned penalties, following formula shall be applicable for Conversion of units from SCM to Kg:

$$G \text{ (in Kg)} = 400 \times RD \times AD$$

Relative Density (RD) = Average Relative Density for the month using Average GC

Data (GA wise) (for calculation **RD shall consider 0.593557**)

Air Density (AD) = 1.22541

Note:

- Gauge Pressure at Station Inlet shall be used as benchmark for imposition of penalties and not suction pressure being displayed at the PLC.

Penalty for Non-Performance during Operation & Maintenance Period

(a) On normal day (i.e., the day other than the schedule maintenance day):

- (i) The party has to ensure that the equipment's are available for operation for minimum 18 hours per day.

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- (ii) If the equipment is down for more than 6 hours on any day. Penalty would be applicable as follows: up to 12 hours: R.S. 20,000/- per day Beyond 12 hours: R.S. 40,000/- per day.

(b) On schedule maintenance day:

- (i) The party would be required to carry out the recommended schedule/preventive maintenance of the equipment for which the party has to indicate the time required for each type of schedule maintenance.
- (ii) If the equipment is down for more than 6 hours & up to 12 hours beyond the time indicated for the agreed schedule maintenance, the party would be penalized R.s. 20,000/- and for more than 12 hours R.s. 40,000/- per day.
- (iii) In any case, the maximum penalty imposed in a month for non-performance of the equipment would be limited to 50% of the Amount of O&M charges to be paid to the party per month per compressor.

(c) Penalty for operator Non-availability & PPE requirements :

In case, the contractor's service personnel are unable to reach the break down site within stipulated time or is unable to complete the maintenance within stipulated time, following penalty will be applicable. This amount will be deducted from the invoice raised by the contractor, at the end of the month.

- 1.1.1 Penalty for 1-hour delay in reaching at all site (i.e., within 3 hours instead of 2 hours) – Rs 500/-
- 1.1.2 Penalty for 2-hour delay in reaching at all site (i.e., within 4 hours instead of 2 hours)- Rs 1000/-.
- 1.1.3 If the shutdown time, which will be calculated from the time the compressor is offered to service person on reaching site, is extended beyond 4 hours, a penalty of Rs 500/- per hour will be applicable. Rs 2500 towards non-availability of mandatory spares at operational sites.
- 1.1.4 Contractor shall not deploy the employee of age less than 18 years in any of the activities. If it is found, then it will be viewed seriously and heavy penalty of Rs. 20000/-per instance and also the termination/blacklist will be done from our approved vendor list.
- 1.1.5 The contractor shall provide full Personal Protective Equipment (PPE) to each individual employee including, soft hat, eye protection, ear plug, and safety shoes. It is mandatory for all personnel to wear said PPE whilst performing their duties, failing which a penalty @ Rs.

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500/- per incidence will be levied in addition to dismissal of the person.

14. SPECIAL TOOLS & TACKLES

Special tools & tackles for erection and commissioning and for operation & maintenance are required to be arranged by successful bidder. Vendor shall maintain sufficient spares to fulfill the warranty & subsequent four years period requirements. In case of additional requirement during the warranty period, if any spare part is taken from OWNER, the same shall be replaced to OWNER with new part supported by necessary document for its authenticity of being new & original spare part.

15. DRAWING & DOCUMENTATION

- The drawings/documents to be submitted by the bidder shall be divided in three categories:
 - a. Drawing documents to be supplied with the offer
 - b. Drawing /document to be submitted for approval (After placement of order)
 - c. Drawing/document to be submitted for information (After placement of order)
- The bidder shall submit list of drawings, which shall be submitted by them in above three categories. All drawing /document shall be submitted in 6 sets.
- Title block of each drawing shall contain at least following information:
 - Name of the Owner:
 - Name of the Consultant:
 - Name of the Project:
 - Name of Bidder:
 - Descriptive title:
 - Drawing No:
 - Revision No:
 - Sheet No. / Total No. of sheet in the drawing:
- Bidder shall furnish relevant calculation and protection relay setting table for the equipment /system being supplied by them, It shall also contain the manufacture catalogue, operation and maintenance manuals for all types of relays/components used,
- The bidder shall submit all drawing within specified time, in requisite number, for each equipment/item for approval.
- Approval of drawings by Owner.

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- The bidder shall submit furnish drawings as indicated/agreed for each item for approval of the Owner/Consultant.
- Owner/Consultant will scrutinize drawing/data furnished by Bidder and comments, if any, will be communicated to the Bidder within 2 weeks from the date of receipt.
- The Bidder shall submit all the drawings/documents in two (2) sets. All the drawing of sub-bidder/bidders etc. shall be checked by Bidder for correctness and compliance with requirement of order/contract and signed before submission to Owner.
- The drawings shall be stamped in either of following category and one print shall be returned to the Bidder.

“APPROVED”

“APPROVED SUBJECT TO INCORPORATION OF COMMENTS”

“NOT APPROVED”

“FOR INFORMATION ONLY”

- Wherever drawings are returned to the Bidder with the marking “Approved subject to incorporation of comments” the bidder shall make the necessary modifications/corrections and resubmit the revised drawings and data for final approval.
- Bidder shall be responsible for correctly incorporating all the points conveyed to him and resubmit the drawings to the Owner for final approval. Specified number of copies of approval drawing.
- Approval of drawing by owner shall not relieve the suppliers of his contractual obligations and responsibility for engineering design, workmanship, materials and performance of equipment, Work shall be carried out exactly as indicated on the approved drawings and data and no alterations shall be made without the written approval of the Owner.
- If any subsequent alterations are found necessary and approved by the Owner, all drawings and data affected by such alterations shall be duly revised and re-submitted for the approval.
- Bidder shall incorporate/cause to incorporate all change made in the drawings from approval stage to the handing over of equipment and submit as built drawings in the requisite sets (these will be in addition to the sets submitted at the time of approval/for information and up to commissioning). The drawing to be submitted shall include all the drawings submitted for approval, information as also the drawings required for normal operations, troubleshooting repair, and maintenance and testing of equipment etc.



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- Bidder shall submit the following drawings/data/document in bound volume prior to submission of final bill to the Owner in soft and hard copies (2 Sets).
- All drawings submitted to Owner for approval information
- Equipment manufacturing drawings submitted for information of Owner.
- Equipment drawings required for operation and maintenance.
- Fault calculations, protection relay setting calculations and recommended settings.
- Inspection reports, factory and site test certificates in bounded volume.
- As built drawings incorporating all site modifications. Instruction manuals
- List of spares for four years operation and maintenance. The list of spares should include ordering specification and manufacture's catalogues.
- Operation and maintenance procedures for individual equipment and total system

16. PARAMETERS FOR SCADA

- Parameters required for SCADA
- We want to monitor / control all parameters available on the PLC through SCADA system Also it should remotely operate from any location.
- Hardware Details: Connector Type, Communication Standard, Communication port Pin details,
- Communication Port Configuration: Baud Rate, data bits, stop bit, parity
- Polling Constraints: Minimum time period between two consecutive poll cycles.
- Protocol Details: Protocol name and message structure for different read / write functions.
- Function codes for reading digital and Analog inputs
- Function code for writing analog values in the IED registers
- List of parameters available in the IED which can be accessed from the IED through serial port.
- Register address of each parameter in the IED.
- A Sample of the details is given below for understanding:

PROTOCOL DETAILS:

Sr. No.	Description	Contents/ Details
1.	Protocol MODBUS RTU	MODBUS RTU
2.	Connection Type	FCC -68 RJ45



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3.	Communication standard	RS485
4.	Signals supplied	TX, Rx, GND
5.	Baud Rate(Speed)	300, 600,1200,4800,9600,19200
6.	Format	Software configurable
7.	Port Configuration	Speed : 19,200 bps Data Bits : 8 Bits Stop Bit : 1 Bit Parity : None
8.	Min. Time period between consecutive Query	2 Sec.

Function Code & Message Structure:

- Function Code: 3 - Read Output Register
- Poll Format:
- Address 1 Byte
- Function 1 Byte
- Start Item 2 Byte
- No. Item 2 Byte

Response Format

- Address 1 Byte
- Function 1 Byte
- Length 1 Byte
- Date Item 1: 4 Byte
- Date Item n: 4 Byte

Function Code & Message Structure:

- Function Code: 2 - Read Input Status

Poll Format:

- Address 1 Byte
- Function 1 Byte
- Start Item 2 Byte
- No. Item 2 Byte
- Response Format:
- Address 1 Byte
- Function 1 Byte
- Length 1 Byte
- Date Item 1 2 Byte
- Date Item n 2 Byte

Memory Mapping, Data Type & Parameter Information:

A. Digital parameters :-

Sr. No.	Description	Address
1	Flow computer malfunction Alarm	8247
2	Mass Flow Low alarm	8245
3	Mass Flow High alarm	8246



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B. Analog parameters :-

Parameter	Register details			
	Engineering Range and Unit	GC Register	Register Format	
Density	Kg/m3 (500-600)	8655-56	Float	
Pressure		8657-58	Float	
Temperature		8659-60	Float	
Mass Flow Rate		8661-62	Float	
Totalized mass flow		8663-64	Float	
Yesterday's Total Mass flow		8665-66	Float	
Today's Total Mass Flow		8667-68	Float	
Corrected volumetric Flow rate		8669-70	Float	
Yesterday's Total corrected Volumetric Flow		8671-72	Float	
Totalized Mass Flow at 6 AM (Snapshot of cumulative)		8673-74	Float	



ENERGISING QUALITY

**PROJECT NUMBER:
VCS-HPCL-C221086**



VENDOR DRAWING & DATA REQUIREMENTS - 400 SCMh BOOSTER COMPRESSOR

DOCUMENT NO

C221086

RJ

ME

VDDR

3040

SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC
BOOSTER COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW
CAPACITY OF 400 SCMh IN GEOGRAPHICAL AREAS OF
RAJASTHAN.

C1	07.04.2026	ISSUED FOR CLIENT REVIEW	AB	SA	AB
REV	DATE	DESCRIPTION	PREP	CHK	APPR



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S. No.	DESCRIPTION	To be submitted with Bid	To be submitted for Approval		To be submitted for Shipment		Submit as certified Final / As Built	
			Required	Days after PO	Required	Days after PO	Required	Days after PO
A.	GENERAL							
1.	PROJECT SCHEDULE	✓	✓					
2.	DULY FILLED-IN "CHECKLIST FOR COMPLETENESS OF BID"	✓						
3.	DULY FILLED-IN "CHECKLIST FOR SCOPE OF SUPPLY"	✓						
4.	DEVIATION LIST (IF ANY) TO THE APPLICABLE SPEC., DATASHEETS	✓						
5.	UTILITIES REQUIREMENT SUMMARY	✓	✓				✓	
6.	FLANGE DETAILS OF PIPING CONNECTION WITH CONNECTION AT BATTERY LIMIT	✓	✓				✓	
7.	GENERAL ARRANGMENT DRAWING OF LCV FILL POST	✓						
8.	DULY FILLED IN EXPERIENCE RECORD PROFORMA	✓						
9.	GUARANTEE PARAMETERS AS SPECIFIED	✓	✓				✓	
10.	TENTATIVE LOAD DATA FOR FOUNDATION DESIGN	✓	✓				✓	
11.	STRUCTURAL STABILITY CERTIFICATES AS PER WL CAPACITY & FEA ANALYSIS REPORT	✓	✓				✓	
12.	LIST OF SUB-VENDORS FOR ALL BOUGHT-OUT ITEMS INCLUDING ELECTRICAL & INSTRUMENTATION ITEMS		✓				✓	
13.	LEAFLET, CATALOGUES FOR ALL ITEMS	✓	✓				✓	
14.	O & M MANUAL	✓	✓		✓		✓	
B.	COMPRESSOR							
1	DATASHEETS FOR THE FOLLOWING							
A	COMPRESSOR	✓	✓				✓	
B	HEAT EXCHANGERS		✓				✓	
C	PRESSURE VESSELS (As per ASME SEC.VIII Div. I		✓		✓		✓	



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S. No.	DESCRIPTION	To be submitted with Bid	To be submitted for Approval		To be submitted for Shipment		Submit as certified Final / As Built	
			Require d	Days after PO	Require d	Days after PO	Require d	Days after PO
D	ELECTRIC MOTOR		✓				✓	
E	GAS DETECTOR		✓				✓	
F	FLAME DETECTOR		✓				✓	
G	RADIATOR COOLER (GAS COOLER)		✓				✓	
2	CATALOGUE FOR COMPRESSOR	✓	✓					
3	OFFERED BOUGHT OUT ITEMS LIST	✓	✓					
3	TYPICAL CROSS-SECTIONAL DRAWING AND LITERATURE TO FULLY DESCRIBE THE DETAILS OF OFFERING							
A	COMPRESSOR	✓	✓					
B	SUCTION VALVE	✓	✓					
C	DISCHARGE VALVE	✓	✓					
D	PISTON ROD GLAND PACKING & PISTON RINGS	✓	✓					
E	LUBE OIL PUMP	✓	✓					
4	V-BELT & PULLEY WITH SELECTIONCHART & CALCULATION		✓		✓		✓	
5	COOLER DATA / DRG WITH THERMAL& MECH DESIGN CALCULATION		✓		✓		✓	
6	DESIGN CALCULATION, GA DRGS FORPULSATION DAMPNER		✓		✓		✓	
7	PIPING & INSTRUMENTATION DIAGRAMS FOR THE FOLLOWING							
A	PROCESS GAS		✓				✓	
B	LUBE OIL		✓				✓	
C	COOLING WATER		✓				✓	

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8	TORQUE ANGLE DIAGRAM, PISTONROD LOAD VS CRANK ANGLE		✓				✓	
9	TORQUE SPEED CHARACTERISTICS		✓				✓	
10	ACOUSTIC / MECHANICAL EVALUATION REPORT		✓				✓	
11	ITEMIZED PRICE LIST OF ESSENTIALSPARES		✓				✓	
12	ITEM LIST OF SPARES WITH PRICEFOR 2 YEARS	✓			✓		✓	
13.	DRG. FOR TESTING ARRANGEMENT & TEST PROCEDURE TO BE ADOPTED						✓	
14.	CERTIFICATE FOR FOLLOWING:							
A	HYDRAULIC TESTING				✓		✓	
B	NON-DESTRUCTIVE TESTING				✓		✓	
C	MATERIAL COMPOSITION & PHYSICALPROPERTIES				✓		✓	
D	LEAK PROOFNESS TEST OF FRAME				✓		✓	
E	LUBE PUMP, FRAME OIL PUMP, HYDOIL PUMP				✓		✓	
15	DESIGN / ACTUAL ASSEMBLYCLEARANCE CHART				✓		✓	
16	TEST RECORDS OF FOLLOWING				✓		✓	
A	MECHANICAL RUNNING				✓		✓	
B	PERFORMANCE TEST / PACKAGE TEST				✓		✓	
C	NOISE LEVEL TEST				✓		✓	
17	LIST OF SPECIAL TOOLS & TACKLES FOR INSTALLATION & MAINTENANCE				✓		✓	
C	ELECTRIC MOTOR							
1	MOTOR DATA SHEET		✓				✓	
2	TECHNICAL LITERATURE / CATALOGUE, SELECTION CHARTS,NOMOGRAPHS ETC.		✓				✓	
3	GA DRAWING		✓				✓	



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4	TERMINAL BOX ARRANGEMENTDRAWING		✓				✓	
5	MOTOR CHARACTERISTIC CURVES		✓				✓	
6	TORQUE SPEED CURVES		✓				✓	
7	CURRENT TIME CURVES		✓				✓	
8	P.F AND EFFICIENCY		✓				✓	
9	TYPE TEST CERTIFICATES						✓	
10	CERTIFICATE FROM THE RELEVANT STATUTORY AUTHORITY (BASED ON THE COUNTRY OF MANUFACTURE) FOR SUITABILITY OF THE OFFERED MOTOR FOR INSTALLATION IN THE SPECIFIED AREA CLASSIFICATION						✓	
11	PRE-COMMISSIONING AND COMMISSIONING PROCEDURE		✓				✓	
D	INSTRUMENTATION							
1	G.A. OF INSTRUMENT PANEL WITH BILL OF MATERIAL & WIRING DIG.FOR LCP		✓				✓	
2	INSTRUMENT DATASHEET		✓				✓	
3	LOGIC DIAGRAM / LADDER DIAGRAM / FUNCTIONAL DIAGRAM		✓				✓	
4	INTERLOCKED P&ID (ALL HYRAULIC, GAS, AIR, WATER, CO2 SYSTEM) FOR HAZOP STUDY		✓				✓	
5	INTERLOCKED P&ID WITH INSTRUMENTATION TAG NO.		✓				✓	
4	LOOP SCHEMATIC							
5	INTER CONNECTING DIAGRAM		✓				✓	
6	OPERATING / CONTROL WRITE UP		✓				✓	
7	ALARM / SHUT DOWN LIST		✓				✓	
8	WIRING DIAGRAM / INTERCONNECTING PIPING		✓				✓	
9	START UP AND SHUT DOWN WRITEUP		✓				✓	



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10	START UP AND SHUT DOWN INLETLOCK DIAGRAM		✓				✓	
11	ALARM AND SHUTDOWN LIST WITHSET POINT		✓				✓	
12	LOAD CONTROL PANEL LAYOUT		✓				✓	
13	TERMINATION DIAGRAM, PANELWIRING DETAIL		✓				✓	
14	LOOP SCHEMATIC		✓				✓	
15.	INTER CONNECTING DIAGRAM		✓				✓	
16.	CABLE SCHEMATIC		✓				✓	
17.	BILL OF MATERIAL		✓				✓	
18.	TEST / INSPECTION CERTIFICATE		✓				✓	
19.	LIST OF RELIEF VALVES WITH SETTINGS		✓				✓	
20.	CERTIFICATE FROM THE RELEVANT STATUTORY AUTHORITY (BASED ON THE COUNTRY OF MANUFACTURE) FOR SUITABILITY OF THE OFFERED INSTRUMENTS FOR INSTALLATION IN THE SPECIFIED AREA CLASSIFICATION		✓				✓	
21.	OPERATION & CONTROL PHILOSOPHY		✓				✓	

	CHECKLIST - TECHNICAL	VCS-SD-CK-001
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**SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC BOOSTER
COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW CAPACITY OF 400 SCM/H IN
GEOGRAPHICAL AREAS OF RAJASTHAN.**

CHECKLIST – TECHNICAL

Bidder confirms following, as a minimum, has been enclosed in the offer.

S.NO.	Requirements	Compiled by Bidder(Tick)
1	Reference List of previous supply of Procured item	
2	Filled – up Data Sheets, duly signed and stamped by bidder enclosed.	
3	List of recommended commissioning spares and accessories for Procured item.	
4	List of recommended spares and accessories for two year normal operation for procured item.	
5	Compliance statement duly filled and stamped enclosed.	
6	GA & assembly drawings, cross section drawings including part list & material list enclosed.	
7	Other technical details & vendor's product catalogues enclosed.	

C1	07.04.2026	ISSUED FOR CLIENT REVIEW	AB		AB
REV	DATE	DESCRIPTION	PREP	CHK	APPR

	COMPLIANCE STATEMENT	VCS-SD-CS-001
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 ENERGISING QUALITY	VENDOR DATA & DRAWING REQUIREMENTS - 400 SCM/H BOOSTER COMPRESSOR	Documents No.	
		C221086-RJ-ME-VDDR-3040	
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**SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC BOOSTER
COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW CAPACITY OF 400 SCM/H IN
GEOGRAPHICAL AREAS OF RAJASTHAN.**

COMPLIANCE STATEMENT

S.No	Requirement	Bidder's Confirmation
1	Bidder confirms that all materials proposed by the bidder are same/ superior to those specified in specification/ data sheets enclosed.	
2	Bidder confirms that the offer is in total compliance with the technical requirements of the Material Requisition. Bidder confirms that deviation expressed or implied anywhere else in the offer shall not be considered valid.	
3	Bidder confirms that all spares and accessories required for two years of normal operation have been quoted separately.	
4	Bidder confirms that prices for start-up/commissioning spares and accessories have been included in the quoted items.	
5	Bidder confirms that in the event of securing order for the requisitioned item(s), good for manufacturing drawings of ordered item(s) shall have complete details with dimensions, part list and material list including back-up calculations in the first submission, failing which the vendor shall be solely responsible for any likely delay in delivery of item(s).	

Bidder's Signature with Stamp

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	VENDOR DATA & DRAWING REQUIREMENTS - 400 SCM/H BOOSTER COMPRESSOR	Documents No.	
		C221086-RJ-ME-VDDR-3040	
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**SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC BOOSTER
COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW CAPACITY OF 400 SCM/H IN
GEOGRAPHICAL AREAS OF RAJASTHAN.**

	DEVIATION SHEET	VCS-SD-DS-001
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**DEVIATION/ EXCEPTION/
CLARIFICATION SHEET**

Sr. No.	Contractor's Inquiry Reference	Contractor's Requirement	Proposed Deviation by Supplier, with Technical Justification	Cost Impact, if any	Contractor's Conclusions

NOTES

- 1- Bidder confirms that apart of from the deviations/exceptions/clarifications listed above, the bid is in full compliance with Inquiry requisition.
- 2- Bidder shall submit this sheet duly filled up and signed by him along with his bid. In case there is no deviation, then also supplier shall submit this sheet along with his bid indicating NIL deviation.

(Contractor's Name and Signature with Seal)

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**SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC BOOSTER
COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW CAPACITY OF 400 SCMH IN
GEOGRAPHICAL AREAS OF RAJASTHAN.**



DRAWINGS & DOCUMENTS

VCS-SD-DD-001

**INFORMATION/ DOCUMENTS / DRAWINGS TO BE SUBMITTED BY SUCCESSFUL
BIDDER**

Successful Bidder shall submit four copies unless noted otherwise, each of the following:

1. Inspection & test reports for all mandatory tests as per the applicable code as well as test reports for any supplementary tests, in nicely bound volumes.
2. Filled in Quality Assurance Plan (QAP) for Purchaser's/ Consultant's approval. These QAPs shall be submitted in two copies within 15 days from LOI/ FOI.
3. Detailed completion schedule activity wise (Bar Chart), within one week of placement of order.

Note : All drawings, instructions, catalogues, etc., shall be in English language and all dimensions shall be metric uni

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 ENERGISING QUALITY	VENDOR DATA & DRAWING REQUIREMENTS - 400 SCMH BOOSTER COMPRESSOR	Documents No.	
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**SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC BOOSTER
COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW CAPACITY OF 400 SCMH IN
GEOGRAPHICAL AREAS OF RAJASTHAN.**



INSTRUCTION TO BIDDER

VCS-SD-ITB-001

INSTRUCTION TO BIDDERS

1. Bidder to note that no correspondence shall be entered into or entertained after the bid submission.
2. Bidder shall furnish quotation only in case he can supply material strictly as per this Material Requisition and specification/data sheet forming part of Material Requisition.
3. If the offer contains any technical deviations or clarifications or stipulates any technical specifications (even if in line with MR requirements) and does not include complete scope & technical / performance data required to be submitted with the offer, the offer shall be liable for rejection.
4. Bidder must submit all documents as listed in checklist with his offer.
5. Supplier must note that stage wise inspection for complete fabrication, testing including the raw material inspected to be carried out.
6. Vendors for bought out items to be restricted to the approved vendor list attached with bid document. Approval of additional vendor if required, for all critical bought out items shall be obtained by the supplier from the purchaser before placement of order. Credentials/PTR of the additional vendor proposed to be submitted by supplier for review and approval of Purchaser/ Purchaser's representative

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**VENDOR DATA & DRAWING REQUIREMENTS -
400 SCMH BOOSTER COMPRESSOR**

Documents No.

C221086-RJ-ME-VDDR-3040

REV-C1

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**SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC BOOSTER
COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW CAPACITY OF 400 SCMH IN
GEOGRAPHICAL AREAS OF RAJASTHAN.**

		LIST OF SPARES		VCS-SD-LS-001	
LIST OF SPARES					
Sl. No.	Part No.	Description	Quantity (Minimum)		
1.					
2.					
3.					
4.					
5.					
6.					
7.					
8.					
9.					
10.					
C1	07.04.2026	ISSUED FOR CLIENT REVIEW	AB		AB
REV	DATE	DESCRIPTION	PREP	CHK	APPR



**SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC BOOSTER
COMPRESSOR PACKAGE OF MIN. AVERAGE FLOW CAPACITY OF 400 SCMH IN
GEOGRAPHICAL AREAS OF RAJASTHAN.**

		REFERENCE LIST				VCS-SD-RL-001							
REFERENCE LIST		Service									Bidders Signature with stamp		
		Size and Rating / thk.											
		Email											
		Client , Address and Contact No.											
		Year of Supply											
		Project											
		SI No.											
C1	07.04.2026	ISSUED FOR CLIENT REVIEW				AB		AB					
REV	DATE	DESCRIPTION				PREP	CHK	APPR					

 Energising Quality		PROJECT NUMBER: VCS-HPCL-C221086					
QUALITY ASSURANCE PLAN BOOSTER COMPRESSOR					TOTAL SHEET NO.		9
DOCUMENT NO.		C221086	RJ	ME	QAP	3040	
<u>SUPPLY OF ELECTRIC MOTOR DRIVEN CNG HYDRAULIC BOOSTER COMPRESSOR PACKAGE OF</u> <u>MIN. AVERAGE FLOW CAPACITY OF 400 SCMH IN GEOGRAPHICAL AREAS OF RAJASTHAN</u>							
C1	07.04.2026	ISSUED FOR CLIENT REVIEW			AB	SA	AB
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S.NO.	OPERATION / PARAMETER	CHARACTERISICS / PARAMETERS	ACCEPTANCE CRITERIA & CERTIFICATION	VENDO R	TPIA	CLIENT/ PMC	REMARKS
Bought Out items & Equipment details							
1.	Canopy & Stationary Frame Dimensional Report	Dimension Reports	As per Technical specification & GAD	P	W/R	R	
2.	Gas Intercooler	Dimensional, Chemical, Mechanical, Hydro test, Physical, Heat Chart	As per Technical specification/Data sheet/Calculation Reports	P	W/R	R	
2.	Water/Air cooled radiator	Dimensional, Chemical, Mechanical, Hydro test, Physical, Heat Chart	As per Technical specification/Data sheet/Calculation Reports	P	W/R	R	
3.	Oil Heat Exchanger	Dimensional, Chemical, Mechanical, Hydro test, Physical, Heat Chart	As per Technical specification/Data sheet/Calculation Reports	P	W/R	R	
4.	Hydraulic Cylinder/Intensifier	Dimensional, Chemical, Mechanical, Hydro test, Physical, Heat Chart	As per Technical specification/Data sheet/Calculation Reports	P	W/R	R	
5.	Hydraulic Pump	Test/compliance/performance/ warranty certificates	As per OEM specifications/Data sheet	P	W/R	R	
6.	Pressure Regulating Valve/Direction & flow control valve	Test/compliance/performance/ warranty certificates	As per OEM specifications/Data sheet	P	W/R	R	
7.	Safety Relief Valve	Test/compliance/performance/ warranty certificates	As per OEM specifications/Data sheet/Sizing calculation report	P	W/R	R	
8.	Temperature Gauge & Thermo well	Test/compliance/performance/warr anty certificates / calibration report	As per Technical specification/Data sheet/wake frequency Calculation Reports	P	W/R	R	
9.	Pressure gauge	Visual, Size verification, Fitment & alignment, Functional & operational, Calibration report, Test certificates for bought out items, Hydro test, pressure test	Technical specification & Approved data sheets	P	W/R	R	
10.	Differential pressure gauge	Visual, Size verification, Fitment & alignment, Functional & operational, Calibration report, Test certificates for bought out items, Hydro test, pressure test	Technical specification & Approved data sheets	P	W/R	R	
11.	Pressure Transducer &	Location as per P&ID, Fitment & alignment, calibration report, Test					

	Transmitter	certificates of bought out items, pressure test, Operational & functional, current consumption w.r.t pressure, IP ratings, supply voltage, Output signal voltage	Technical specification & Approved data sheets	P	W/R	R	
12.	Temperature Transducer & Transmitter	Location as per P&ID, Fitment & alignment, calibration report, Test certificates of bought out items, pressure test, Operational & functional, current consumption w.r.t pressure, IP ratings, supply voltage, Output signal voltage	Technical specification & Approved data sheets	P	W/R	R	
13.	Water pump	Performance report, Fitment & alignment , Test certificates, Compliance report , chemical & physical reports	Technical specification & Approved data sheets	P	W/R	R	
14.	SS Needle / Block & Bleed valve	Chemical, Mechanical, Hydro test, Physical, Heat Chart	As per Technical specification/Data sheet	P	W/R	R	
15.	Check Valve	Chemical, Mechanical, Hydro test, Physical, Heat chart	As per Technical specification/Data sheet/ calculation report	P	W/R	R	
16.	SS Tube	Chemical, Mechanical, Hydro test, Physical, Heat Chart	As per technical specifications/Data sheet	P	W/R	R	
17.	CS tube	Chemical, Mechanical, Hydro test, Physical, Heat Chart	As per OEM specifications/Data sheet	P	W/R	R	
18.	Isolation Ball valve	Visual, Dimensional, Fitment & Alignment, Physical Test, Chemical Test, Operational & functional, Leak test, Test certificates of bought out items, Cv verification w.r.t data sheet	Technical specification & Approved data sheets	P	W/R	R	
19.	SS Tubing & Fittings	Visual, Dimensional, Fitment & Alignment as per P&ID shown, Physical Test, Chemical Test, Operational & functional, Leak test, Test certificates of bought out items	Technical specification & Approved data sheets	P	W/R	R	
20.	SS Manifold	Visual, sizing verifications, Fitment & Alignment, Operational & functional, Leak test, Test certificates of bought out items, Flow direction as per P&ID	As per OEM specification & design	P	W/R	R	
21.	Inlet & Outlet filter	Visual, Dimensional, Fitment & Alignment, Physical Test, Chemical Test report, Operational & functional, Leak test, Hydro test,	Technical specification / Approved data sheets/ Filter Sizing calculation sheet	P	W/R	R	

		filter sizing calculation					
22.	Filter Elements	Visual, Dimensional, Fitment & Alignment, Test certificates of bought out items Operational & functional, Leak test, Hydro test, filter sizing & pressure drop calculation with elements	Technical specification / Approved data sheets/ Filter Sizing calculation sheet	P	W/R	R	
23.	Pneumatic Actuators	Performance report, Fitment & alignment , Test certificates, Compliance report , chemical & physical reports	As per OEM specifications/Data sheet	P	W/R	R	
24.	Air compressor Block & vessels	Performance report, Fitment & alignment , Test certificates, Compliance report , chemical & physical reports	As per OEM specifications/Data sheet	P	W/R	R	
25.	Pressure switch	Performance report, Fitment & alignment , Test certificates, calibration, Compliance report , chemical & physical reports	As per OEM specifications/Data sheet	P	W/R	R	
26.	ESD & Hooter	Fitment & alignment , Test certificates, Compliance report , chemical & physical reports	As per OEM specifications/Data sheet	P	W/R	R	
27.	Solenoid Valve & manifold	Performance report, Fitment & alignment , Test certificates, Compliance report , chemical & physical reports	As per OEM specifications/Data sheet	P	W/R	R	
28.	Main Motor & Auxiliaries Motor	Routine Test, Functional Test, Compliance report , Alignments test	Technical specification / Approved data sheets	P	W/R	R	
29.	Flameproof Junction Box	Test report, compliance report, Physical & chemical report	As per OEM specifications/Data sheet	P	W/R	R	
30.	Mass Flow meter	Performance report, Fitment & alignment , Test certificates, Compliance report , calibration report, chemical & physical reports	Technical specification / Approved data sheets	P	W/R	R	
31.	Gas Detector	Performance report, Fitment & alignment , Test certificates, Compliance report , calibration report, chemical & physical reports	Technical specification / Approved data sheets	P	W/R	R	

32.	Flame Detector	Performance report, Fitment & alignment , Test certificates, Compliance report , calibration report, chemical & physical reports	Technical specification / Approved data sheets	P	W/R	R	
33.	Control panel	Test report, compliance report, Routine test ,Physical & chemical report	As per OEM specifications/Data sheet	P	W/R	R	
34.	Tube light & Aviation lamp	Test report, compliance report, Routine test ,Physical & chemical report	As per OEM specifications/Data sheet	P	W/R	R	
35.	PLC/HMI	Test report, compliance report, Routine test ,Physical & chemical report	As per OEM specifications/Data sheet	P	W/R	R	
36.	FRLS cable	Test report, compliance report, Routine test ,HV-IR test, Physical & chemical report	As per OEM specifications/Data sheet	P	W/R	R	
37.	CO2 cylinder	Hydro stretch test, leak test, Physical & chemical report, Filling permission(PESO) report	Technical specification / Approved data sheets	P	W/R	R	
38.	Air Dryer	Test report, compliance report, Routine test ,Physical & chemical report	As per OEM specifications/Data sheet	P	W/R	R	
39.	Flame arrester	Test report, compliance report, Routine test ,Physical & chemical report	As per OEM specifications/Data sheet	P	W/R	R	
40.	Package CCOE approval certificates	Valid PESO package approval certificates, Number plate embossing with PESO no. & other details.	As per Technical specification & approved Name plate	P	W/R	R	
41.	All electrical PESO certificates	Valid PESO certificates of all electrical & hazardous components	As per PESO certificates	P	W/R	R	
42.	Fire & flame retardant PU foam	Test certificates	As per PESO & PNGRB guidelines	P	W/R	R	
43.	LCV Fill Post Hoses	Test certificates/Hydro test	As per OEM specifications/Data sheet	P	W/R	R	
44.	LCV fill post	Dimensional check / Pneumatic Leak test	As per OEM specifications/Data sheet	P	W/R	R	

Complete Assembly Simulation & Testing Details							
43.	Complete Assembly @ pulsating pressure Leak Test	Leak Test Throughout on all Joints By using Soap solution and Holding Time up to 4hrs @ Pressure 200~250 Bar(g) Media-Nitrogen	Technical specifications & OEM test procedure.	P	W	W/R	
44.	Complete Assembly @ static pressure Leak Test	Leak Test Throughout on all Joints By using Soap solution and Holding Time up to 4hrs @ Pressure 200~250 Bar(g) Media-Nitrogen	Technical specifications & OEM test procedure.	P	W	W/R	
45	Main Motor overload relay simulation	Tripping test during routine test	As per Approved control philosophy	P	W	W/R	
46	All radiator fan & water pump overload relay simulation	Tripping test during routine test	As per Approved control philosophy	P	W	W/R	
47	Air compressor overload relay simulation	Tripping test during routine test	As per Approved control philosophy	P	W	W/R	
48	Air compressor pressure failure test	Tripping test during routine test	As per Approved control philosophy	P	W	W/R	
49	Over & under voltage simulation test	Simulation check	As per Approved control philosophy	P	W	W/R	
50	Earth leakage failure	Simulation check	As per Approved control philosophy	P	W	W/R	
51	Failure of any wire break	Crimping connection check	As per Approved control philosophy	P	W	W/R	
52	Failure Barriers	Simulation check	As per Approved control philosophy	P	W	W/R	
53	Low & High oil level alarms check	Simulation check	As per Approved control philosophy	P	W	W/R	
54	Low & High oil temperature alarm check	Simulation check	As per Approved control philosophy	P	W	W/R	
55	Low & High Water temperature alarm check	Simulation check	As per Approved control philosophy	P	W	W/R	
56	Low & High 1 st stage Discharge Gas temperature alarm check	Simulation check	As per Approved control philosophy	P	W	W/R	
57	Low & High 2 nd stage Discharge Gas temperature alarm check	Simulation check	As per Approved control philosophy	P	W	W/R	
58	Low & High suction pressure simulation test	Simulation check	As per Approved control philosophy	P	W	W/R	
59	1 st stage Low & high pressure fault	Simulation check	As per Approved control philosophy	P	W	W/R	

60	2 nd stage Low & high pressure fault	Simulation check	As per Approved control philosophy	P	W	W/R	
61	Low bank cascade Low & high pressure fault	Simulation check	As per Approved control philosophy	P	W	W/R	
62	Med bank cascade Low & high pressure fault	Simulation check	As per Approved control philosophy	P	W	W/R	
63	High bank cascade Low & high pressure fault	Simulation check	As per Approved control philosophy	P	W	W/R	
64	Low & High Emergency pressure fault	Simulation check	As per Approved control philosophy	P	W	W/R	
65	Low & High water pressure fault	Simulation check	As per Approved control philosophy	P	W	W/R	
66	3 phase Relay failure	Simulation check	As per Approved control philosophy	P	W	W/R	
67	Compressor auto start simulation test	Simulation check	As per Approved control philosophy	P	W	W/R	
68	Compressor auto stop simulation test	Simulation check	As per Approved control philosophy	P	W	W/R	
69	Compressor auto start/stop bank simulation	Simulation check	As per Approved control philosophy	P	W	W/R	
70	Gas Detector-1 & 2 simulation test	Simulation check	As per Approved control philosophy	P	W	W/R	
71	Flame Detector-1 & 2 simulation test	Simulation check	As per Approved control philosophy	P	W	W/R	
72	Co2 flooding system alarm simulation test	Simulation check	As per Approved control philosophy	P	W	W/R	
73	Co2 weight loss Monitoring test	Simulation check	As per Approved control philosophy	P	W	W/R	
74	Tube light functional test	Functional test	As per Approved control philosophy	P	W	W/R	
75	Aviation light functional test	Functional test	As per Approved control philosophy	P	W	W/R	
76	Door limit & selector switch functional test	Functional test	As per Approved control philosophy	P	W	W/R	
77	All ESD functional test	Functional test	As per Approved control philosophy	P	W	W/R	
78	Failure of Mass flow meter meeting	Functional test	As per Approved control philosophy	P	W	W/R	
79	Auto change over from primary CPU to secondary By failure due to any reason	Redundant CPU auto changeover test by manual failure	As per Approved control philosophy	P	W	W/R	
80	Canopy structure Painting inspection at works, surface preparation to be	Functional test	As OEM procedure	P	W	W/R	

	inspected after cleaning and before application of first coat of primer						
81	No load Mechanical run test (NLMRT) of the compressor with rated speed and shop driver (4 hours minimum)	Functional test	As OEM procedure	P	W	W/R	
COMPRESSOR PERFORMANCE TEST							
82	Delivery test (200~240 kgcm2) Media Nitrogen / NG @ suction-30 ~110 kg/cm2	Performance test (4 Hrs.)	As technical specification & OEM procedure	P	W	W/R	
83	Noise test	Performance test	As technical specification & OEM factory procedure	P	W	W/R	
84	Vibration test as per standard at site	Performance test	As technical specification & OEM factory procedure	P	W	W/R	
85	Overall power consumption (KWH)	Performance test	As technical specification	P	W	W/R	
86	Input power @ full rated load	Performance test	As technical specification & OEM procedure	P	W	W/R	
87	Three phase power in (KW)	Performance test	As technical specification & OEM procedure	P	W	W/R	
88	Compressor Load %	Performance test	As technical specification & OEM procedure	P	W	W/R	
89	Specific power consumption of gas in KW/KG	Performance test	As technical specification & OEM procedure	P	W	W/R	

Notes:

1. The above testing and acceptance criteria are minimum requirements; however, manufacturer shall ensure that the product shall also comply to the additional requirements as per particular Technical Specification (PTS) and Data Sheet.
2. The supplier shall submit their own detailed QAP prepared on the basis of above / Technical specification for approval of Owner/Owner's representative.
3. Supplier shall submit calibration certificates of all instruments/Equipment to be used for inspection and Testing to TPIA with relevant procedures and updated standards for TPIA review/Approval. All reference codes / documents shall be arranged by vendor for reference of TPIA at the time of inspection.
4. Owner / Owner's representative include TPIA will have the right to inspect activity of manufacturing at any time.
5. TPIA along with Owner / Owner's representative shall review/approval all the documents related to QAP/Quality manuals/Drawings etc., submitted by supplier.
6. Contractor shall in coordination with supplier/sub vendor shall issue detailed production and inspection schedule indicating the dates and the location of facilities Owner/Owner's representative and TPIA to organize inspection.
7. Special manufacturing procedure have to be specially approved or only previously approved procedures have to be used, in case of conflict between specification more stringent condition shall be applicable.
8. The supplier shall submit separate control logic sheet along with actuator/SOV operation philosophy.



LIST OF RECOMMENDED VENDORS FOR BOUGHT OUT ITEMS

**TOTAL
SHEETS**

61

DOCUMENT NO

VCS-00-00-VL-0001

LIST OF RECOMMENDED VENDORS FOR BOUGHT OUT ITEMS

16

31.12.2025

Issued as Standard

Ramveer Singh

Safdar Ali

Rachna Shukla

REV

DATE

DESCRIPTION

PREP

CHK

APPR

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A. MECHANICAL & PIPELINE

1.0 CARBON STEEL PIPES

1.1 PIPE CARBON STEEL TO INDIAN STANDARDS

- a. A.S.T. Pipes Pvt. Ltd. (AST Group)
- b. Advance Steel Tube Ltd.
- c. Apl Apollo Tubes Ltd. (Er. Bihar Tubes Ltd.
- d. Asian Mills Pvt. Ltd.
- e. Asrani Tubes Limited
- f. Dadu Pipes (P) Ltd.
- g. Essar Steel Limited(Er Hazira Pipes Mill)
- h. Gaurang Products Pvt Ltd. (Ast Group)
- i. Goodluck Steel Tubes Ltd.
- j. Hi-Tech Pipes Limited
- k. Indus Tube Limited
- l. Jindal Industries Ltd
- m. Jindal Pipes Ltd.
- n. Jindal Saw Ltd (Kosi Works)
- o. Jotindra Steel & Tube Ltd
- p. Lalit Pipes And Pipes Ltd.
- q. Maharashtra Seamless Ltd.
- r. Man Industries (India) Ltd. – Pithampur
- s. Man Industries (India) Ltd. Anjar
- t. Mukat Tanks & Vessels Ltd.
- u. Nezone Tubes Limited
- v. North Eastern Tubes Limited
- w. Pratibha Industries Limited
- x. Pratibha Pipes & Structural Ltd.



- y. Psl Ltd (Chennai)
- z. Psl Ltd (V1, V2 & NC)
- aa. Rama Steel Tubes Ltd.
- bb. Ratnamani Metals And Tubes Ltd.
- cc. Ravindra Tubes Limited
- dd. Samshi Pipe Industries Limited
- ee. Surya Roshni Ltd.
- ff. Swastik Pipes Ltd.
- gg. Utkarsh Tubes & Pipes Ltd. (Formly Bmw)
- hh. Welspun Corp. Limited (Dahej)
- ii. Zenith Birla (India) Limited

1.2 PIPE & TUBULARS TO A.P.I. STANDARDS

- a. Arcelormittal Tubular Products Roman Sa, Romania
- b. Bhel (Trichy), India
- c. Dalmine Spa (Enquiry To Tenaris), Uae
- d. Eewkorea Co. Ltd (Germany), Korea
- e. Eew Korea Co. Ltd. (Korea), Korea
- f. Eisenbau Kramer Gmbh, Germany
- g. Hyundai Rb Co. Ltd. South Korea
- h. Ilva Lamiere E Tubi Srl (Enq To Ilva Spa, Italy)
- i. Inox Tech. Spa, Italy
- j. ISMT Ltd. Ahmednagr, India
- k. TATA Steel, India
- l. PSL
- m. Jindal Pipes Ltd., India
- n. Jindal Saw Ltd. (Kosi Works), India
- o. Jindal Saw Ltd. (Nashik Works), India
- p. Lalit Pipes And Pipes Ltd. India
- q. Maharashtra Seamless Ltd., India
- r. Man Industries (I) Ltd. (Pithampur), India
- s. Mukat Tanks & Vessels Ltd., India
- t. Pratibha Industries Limited, India
- u. Ratnamani Metals And Tubes Ltd., India
- v. Siderca S.A.I.C (Enquiry Totenaris), Uae



- w. Sumitomo Metal Ind. Ltd., India
- x. Surya Roshni Ltd., India
- y. Swastik Pipes Ltd, India
- z. Tata Steel Uk Limited (Formerly C702)
- aa. Tubos De Acero De Mexico Sa (Enq. Tenaris), Uae
- bb. Tubos Reunidos Sa Spain
- cc. Umran Steel Pipe Inc (Turkey), Turkey
- dd. Valcovny Trub Chomutov, Czech Republic
- ee. Vallourec And Mannesmann Tubes, France
- ff. Welspun Corp Limited (Dahej), India

1.3 PIPE/TUBE CS (SEAMLESS) TO ASTM STANDARDS

- a. Arcelormittal Tubular Products Roman Sa, Romania
- b. Bhel (Trichy), India
- c. Changshu Seamless Steel Tube Co. Ltd., China
- d. Dalmine Spa (Enquiry To Tenaris, Uae
- e. Heavy Metals & Tubes Limited (Mehsana), India
- f. Ismt Ltd. Ahmedngr, India
- g. Ismt Ltd. Baramati India
- h. Jfe Steel Corporation, Uae
- i. Jindal Sdaw Ltd (Nashik Works) India
- j. Klt Automotive And Tubular Products Ltd., India
- k. Mahalaxmi Seamless Limited, India
- l. Maharashtra Seamless Ltd, India
- m. Products Tubulares S.A.U, Spain
- n. Ratnadeep Metal Tubes Ltd., India
- o. Staineest Tubes Pvt Ltd., India
- p. Sumitomo Metal Ind. Ltd., India
- q. Tubos Reunidos Sa Spain
- r. Valcovny Trub Chomutov, Czech Republic
- s. Vallourec Andmannesmann Tubes France
- t. Yangzhou Chengde Steel Pipe Co. Ltd Dubai (UAE)
- u. Jindal SAW Ltd Kosi Kalan Mathura

1.4 PIPE CARBON STEEL (WELDED) TO ASTM STANDARDS

- a. Eew Korea Co. Ltd. (Germany), Korea



- b. Eew Korea Co. Ltd. (Korea), Korea
- c. Eisenbau Kramer Gmbh, Germany
- d. Hyundai Rb Co. Ltd., South Korea
- e. Inox Tech. Spa, Italy
- f. Jindal Saw Ltd (Kosi Works), India
- g. Lalit Pipes and Pipes Ltd., India
- h. Man Industries (I) Ltd.(Pithampur), India
- i. Man Industries (India) Ltd. Anjar, India
- j. Mukat Tanks & Vessels Ltd., India
- k. Ratnamani Metals And Tubes Ltd., India
- l. Sumitomo Metal India Ltd., India
- m. Tata Steel Uk Limited

2.0 VALVES

2.1 GLOBE VALVES

- a. BDK (New Delhi)
- b. Datre Corpn (Calcutta)
- c. KSB Pumps (New Delhi)
- d. L&T (New Delhi)
- e. Neco Schuber & Salzer Ltd. (New Delhi)
- f. Niton Valve (Mumbai)
- g. Ornate Valves (Mumbai)
- h. Panchavati Valves (Mumbai)
- i. AV Valves Ltd.
- j. BHEL (Trichy), India
- k. Econo Valves Pvt Ltd, India
- l. Fouress Engg (I) Ltd (Aurangabad)
- m. Guru Industrial Valves Pvt Ltd
- n. Leader Valves Ltd, India
- o. NSSL Ltd. (Neco Schubert & Salzerltd)
- p. Oswal Industries Ltd, India
- q. Petrochemical Engineering Enterprises, India
- r. Sakhi Engineers Pvt Ltd
- s. Shalimar Valves Pvt Ltd
- t. Steel Strong Valves India Pvt Ltd, India



- u. Petro Valves Pvt. Limited, Ahmedabad
- v. Hawa Engineers Limited, Ahmedabad (Non API)
- w. G M Engineering PVT LTD (Non API)

2.2 CHECK VALVES

- a. Advance Valves Pvt. Ltd., Noida
- b. Aksons & Mechanical Enterprises, Mumbai
- c. Larsen & Toubro Limited (Audco India Limited, Chennai)
- d. AV valves Ltd., Agra
- e. BDK engineering India Ltd., Hubli
- f. BHEL, OFE&OE Group, New Delhi
- g. Datre Coroportion Limited, Calcutta
- h. Leader Valves Ltd., Jalandhar
- i. Neco schubert & Salzer Ltd., New Delhi
- j. Niton Valves Industries (P) Ltd., Mumbai
- k. Precision Engg. Co., Mumbai
- l. Econo Valves Pvt Ltd, India
- m. Fouress Engg (I) Ltd (Aurangabad)
- n. KSB Pumps Ltd (Coimbatore), India
- o. NSSL Ltd. (Neco Schubert & Salzer Ltd)
- p. Oswal Industries Ltd, India
- q. Panchvati Valves & Flanges Pvt Ltd, India
- r. Petrochemical Engineering Enterprises, India
- s. Sakhi Engineers Pvt Ltd
- t. Shalimar Valves Pvt Ltd
- u. Steel Strong Valves India Pvt Ltd, India
- v. Shiv Shakti Engineering Works (Approved up to 06 inch)
- x. Hawa Engineers Limited, Ahmedabad
- y. G M Engineering PVT LTD (Non API)

2.3 PLUG VALVES

- a. Breda Energia Sesto Industria Spa, Italy
- b. Fisher Sanmar Ltd., Chennai
- c. Larsen & Toubro Ltd., New Delhi
- d. Nordstrom Valves, USA



- e. Serck Audco Valves, UK
- f. Sumitomo Corporation India Pvt. Ltd., New Delhi
- g. Z Corporation, Korea
- h. Hawa Valves (India) Pvt. Ltd., Mumbai
- i. Steel Strong Valves India Pvt. Ltd., Navi Mumbai
- j. Econo Valves
- k. Flow-Serve PTE (Mfr. SERCK), India

2.4 BALL VALVES

- a. Hawa Valves (India) Pvt. Ltd, Navi Mumbai
- b. Larsen & Toubro, Delhi
- c. Microfinish Valves Pvt. Ltd., Noida
- d. Oswal Industries Ltd., Gandhi nagar
- e. Virgo Engineers Ltd., Delhi
- f. Boteli Valve Group Co. Ltd., China
- g. Cameron (Malaysia) SDN BHD, Malaysia
- h. Dafram S.P.A., Italy
- i. Fangyuan Valve Group Co. Ltd., China
- j. Franz Schuck GmbH, Germany
- k. O.M.S. Saleri (Italy)
- l. Pibi Viesse S.P.A (Italy)
- m. Nuovo Pignone (Italy)
- n. Perar S.P.A (Italy)
- o. Pietro Fiorentini (Italy)
- p. Cooper Cameron Valv Italy SRL-FRM, Italy
- q. Petrol Valves SRL
- r. Tormene Gas Technology S.P.A (VALVITALIA)
- s. Petro Valves Pvt. Limited, Ahmedabad
- t. Shiv Shakti Engineering Works (Approved up to 08 inch)
- u. Hawa Engineers Limited, Ahmedabad
- v. G M Engineering PVT LTD.
- w. DelVal Flow Controls Pvt. Ltd

**3.0 TEE****3.1 FLOW TEE**

- a. Coprosider SPA, Italy
- b. GEA Energy System India Limited, Chennai
- c. Multitex Filtration
- d. Pipeline Engineering, UK
- e. Scomark Engg. Limited (U.K.)
- f. Skeltonhall Limited, Engaland(U.K.)
- g. Technospecial SPA, Italy
- h. Tectubi SPA, Italy
- i. RMA Germany
- j. Pipefit Engineers Pvt. Ltd.
- k. Vee Kay Vikram & Co.

3.2 SPLIT TEE

- a. IPSCO, Canda
- b. TD Willamsons, USA
- c. Plant-Tech Power Technical Services Pvt Ltd
- d. Teemans, UK
- e. Vee Kay Vikram & CO.

4.0 FLANGES

- a. Aditya Forge Ltd., Vadodara
- b. Amforge Industries Ltd., Mumbai
- c. CD Engineering Co., Ghaziabad
- d. Echjay Forgings Pvt. Ltd. (Bombay), Mumbai
- e. Echjay Industries Ltd., Rajkot
- f. Forge & Forge Pvt. Ltd., Rajkot
- g. Golden Iron & Steel Works, New Delhi
- h. JK Forgings, New Delhi
- i. Metal Forgings Pvt. Ltd., Mumbai
- j. Perfect Marketings Pvt. Ltd., New Delhi
- k. Sky Forge, Faridabad
- l. S&G, Faridabad
- m. Chaudhry Hammer Works Ltd, India



- n. JAV Forgings (P) Ltd, India
- o. Kunj Forgings Pvt Ltd, India
- p. MS Fittings
- q. R.N. Gupta & Co. Ltd, India
- r. R.P. Engineering Pvt Ltd, India
- s. Sanghvi Forgings & Engineering Ltd
- t. Shri Ganesh Forgings Ltd., India
- u. Uma Shankar Khandelwal & Co., India
- v. Sawan Engineers, Baroda
- w. Stewarts & Lloyds of India Ltd., Kolkata
- x. Engineering Services Enterprises
- y. Pipefit Engineers Pvt. Ltd.
- z. Jindal Forging
- aa. Vivial Forges
- bb. N J Engineers (Up to 12 Inch Class 600 Only)

5.0 FITTINGS

- a. Commercial Supplying Agency, Mumbai
- b. Dee Development Engineers Ltd.
- c. Eby Industries, Mumbai
- d. Flash Forge Pvt. Ltd., Vishakhapatnam
- e. Gujarat Infra Pipes Pvt. Ltd., Vadodara
- f. M.S. Fittings Mfg. Co. Pvt. Ltd., Kolkata
- g. Stewarts & Lloyds of India Ltd., Kolkata
- h. Teekay Tubes Pvt. Ltd., Mumbai
- i. Pipe Fit, Baroda
- j. Sky Forge, Faridabad
- k. S&G, Faridabad
- l. Sawan Engineers, Baroda
- m. Eby Fasteners, India
- n. Leader Valves Ltd, India
- o. R.N. Gupta & Co. Ltd, India
- p. Exten Engg Pvt Ltd
- q. Sivananda Pipe & Fittings Ltd
- r. Jindal Forging



- s. Vivial Forges
- t. PK Tubes Rajasthan
- u. CSA Fitting
- v. Gujarat Infrapipes pvt ltd, Vadodara
- w. KS Pipes Fitting Pvt Ltd, Palwal
- x. Tube Bend, Kolkata

6.0 PIG LAUNCHERS/ RECEIVERS/ PIG SIGNALERS

- a. Bassi Luigi Fittings B.V., Holland
- b. BRAUN STAHL PIPE TEC, GERMANY
- c. FORAIN, ITALY
- d. Fluidel SRL, ITALY
- e. RMA Maschinen- und, GERMANY
- f. Siirtec Nigi, Itlay
- g. SCHUCK ARMATUREN, GERMANY
- h. T.D. Williamson Inc., USA
- i. Tectubi SPA, Italy
- j. Taylor Forge Engineering System INC, USA
- k. Tormene Americana S.A. (Argentina)
- l. Tormene Gas Technology S.p.A., Italy
- m. PIPELINE ENGINEERING, UNITED KINGDOM
- n. Krohne, Oil & Gas BV, Drive Houston,
- o. Multitex Filtration Engrs. Ltd, New Delhi
- p. BGR ENERGY SYSTEMS LIMITED New Delhi
- q. Glapwell Contracting Services Ltd. UK
- r. FULGOSI GIOVANNI S.n.c di Corrado & C, ITALY
- s. VEEKAY VIKRAM & CO, GUJRAT
- t. GBM S.R.L, ITALY
- u. Cardew Ltd., Alexeander
- v. Forain S.R.L.
- w. GD Engineering, India
- x. Pipeline Engineering, UK
- y. Siirtec Nigi SPA
- z. Control Plus



- aa. Oswal Infrastructure

7.0 LONG RADIUS BENDS

- a. Jindal Saw Ltd. (Kosi Works), India
- b. PSL Limited (Gandhidham – Mfrg), India
- c. BHEL, Trichy, Tamilnadu
- d. Welspun, Gujarat
- e. Sawan
- f. Gujarat Infra
- g. P K Tubes
- h. DEE Development
- i. Pipefit Engineers Pvt. Ltd.

8.0 CLEAN AGENT SYSTEM

- a. ADN Fire Safety Pvt Ltd (Vashi East, Thane)
- b. Chetan Corporation (Ahmedabad)
- c. Chetan Engineers (Ahmedabad)
- d. Mx Systems International Pvt. Ltd. (Mumbai)
- e. New Fire Engineers (P) Ltd (Sil Vassa)
- f. Nitin Fire Protection Industries Ltd (New Bombay)
- g. Nohmi Bosai (India) Private Limited
- h. Tyco Fire & Security India Pvt. Ltd (Bangalore)
- i. Vimal Fire Controls Pvt Ltd (Vadodara)

9.0 INSULATING JOINTS

- a. IGP Engineers
- b. V K Vikram
- c. Advance Electronics
- d. Nupros INC

10.0 GASKETS

- a. IGP Engineers (P) Ltd., Madras
- b. Madras Industrial Products, Madras
- c. Dikson & Company, Bombay
- d. Banco Products (P) Ltd., Vadodara
- e. Goodrich Gaskets Pvt Ltd
- f. Starflex Sealing India Pvt Ltd, India



- g. Teekay Meta Flex Pvt Ltd
- h. UNIKLINGER Ltd
- i. HEM Engg. Corp.
- j. Unique Industrial Packing Pvt. Ltd.

11.0 FASTENERS

- a. Nireka Engg. Co. (P) Ltd., Calcutta
- b. Precision Taps & Dies, Bombay
- c. AEP Company, Vithal Udyoug Nagar
- d. Fix Fit Fasteners, Calcutta
- e. Precision Engg. Industries, Baroda
- f. Echjay Forgings Pvt. Ltd., Bombay
- g. Capital Industries, Bombay
- h. Boltmaster India Pvt Ltd, India
- i. Deepak Fasteners Limited, India
- j. Fasteners & Allied Products Pvt Ltd, India
- k. Hardwin Fasteners Pvt Ltd, India
- l. J.J. Industries, India
- m. Multi Fasteners Pvt Ltd, India
- n. Nexo Industries, India
- o. Pacific Forging & Fasteners Pvt Ltd, India
- p. Pioneer Nuts & Bolts Pvt Ltd, India
- q. Precision Auto Engineers, India
- r. President Engineering Works, India
- s. Sandeep Engineering Works, India
- t. Syndicate Engineering Industries, India

12.0 WELDING ELECTRODES FOR PIPELINE/PIPING WORK

- a. For Mainline – Lincoln/ Bohler make
- b. For Terminal – For root pass –Lincoln/ Bohler make
For other passes – Lincoln, D&H or equivalent make Lincon
- c. D & H SECHERON ELECTRODE PVT. LTD. (For Mainline & Terminal- Root & Other pass)

13.0 STRAINERS

- a. Bombay Chemical Equipments
- b. Gujarat Auto filed



- c. Multitex Filtration Engineering Limited
- d. Grand Prix Engineering Limited

14.0 COLD APPLIED TAPES

- a. Denso GmbH
- b. Raychem

15.0 HEAT SHRINKABLE SLEEVE/ FIBREGLAS REINFORCED SLEEVE

- a. Covalence - Seal For Life India Pvt. Ltd. (Formerly Covalence Raychem- Berry Plastics Corporation)
- b. Canussa-CPS
- c. Rigil Techno India(Distributor for M/s CYG Changtong New Material Co Ltd,China)
- d. M/s Shri Narayan Impac-India LLP

16.0 STUD BOLTS WITH NUTS

- a. Multi Thread Fasteners, Baroda
- b. Darukhanwala
- c. Precision Engineers, Baroda
- d. Unbrako
- e. TVC

17.0 WARNING MAT

- a. Sparco Multiplast Pvt. Ltd., Ahmedabad
- b. Singhal Industries , Ahemdabad
- c. Puja Packing, Mumbai
- d. Bina Enterprises, Mumbai
- e. Shree Vijay Wire & Cable Industries

18.0 HDPE PIPES/DUCT

- a. Climax Synthetics (P) Ltd., Vadodra
- b. Indian Poly Pipes, Calcutta
- c. Jain Irrigation Systems Ltd., Jalgaon
- d. Kirti Industries (India) Ltd., Indore
- e. Ori Plast Limited, Calcutta
- f. Phoel Industries Limited, Delhi
- g. Sangir Plastics (P) Ltd., Mumbai
- h. Veekay Plast,Jaipur



- i. Kisan Irrigation
- j. Dutron Polymers Ltd.
- k. Manikya Plastichem (P) Ltd
- l. Himalyan Pipe Industries
- m. NorthEast Polypipes LLP
- n. Mukand Poly Products (for Duct)

19.0 DRY GAS FILTER & FILTER SEPERATOR

- a. Grand Prix Fab (Pvt.) Ltd. (New Delhi)
- b. Perry Equipment, USA
- c. Faudi Filter, Germany
- d. Forain S.r.l., Italy
- e. ABB, Faridabad
- f. Burgess Manning, USA
- g. Multitex Filtration Engineers India
- h. Triveni Plenty Engg. Ltd. (New Delhi)
- i. Siirtec International Contractor S.P.A (Italy)
- j. Flashpoint, Pune india
- k. Filtration Engineers (I) Pvt Ltd, India
- l. Gujarat Otofilt, India
- m. Tormene Gas Technology
- n. Ultrafilter (India) Pvt Ltd, India
- o. Ravi Techno Systems Pvt Ltd, India
- p. Siirtec Nigi S.P.A
- q. Filtan Filter Anlagenbau GmbH
- r. Fairley Arlon BV
- s. PECO Facet
- t. EPE Epenstenner GMBH
- u. Filtrex srl
- v. Petromar Engineered Soln
- w. Plenty Filter
- x. Eurofiltec
- y. PTI Technologies Inc
- z. R S Instrumentation and Services
- aa. Inel Gas Control Pvt Ltd

**20.0 FILTER ELEMENT**

- a. Peco – Facet
- b. Velcon
- c. Pall – Filterite
- d. Burgess Manning

21.0 NDT AGENCY

- a. NDT Services, Ahmedabad
- b. GEECY Industrial Services Pvt. Ltd., Mumbai
- c. Corrosion Control Services, Mumbai
- d. Perfect Metal Testing & Inspection Agency, Calcutta
- e. Inter Ocean Shipping Co., New Delhi
- f. RTD, Mumbai
- g. Sievert, Mumbai
- h. X-Tech, Vizag
- i. Industrial X Ray and Allied Radiographers (I) Pvt. Ltd.
- j. Inspection Technology
- k. IXAR
- l. Aditya NDT Services
- m. Riya NDT Engineers
- n. Star NDT Services
- o. Technical testing & Inspection Services
- p. PD Engineering Inspection Services
- q. Victor Engineering & Testing

22.0 Cold Applied Tapes

- a. Denso GmbH
- b. Polyken (Berry Plastics Corporation)

22.1 End Seal Tape-PVC

- a. MAASH Industries

23.0 PUR Coating

- a. Powercrete (Berry Plastics Corporation)

24.0 Casing End Closure



- a. Raci, Italy
- b. Raychem RPG Limited

25.0 Casing Insulators

- a. Raci, Italy
- b. Raychem RPG Limited
- c. VeekayVikram

26.0 FIRE FIGHTING EQUIPMENT**26.1 FIRE EXTINGUISHERS**

- a. Avon Services (Production & Agencies) Pvt. Ltd., Bombay
- b. Kooverji Devshi & Co., Bombay
- c. Reliable (Fire Protection) India Ltd., Bombay
- d. Zenith Fire Services, Bombay
- e. Safex Fire Services, Bombay
- f. Brij Basi Hi
- g. tech Udyog
- h. Bharat Engg Works, India
- i. Gunnebo India Ltd
- j. Nitin Fire Protection Industries Ltd, India
- k. Supremex Equipments, India
- l. Vimal Fire Controls Pvt Ltd., India

26.2 FIRE HYDRANTS, MONITORS, DELUGE VALVE, NOZZLES

- a. Zenith
- b. Minimax
- c. Newage
- d. HD Fire
- e. Vijay Fire
- f. Asco Strumech Pvt Ltd, India
- g. Brij Basi Hi
- h. tech Udyog
- i. Gunnebo India Ltd
- j. Nitin Fire Protection Pvt Ltd
- k. Shah Bhogilal Jethamal & Brothers
- l. Pumps & Engineering Works

**26.3 FIRE & GAS DETECTION & ALARM**

- a. Ambetronics Engineers PVT LTD.

26.4 RRL Hose

- a. Jayshree
b. Newage

26.5 HOSES

- a. Ashit Sales Corporation, Bombay
b. Royal India Corporation, Bombay
c. Gayatri Industrial Corporation
d. Simplex Rubber Products Ltd., Ahmedabad
e. Zaverchand Marketing Pvt. Ltd., Baroda
f. Presidency Rubber Mill, Calcutta
g. The Cosmopolite, Calcutta
h. Simplex Rubber Products, Thane

NOTE:

- 1) For procuring bought out items from vendors other than those listed above, the same may be acceptable subject to the following: -
 - a) The vendor/ supplier of bought out item(s) is a manufacturer/ supplier of said item(s) for intended services and the sizes being offered is in their regular manufacturing supply range.
 - b) The vendor / supplier should not be in the Holiday list of CLIENT / VCS / other PSU.
 - c) Should have supplied at least one single random length (i.e. 5.5 meters to 6.5 meters) for item assorted pipes / tubes and for other items, which are to be supplied in quantity on number-basis (other than assorted pipes / tubes) minimum 01 (One) number of same or higher in terms of size and rating as required for intended services. The bidder should enclose documentary evidences i.e. PO copies, Inspection Certificate etc. for the above, along with their bids.
- 2) For any other item(s) for which the vendor list is not provided, bidders can supply those item(s) from vendors/ suppliers who have earlier supplied same item(s) for the intended services in earlier projects and the item(s) offered is in their regular manufacturing/ supply range. The bidder is not required to enclose documentary evidences (PO copies, Inspection Certificate etc.) along with their offer, however in case of successful bidder, these documents shall require to be submitted by them within 30 days from date of Placement of Order for approval to CLIENT / VCS.
- 3) The details of vendors indicated in this list are based on the information available with VCS, Contractor shall verify capabilities of each vendor for producing the required quantity with. PMC does not guarantee any responsibility on the performance of the vendor. It is the contractor's responsibility to verify the correct status of vendor and quality control of each parties and also to expedite the material in time.

**B. CIVIL AND STRUCTURAL**

Sr. No.	Items/ Name of Products	Make/ Brands/ Manufactures
1.	Reinforcement Steel	TATA, SAIL, RINL, JSW.
2.	Cement	Ambuja, ACC, JK, Grasim, Ultratech, Birla, L&T, Cement Corporation of India
3.	Structural Steel	TATA, SAIL, RINL, IISCO, ESSAR
4.	Structural Steel Tubes ISI Marked	TATA, JINDAL, SURYA
5.	Mineral wool for thermal insulation of ceilings (Under deck insulation)	Rock wool (india) Ltd. Minwool Rock Fibres Ltd., Lloyd Insulation.
6.	Synthetic Enamel Paint (1st quality only)	ICI Paint (Dulux), Asian Paint (Apolite), Berger Paints (Luxol). Goodlass Nerolac Paints (Nerolac), Jenson & Nicholson Paints Ltd (Borolac)
7.	G.I SHEET	ESSAR, JSW, SAIL
8.	Sheeting Screw	Corroshield, Buildex,
9.	Chemical for Antitermite treatment	DE- NOCIL Bombay, Pest Control of India, Trishul
10.	Factory made Panelled Door shutter	Century; Godrej ; M/s Hindustan Housing factory Ltd., New Delhi ; M/s Delhi Construction Eqp, Sadar Bazar, Delhi ; M/s Joinery manufacturing Co., Calcutta;
11.	PVC Panel Door (Solid Core)	Rajshri Plastiwood Limited, Sintex, Hindopan, Marino
12.	Pressed steel door frames/ cupboard and window frames (manufacturers)	M/s SAIL, M/s TATA
13.	Pressed steel door frames/ cupboard and window frames (fabricators)	M/s Loyal safe works Mayapuri, N/Delhi M/s Multiwyn Industrial Corpn Calcutta M/s Metal Window Corpn N/Delhi M/s Chhabra Steel Udyog, 260 Sadar Bazar, Meerut Cantt. M/s Delite safe works, Rani Jhansi Road, N/Delhi
14.	Steel Windows, Ventilators (as per IS- 1038 of 1983) & frames pressed steel door/window	M/S Multiwyn Industrial Corpn, Calcutta ; M/S Metal Window Corp, N/ Delhi ; M/S Chhabra Steel Udyog 260, Sadar Bazar, Meerut Cantt ; Agent steel MFG Pvt Ltd, Ahmedabad ; Godrej ;
15.	AI Section for AI Door/ Window/ Partitions	Hindalco, Ajit India, Jindal
16.	AluminumI Door/ Window/ Glazing Fabricated and Anodized	M/s Alumilite Pvt Ltd, M/s Ajit India Pvt Ltd, M/s Ramniklal S Raste Agra, Argent Industries, M/s Aluminium Tech Industries, I-2249 DSIDC Narela, Delhi, M/s VR Associates, GH-14/242 Paschim Vihar,



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		Delhi
17.	Aluminium door and windows Fittings	M/s Elite Enterprises C/6 Shalimar Hardware 133, Jarg Mahal, Dhobitalao Mumbai 400002. M/s Mohan Metal Industries 178/2-A, Bhole Nath Nagar, Shahadara, Delhi 110032. Mepro, Argent New Delhi, Classic, New Delhi. Jindal, Argent New Delhi, Golden Industries Pvt. Ltd. ECIE
18.	Aluminium Grill	Alu Grill, Arihant Aluminium Corporation, Decogrille
19.	Door Closer	Everite, Golden, Gandhi,
20.	Floor Spring	Prabhat, Everite
21.	Plywood for general purpose (IS-303)	National Plywood Inds Pvt Ltd, S Fancy lane, 8th floor, Calcutta-700001, Merino Plywood, Archid Ply, Ply, Swastik, Universal, Century, Greenply, National.
22.	Pre laminated Particle board	Kitply, Bhutan board, Ecoboard, Novapan, Archid ply, Merinova, Merino
23.	Laminated Sheets	Formica, Merino Lam, Greenlam, National, Century
24.	Modular Partitions	Godrej, Blowplast
25.	False Ceiling (Mineral Fibre Board)	Armstrong, Daiken, Luxalon, Llyods, Gypboard, Trac, Aerolite
26.	Aluminium False Ceiling	Lloyds, Armstrong, Luxlon, Trac
27.	Flooring Tiles (Mosaic / Terrazzo / PCC) (1st quality only)	Kajaria Ceramics, NITCO, Royal Tiles, Gem Tiles, Hindustan Tiles, M/S National Tiles & Industries, Ultra Tiles
28.	Glazed Ceramic Tiles, Non-Skid (Floor/Wall), (1st quality only)	Kajaria, Somany, NITCO. Murudeshwar Ceramic Ltd (Navin Diamond tile), Johnson (Marbonite),
29.	Vitrified/ Designer Vitrified Tiles (1st quality only)	Asian, Marbonite (Johnson), Kerrogres (Kajaria), NITCO, Orient
30.	PVC Tiles/Flooring (IS 3461) (1st quality only)	Marblex Tiles, Krishna Tiles, Polyfin, Armstrong, Wonder floor.
31.	False Flooring	Godrej or equivalent
32.	Glass Mosaic Tiles (1st quality only)	Paladio, Coral, Accura, Bisazza, Italia, Mridul.
33.	Designer Paver Tiles/ Interlocking tiles ISI marked/ Grass-jointed Tiles. (1st quality only)	Pavit, Ultra, Hindustan, Eurocon, Vyara, National Tiles, Gem, Unistone, Konkrete, Unitile
34.	Wall care Putty for Base preparation (1st quality only)	Birla Wall care putty, Berger, Jenson & Nicholson, JK White



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35.	White Cement (1st quality only)	Birla, JK
36.	Cement based Paints (1st quality only)	Super Snowcem, Duracem, Super Acrocem.
37.	Dry Distemper / Oil bound Distemper (1st quality only)	Goodlass Nerolac Paint, Shalimar Paint, Jenson & Nicholson, Asian Paint, Berger. ICI Dulux
38.	Acrylic Washable Distemper (1st quality only)	Asian, Berger, ICI Dulux, Jenson & Nicholson, Nerolac, Shalimar, Garware & Goodlass
39.	Plastic Emulsion Paint (1st quality only)	Asian, Berger, ICI, Nerolac, Jenson & Nicholson, Shalimar, Garware & Goodlass
40.	Exterior Acrylic Emulsion (1st quality only)	ICI (Weathercoat), Excel (Nerolac), Apex (Asian), Berger, Jenson & Nicholson, Shalimar, Garware & Goodlass
41.	Polymer based Paint	STP, CICO
42.	Textured Paint / Wall Tile (1st quality only)	Unitile, Heritage, Spectrum, Iokos, Acropaints, Asian
43.	Flexible board for Expansion joint	STP or equivalent
44.	Grout	Shrinkomp, Fosroc, Fairmate
45.	Integral water proofing compound	STP, Pidilite, Fosroc, CICO, Sika.
46.	Concrete Admixture	Pidilite, Fosroc, CICO, Sika.
47.	Water proofing for cementations surface IS-2645	Acrocrete & Acrocote, CICO, Fosroc, STP
48.	Bituminous Product	M/s Faridabad Spinning & Woolen Mills Pvt Ltd, 837, SP Mukherjee Marg Delhi, M/s STP Ltd (Formerly Shalimar Tar Products) M/s Bitufelt Pvt Ltd 123/377 Fazalm Ganj Kanpur 208012, Texas, Texas India Ltd, Multiplas
49.	Hardeners	Ironite, Ferrok, Hardonate
50.	Construction Chemicals	Choksey, CICO, Forsroc, Sika
51.	Stainless Steel Cladding	Jindal
52.	Punch Tape Concertina Coil	Global Technocrat, S.G. Engineers Delhi
53.	Stainless Steel Railing	Jindal
54.	FRP/ HDPE Garbage Bins	Sintex, Swift, Nutech, Sheetal
55.	Sanitary ware	Neycer Kermag (standard), Hindustan Sanitary Ware (1st quality), Parryware (superfine), Cera (1st quality), Classica (1st / standard)



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56.	WC seat cover ISI Marked	Parryware, Neycer Kermag (standard), Hindustan Sanitary Ware (Ist quality), Cera (Ist quality), Classica (Ist / standard)
57.	PVC Flushing Cistern IS: 774-1984 (ISI Certified)	Parryware, Hindustan Sanitary Wares, Cera.
58.	Faucets & Taps, Stop Valves & Pillar Taps, Surgical basin mixer, Shower rose etc.	Gem, Parko, Parryware, HSW, Jaquar
59.	Kitchen Stainless Steel Sink	Diamond, Nirali, Neel Kanth, Jayna
60.	Looking Mirror	Saint Gobain, Modi Float, Triveni Float Glass, Crown, Atul.
61.	Readymade Bathroom Cabinets	Commander Gratings (I) Pvt Ltd, Gratolite Cabinet, A- 4 Sector VIII Noida-202701, Alpina, Cera.
62.	Float Valve	Leader, Bombay Metal & Alloy Co, Bombay superflow.
63.	SGSW Pipes (IS-651) ISI Marked	Perfect Agra, Devraj Ind Gaziabad, Buran, RK, Prince,
64.	CI (Centrifugally Cast) Pipes for sewage disposal ISI marked	NICCO, SRIF, A-1 Singhal Casting Co Agra, Jindal Saw, Kesoram, NECO
65.	PVC rain water/sewage pipes (IS-4985)	Reliance, Finolex, Supreme, Kisan, Prince, Hindustan Plastic & machine corporation, Polypack industries (P) Ltd.
66.	HDPE Water storage tanks (Rotational Moulded)	Sintex, Swift, Nutech, Sheetal
67.	Cast Iron Pipes and Fittings	Hindustan Engineering Products Company Calcutta, S.L.C., Standard approved manufacturers of any other brand of fittings having ISI marking, RIF, BIS
68.	RCC Pipes	Indian Hume Pipe Company, Delhi / Allahabad / Chandigarh / Lucknow; Hindustan Pressure Pipes, Kolhapur; Dhere Concrete Products, Pune or any other approved manufacturer conforming B.I.S. Standard
69.	Brass Fittings	Leader Engineering Works, Jalandhar; L & K Mathura; Luster Sanitary, Jalandhar; Annapurna Metal Works, Calcutta; Neta Metal Works, Jalandhar
70.	C.P. Fittings	Ego Metal Works, Ballabhgarh; Jaquar Industries, Delhi; Soma Plumbing Fixtures Limited, Calcutta; Gem Sanitary Appliances Pvt. Ltd., Delhi; Essco Sanitations, Delhi.
71.	Stone Ware (Salt-Glazed) Pipes	Hind Ceramics Limited, Orissa; Ceramic Industries Limited, Sambalpur; Shrikamakshi Agencies, Madras; Binary Udyog Pvt. Limited, Howrah; Tirumati Moulds Limited, Nagpur.
72.	Asbestos Cement Pipes and Fittings	Ganga Asbestos Limited, U.P.; Hyderabad Asbestos Cement Products Limited; J.K. Super Pipe Industries, Nanded; Konark Cement and Asbestos Limited, Orissa; Maharashtra Asbestos Limited, Bombay.



List of Recommended Vender/Suppliers of Major Bought-Out Items: Unless otherwise specifically mentioned in the Schedule of Items, Contractor has to use materials as listed below, of only these brand names/Company's names, which are mentioned in the RECOMMENDED list for structural items thereon.

Sl. No.	Items/Name of Products	Makes/Brands/Manufactures
1	Structural Steel	SAIL / TATA / RINL / IISCO / ESSAR / ISPAT
2	Structural Steel Tubes ISI Marked	TATA / JINDAL / SURYA / SWASTIK
3	Synthetic Enamel Paint 1st Quality only	ICI Paint (Deluxe), Asian Paint (Apolite), Shalimar Paint (Superlac), Goodlass, Nerolac Paint (Nerolac), Berger Paints

Any materials not fully specified in these specifications and which may be offered for use in the works shall be subject to approval of Engineer, without which it shall not be used anywhere in the construction works.

**C. ELECTRICAL****LIST OF SUPPLIERS OF MAJOR BOUGHT-OUT ITEMS****1.0 AIR CONDITIONER**

- a. Blue Star
- b. O General.
- c. Daikin.
- d. Carrier
- e. Hitachi.
- f. LG.
- g. Lloyds
- h. Mitsubishi
- i. Panasonic
- j. Sharp
- k. Samsung.
- l. Blue star.
- m. Haier.
- n. Voltas.
- o. Videocon.

2.0 BATTERIES (LEAD ACID)

- a. Amco Batteries Ltd.
- b. Exide Industries Ltd.
- c. HBL Power System Ltd.
- d. Amara Raja Batteries Ltd.
- e. Su-Kam Power Systems Ltd.
- f. Base Corporation Ltd.
- g. Okaya Power Ltd.
- h. Southern Batteries Pvt Ltd.
- i. True Power International Ltd.
- j. Evolute Solutions Pvt Ltd.
- k. Greenvision Technologies Pvt Ltd.
- l. Artheon Electronics Ltd.

**3.0 BATTERIES (NICKEL CADMIUM)**

- a. Amco Saft Batteries Ltd.
- b. HBL Power Systems Ltd.

4.0 BATTERIES (Lithium-ion)

- a. PROSTARM INFO SYSTEMS LTD.

5.0 BATTERY CHARGER/DC-DC CONVERTER

- a. Amara Raja Power System(P)Ltd.
- b. BCH.
- c. Chhabi Electricals Pvt. Ltd.
- d. Caldynes Automatics Limited.
- e. HBL Nife Power Systems Ltd..
- f. Universal Instrument Mfg. Co Pvt Ltd.
- g. Hitachi HI-REL Power Electronics P. Ltd
- h. Mass-Tech Controls Pvt Ltd
- i. Dubas Engineering Pvt Ltd
- j. Chloride Power Systems & Solutions Ltd
- k. Synergee
- l. Enertech
- m. Vertiv

6.0 CABLE – FIRE ALARM & COMMUNICATION CABLES

- a. Cords Cable Industries Ltd.
- b. CMI.
- c. Delton cables Ltd.
- d. ELKAY Telelinks.
- e. KEI Industries Ltd.
- f. Elkay Telelinks

7.0 CABLE – HT(XLPE)

- a. Universal Cable Ltd.
- b. KEI Industries Ltd.
- c. Industrial Cables.
- d. NICCO Corporation Ltd.



- e. Uniflex.
- f. Polycab.
- g. Torrent cables Ltd.

8.0 CABLE – LT / MV POWER AND CONTROL

- a. Cords Cable Industries Ltd.
- b. Universal Cable Ltd.
- c. KEI Industries Ltd.
- d. Havells.
- e. Delton.
- f. Elkay Telelinks.
- g. Evershine Electricals.
- h. Ecko.
- i. Ravin.
- j. Rallison.
- k. Suyog.
- l. Netco.
- m. Uniflex.
- n. Paramount.
- o. Gloster.
- p. Associated cables Pvt Ltd.
- q. CMI.
- r. Gemscab.
- s. Industrial cables.
- t. NICCO.
- u. Polycab.
- v. Torrent.

9.0 CABLE – GLAND

- a. Baliga.
- b. Comet.
- c. Flexpro.
- d. Flameproof.
- e. FCG.



- f. Electro Werke.
- g. Dowels.
- h. CCI.
- i. Sudhir Switchgear
- j. Keyson Techno Equipments,
- k. RAY Flameproof Pvt Ltd

10.0 CABLE – LUGS & TERMINAL BLOCKS

- a. Dowels.
- b. Jainson.
- c. Sharma Electrical
- d. Punitam
- e. Yamuna Powers
- f. Rapid Manufacturer
- g. Varun Controls.

11.0 CABLE – TRAY

- a. Ercon Composites.
- b. Yamuna Power & Infrastructure Ltd.
- c. MEM
- d. Bharti
- e. Profab.
- f. Ratan.
- g. Slotco.

12.0 CABLE TERMINATION AND JOINTING KIT

- a. CCI.
- b. Raychem.
- c. M-Seal.

13.0 CEILING/EXHAUST/PEDESTAL FANS & CIRCULATORS

- a. Bajaj Electricals Ltd.
- b. Crompton Greaves Ltd.
- c. Khaitan Electricals Ltd.



- d. Havell's.
- e. Philips
- f. Usha
- g. Orient

14.0 CONTRACTORS – AC POWER

- a. Andrew Yule.
- b. ABB.
- c. BHEL.
- d. C&S.
- e. Havell's.
- f. L&T.
- g. Schneider.
- h. Siemens Ltd.
- i. Telemecanique.
- j. GE India Industrial

15.0 CONTROL TRANSFORMER

- a. AE.
- b. Indushree.
- c. Intra Vidyut.
- d. Kalpa Electricals.
- e. Transpower Industries Ltd.
- f. Siemens.
- g. PROSTARM INFO SYSTEMS LTD.

16.0 GAS GENERATOR/DIESEL GENERATOR SET

- a. Sterling and Wilson.
- b. Jackson Limited.
- c. Sudhir Gensets.
- d. Power Engineering (India) Pvt Ltd.
- e. Prasha Technologies Limited.
- f. Kumar Generator house.
- g. Ashok Leyland Ltd.



- h. Powerica Limited.
- i. Supernova Engineers Limited.
- j. Bhaskar Power Products (P) Ltd.
- k. Caterpillar India (P) Ltd.
- l. Cummins India Ltd.
- m. Escorts Ltd.
- n. Greaves Cotton Ltd.
- o. Kirloskar Ltd.
- p. Mahindra & Mahindra Ltd.
- q. Honda.
- r. Perkins.
- s. Eicher.
- t. Tata Motors.
- u. Ashok Leyland.

17.0 FLAME PROOF LDB'S/ JB'S/CONTROL STATION/ SWITCHES

- a. FCG.
- b. Sudhir switchgears.
- c. Prompt Engineering Works
- d. Flame Proof equipments pvt. Ltd.
- e. Baliga Lighting Equipments Pvt. Ltd.
- f. Flexpro Electricals Pvt. Ltd.
- g. Exprotecta, Beroda.
- h. FFLP Control Gears.
- i. Sterling.
- j. Kaysons Techno Equipment
- k. Bajaj
- l. Phoenix Contact
- m. RAY Flameproof Pvt Ltd

18.0 HIGH MAST

- a. Bajaj Electricals Limited.
- b. Crompton Greaves Limited.
- c. Philips India Limited.



- d. Surya Roshani.
- e. Transrail

19.0 HIGH VOLTAGE PCC/ MCC PANELS

- a. BHEL.
- b. Control and Switchgear.
- c. Siemens.
- d. Tricolite Electrical Industries.
- e. Schneider.
- f. CGL.
- g. L&T.
- h. ABB.

20.0 INDICATING LAMPS

- a. Alstom Ltd.
- b. BCH.
- c. L&T Ltd.
- d. Siemens Ltd.
- e. Vaishno Electricals.
- f. Tecknik
- g. ABB

21.0 INDICATING METERS

- a. L&T
- b. ABB.
- c. AMCO.
- d. AE
- e. Alstom Ltd. (EE).
- f. Conzerv/Schneider
- g. Elecon Measurement Pvt. Ltd.
- h. HPL Electric & Power Pvt. Ltd.
- i. MECO Instruments Ltd.
- j. Minilec.



- k. Rishabh Instruments Pvt. Ltd.
- l. Trinity energy system.
- m. Kaycee.
- n. Salzer.
- o. Trinity
- p. Secure Meters Limited

22.0 LIGHTING FIXTURES

- a. GE Lighting Pvt. Ltd.
- b. Bajaj Electricals Ltd.
- c. Crompton Greaves Ltd.
- d. Philips India Ltd.
- e. Havells
- f. Wipro
- g. Osram
- h. Siska
- i. Halonix
- j. Surya
- k. Polycab
- l. Kawlity Photonix

23.0 LIGHTING FIXTURES – FLAMEPROOF

- a. Bajaj Electricals Ltd.
- b. Baliga Lighting Equipment Pvt. Ltd.
- c. Crompton Greaves Ltd.
- d. CEAG Flameproof Controlgear Pvt. Ltd.
- e. Flexpro Electricals Pvt. Ltd.
- f. Philips India Ltd.
- g. Sudhir Switchgears Pvt. Ltd.
- h. FCG.

24.0 MINIATURE CIRCUIT BREAKERS (MCBS) AND LIGHTING DB

- a. ABB.
- b. Hagger.
- c. Havell's India Ltd.



- d. Indo Asian Fusegear Ltd.
- e. Legrand.
- f. MDS Switchgear Ltd.
- g. Schneider.
- h. Siemens Ltd..
- i. HPL.
- j. L & T
- k. Siemens

25.0 MOULDED CASE CIRCUIT BREAKER (MCCBS)

- l. ABB.
- m. Andrew Yule.
- n. Larsen & Toubro.
- o. Schneider.
- p. Siemens.
- q. Control and Switchgear.
- r. Indo Asian,
- s. Hager.
- t. Merlin Gerin.
- u. Havell's India Ltd
- v. General Electric

26.0 PROTECTION RELAYS

- a. BCH.
- b. L&T Ltd.
- c. Siemens Ltd.
- d. Tele-mechanique & Controls (India) Ltd.
- e. Eaton
- f. Alstom
- g. ABB
- h. Ashida
- i. Novatek Electronics
- j. Cantronics
- k. Control & Switchgear
- l. Frick Electronics



- m. Indian Technologies
- n. Rao Electromechanical
- o. Kudam Corporation

27.0 LOW/MEDIUM VOLTAGE POWER CONTROL CENTER (PCC)/ MCC/ PDB/ MLDB/ LDB

- a. ABB.
- b. BCH.
- c. BHEL.
- d. C & S.
- e. Elecmech Switchgear & Instrumentation.
- f. KMG ATOZ.
- g. L&T.
- h. Pyrotech Electronics Pvt. Ltd.
- i. Risha Control Engineers Pvt. Ltd.
- j. UDKAM PROCESS EQUIPMENT INDIA PVT. LTD
- k. Tricolite Electrical Industries.
- l. Unilec Engineers Ltd.
- m. Vidyut Control India Pvt. Ltd.
- n. Control and Schematic.
- o. Zenith Engineering.
- p. Schneider Electric,
- q. AEG,
- r. HAVELL'S,
- s. MDS
- t. Synergy System
- u. Expert Engineers
- v. R K Engineering & Infrastructure

28.0 PUSH BUTTONS

- a. BCH.
- b. Alstom Ltd.
- c. L&T.
- d. Siemens Ltd.
- e. Tele-Menchanique & Controls (India) Ltd.



- f. Vaishno Electricals.

29.0 SWITCHES-CONTROL

- a. BCH.
- b. Easum Reyrolle Relays & Devices Ltd.
- c. Alstom.
- d. Kaycee Industries Ltd..
- e. L&T.
- f. Siemens Ltd.

30.0 SWITCHES – 5/15A PIANO/ PLATE, SWITCH SOCKET

- a. Anchor Electronics & Electricals Pvt. Ltd.
- b. Kingal Electricals Pvt. Ltd.
- c. North-West Switchgear Ltd.
- d. Schneider
- e. Havells
- f. Cona
- g. Orient
- h. Panasonic

31.0 SWITCH SOCKET OUTLETS (INDUSTRIAL)

- a. Alstom Ltd.
- b. Best & Crompton Engineering Ltd.
- c. BCH.
- d. Crompton Greaves Ltd.
- e. Essen Engineering Company Pvt. Ltd.

32.0 SOLAR POWER SYSTEM MODULES

- a. Tata Power Solar Systems Ltd
- b. REIL,
- c. CEIL,
- d. HBL Power.
- e. Vikram Solar.
- f. Waaree Solar.
- g. Solar Semiconductor.



- h. Sonali.
- i. Mundra Solar
- j. Vikram Solar
- k. Bharat Electronics Limited
- l. Waaree
- m. Lumipex
- n. Independent Qualitative Accessories
- o. Statcon Energy
- p. Volks Energy
- q. Green Engineering

33.0 SOLAR PCU/CHARGE CONTROLLER

- a. Statcon Energy
- b. Enertech UPS Pvt Ltd
- c. Consul Nowatt
- d. OPS India
- e. Schneider
- f. Delta Electronics
- g. ABB Ltd
- h. KACO
- i. SMA
- j. HBP Power
- k. Universal Instrument

34.0 TERMINALS BLOCKS

- a. Connectwell.
- b. Controls & Switchgear Co. Ltd.
- c. Elmex Controls Pvt. Ltd.
- d. Essen Engineering Co. Pvt. Ltd.
- e. Phoenix Contact

35.0 DISTRIBUTION TRANSFORMERS

- a. ABB.
- b. Andrew Yule.
- c. Areva.



- d. BHEL.
- e. Bharat Bijlee
- f. Crompton Greaves.
- g. EMCO Ltd.
- h. Intra Vidyut.
- i. Indushree.
- j. Indcoil
- k. Kirloskar.
- l. Skippers Electricals.
- m. Transformers & Rectifiers (I) Ltd.
- n. Voltamp.

36.0 UPS SYSTEM AND INVERTER

- a. DB Power.
- b. Keltron.
- c. Hi-Rel/HITACHI.
- d. Dubas.
- e. Toshiba Corporation.
- f. Fuzi Electric Co Ltd.
- g. Emerson.
- h. Synergy System.
- i. Eaton.
- j. Enertech
- k. PROSTARM INFO SYSTEMS LTD.

37.0 GI-OCTAGONAL POLE

- a. Bajaj.
- b. Transrail.
- c. Wipro.
- d. K.L. Industries.

38.0 ELECTRICAL MOTORS

- a. Siemens.
- b. Crompton Greaves.
- c. Kirloskar.
- d. BHEL.



- e. Bharat Bijlee.
- f. Hindustan motors.
- g. Alstom.
- h. Texmo.
- i. GE India.
- j. National Motors.
- k. ABB.

39.0 ELECTRICAL PROCESS HEATER

- a. Escorts Limited, Faridabad, Haryana.
- b. Spherehot / Kanti Lal Chuni Lal & Sons Appliances Pvt Ltd.Surat.
- c. Kerone, Bhayander(E), Thane – 401105.
- d. Excel Heaters, Andheri (West), Mumbai - 400 053, India.
- e. Nirmal Industrial Controls Pvt. Ltd., Mulund(W), Mumbai - 400 080.

40.0 CATHODIC PROTECTION AGENCIES/CONTRACTOR/ VENDERS

- a. CALTECH Engineering Service.
- b. Universal Corrosion Prevention India.
- c. Cathodic Technology Limited.
- d. Cathodic Control Company Pvt. Ltd.
- e. CORRTECH International Pvt Ltd.
- f. MITCORR Cathodic Protection Pvt Ltd.
- g. Underground Pipeline & NDTs Pvt. Ltd.
- h. JG Corrosion Solution.
- i. Mercury Cathodic Protection Service.
- j. UNDTs Corrosion Service.
- k. SARK EPC Projects Pvt. Ltd.

41.0 BACKUP AGENCY FOR INTERFERENCE SURVEY & MITIGATION

- a. PLE Germany
- b. Vendor Velde
- c. Nippon Japan
- d. SSS India CIPL / interference survey.
- e. Balslev Denmark, .
- f. SSS Germany



- g. Jef Techno Slutions Pvt. Ltd.
- h. SARK EPC Projects Pvt. Ltd.

42.0 PERMANENT REFERENCE CELL

- a. PERMACELL/ HARCO (USA)
- b. CORRTECH (ZULU), INDIA
- c. TINKER RASOR, USA
- d. SILVION, UK

43.0 CP CABLES

- a. Brooks Cables.
- b. Nicco Corporation Ltd.
- c. CCI Ltd.
- d. Delton Cables Ltd.
- e. KEI Industries.
- f. Torrent Cables.
- g. Universal cables.
- h. Victor Cables.
- i. Associated Flexible & Wires Pvt Ltd.
- j. Asain Cables (RPG Cables).
- k. Fort Gloster (Gloster Cables Ltd).
- l. Finolex Cable.
- m. Rediant Cables.
- n. NETCO Cables Pvt Ltd.
- o. Havells Ltd.

44.0 CP SACRIFICIAL ANODES

- a. Scientific Metals Engineers Pvt. Ltd., Karaikudi
- b. PSL Holding Pvt. Ltd., Mumbai.
- c. Cathodic Controls, Bangalore.
- d. BHEL, Bhopal.
- e. Nippon Corrosion, Japan.
- f. AFIC, KSA.
- g. Platt Bros. and Company, USA



- h. Wilson Walton International.
- i. Impalloy International.
- j. Corrpro International.
- k. HOCKWAY, UK
- l. NAKABOHTEC, Japan .
- m. Cortech International
- n. Titanor Component
- o. Shakti Metals, Ahmedabad
- p. Shaktianodes Private Limited, Ahmedabad

45.0 CP PORTABLE /PERMANENT REFERENCE CELL

- a. M/S PERMACELL/HARCO, USA
- b. M/S BORIN MANUFACTURER, USA
- c. M/S M.C.MILLER, USA
- d. M/S TINKER RASOR

46.0 CPTR (AC OPERATED)

- a. Canara Electric
- b. CATHODIC CONTROL COMPANY PVT LTD.
- c. Kriston Systems

47.0 PIN BRAZING / THERMITWELD

- a. SAFETRACK, SWEDEN
- b. CadWeld
- c. BAC, UK
- d. ERICO, USA
- e. THERMOWELD, USA
- f. ERICO, EUROPE

48.0 CP SURGE DIVERTER/SPARK GAP ARRESTOR (EX-D)

- a. Dhen,
- b. OBO
- c. Corrpro system
- d. Sohne
- e. Dairyland

49.0 CP SOLID STATE POLARISATION CELL.

- a. Dairyland



- b. Metricorr
- c. Rustol
- d. Dhen
- e. Kriston Systems

50.0 PETROLEUM COKE BREEZE:

- a. M/S LORESCO, USA
- b. M/S ASBURY, USA
- c. M/S GOA CARBON, GOA
- d. M/S INDIA CARBON, CALCUTTA
- e. M/S PETROCARBON & CHEMICAL COMPANY, HALDIA

51.0 CP ANODE (MMO TYPE):

- a. Corrttech
- b. Scientific Metal Engineers Karaikudi
- c. Titanor Component Ltd., Goa, India.
- d. Denora Permelic S.P.A (Italy). .
- e. Oronzio De Nora S.A. Ingano Switzerland
- f. CER Anode Technologies International USA
- g. ACTEL, UK
- h. ELTECH System Corporation, Texas
- i. MAGNETO-CHEMIE, Netherlands
- j. MATCOR (USA)

52.0 HEAT SHRINK CAP FOR CP ANODE

- a. MATCOR (USA) To Cable Joint
- b. Seal For Life

**53.0 ER- PROBE (EXTERNAL CORROSION) ER- PROBE & CORROSION
COUPON ASSEMBLY**

- a. Rose Corrosion Services UK
- b. Metal Samples, USA. .
- c. Roharbak Cosasco USA
- d. Caproco UK
- e. Emetricorr, Denmark

54.0 HEAT SHRINK CAP FOR ANODE TO CABLE JOINT



- a. Raychem, USA
- b. Matcor (USA)

55.0 MMO WIRE ANODES (WITH FACTORY PRE-PACKED COKE BREEZE)

- a. Matcor (USA)
- b. Covalence (USA)
- c. Berry Plastics (USA) – (Seal for Life Industries)

56.0 MMO WIRE ANODES (WITHOUT FACTORY PRE-PACKED COKE BREEZE)

- a. GROUPPO DENORA, GOA, INDIA
- b. CERANODE TECHNOLOGIES, USA
- c. TELPRO, USA

57.0 MMO TUBULAR/ STRIP/ RIBBON ANODES

- a. GROUPPO DENORA, GOA, INDIA
- b. ORANZIO DE NORA, ITALY
- c. MAGNETOCHEMIE, HOLLAND
- d. ACTEL LTD., U.K.
- e. ELTECH SYSTEMS CORPORATION, USA
- f. CERANODE TECHNOLOGIES, USA
- g. MATCOR (USA)

Note: -

For any other brought out item(s) for which the vendor list is not provided in the tender , bidders can supply those item(s) from vendors/ suppliers who have earlier supplied similar item(s) for the intended services in earlier Oil and Gas projects and the item(s) offered is in their regular manufacturing/ supply range.

- 1) The vendor/supplier should not be in the Holiday list of OWNER/ CONSULTANT/other PSU
- 2) The bidder is not required to enclose documentary evidences (PO copies, Inspection & Completion with satisfactory working certificates etc.) along with their offer, however in case of successful bidder, these documents shall required to be submitted by them within 30 days from date of Placement of Order for approval to OWNER / CONSULTANT.

**D. INSTRUMENTATION****LIST OF RECOMMENDED VENDER/SUPPLIERS OF MAJOR BOUGHT-
OUT ITEMS****1.0 PRESSURE GAUGES**

- a. AN Instruments Pvt Ltd
- b. Badotherm Process Instruments B.V.
- c. Baumer Bourdon Haenni S.A.S
- d. British Rototherm Co Ltd
- e. Budenberg Gauge Co Ltd
- f. Dresser Inc
- g. Forbes Marshall (Hyd) Pvt Ltd
- h. General Instrument Consortium
- i. H. Guru Instruments (South India) Pvt Ltd
- j. Manometer (India) Pvt Ltd
- k. Nagano Keiki Seisakusho Ltd
- l. Hirlekar Precision, India
- m. Waaree Instruments Ltd
- n. Walchandnagar Industries Ltd (Tiwac Divn)
- o. Wika Alexander Wiegand & Co GmbH
- p. Wika Instruments India Pvt Ltd
- q. Ashcroft India Pvt Ltd.
- r. ITEC Measures Pvt Ltd

2.0 TEMPERATURE GAUGES

- a. AN Instruments Pvt Ltd.
- b. Badotherm Process Instruments B.V.
- c. Bourdon Haenni S.A.
- d. Dresser Inc.



- e. General Instruments Consortium
- f. H. Guru Instruments (South India) Pvt Ltd
- g. Nagano Keiki Seisakusho Ltd
- h. Solartron ISA
- i. Walchandnagar Industries Ltd (Tiwac Divn)
- j. Wika Alexander Wiegand & Co GmbH
- k. Wika Instruments India Pvt Ltd
- l. Pyro Electric, Goa
- m. Ashcroft India Pvt Ltd.
- n. ITEC Measures Pvt Ltd

3.0 TEMPERATURE ELEMENTS INCLUDING SKIN TYPE

- a. ABB Automation Ltd
- b. Altop Industries Ltd
- c. Bourdon Haenni S.A.
- d. Detriv Instrumentation & Electronics Ltd
- e. General Instruments Consortium
- f. Japan Thermowell Co Ltd
- g. Tecnomatic S.P.A
- h. Tempsen Instrument India Ltd
- i. Thermo Electric Co. Inc.
- j. Thermo-Couple Products Co
- k. Thermo-Electra B.V.
- l. Wika Alexander Wiegand & Co GmbH
- m. Altop Industries Ltd., Baroda
- n. Nagman Sensors (Pvt.) Ltd.
- o. Pyro Electric, Goa
- p. ITEC Measures Pvt Ltd

4.0 POSITIVE DISPLACEMENT FLOW METERS

- a. RMG (Germany)
- b. Elster Instromet
- c. Romet
- d. Dresser
- e. Itron



- f. FMG
- g. Common
- h. Metreg
- i. Raychem RPG
- j. Vemmtec

5.0 TURBINE FLOW METER

- a. Daniel
- b. Elster Instromet
- c. Itron
- d. RMG
- e. Rockwin

6.0 ELECTRONIC VOLUME CORRECTOR

- a. Elgas
- b. Itron
- c. Plum
- d. Pietro Fiorentini

7.0 ORIFICES (METER RUN, FLOW CONDITIONER, ORIFICE PLATE AND ASSEMBLY)

- a. Emerson
- b. FMC, USA
- c. Pietro Fiorentini S.P.A (Italy)
- d. Canalta Controls, Canada

8.0 ULTRASONIC FLOW METERS

- a. Daniel (USA)
- b. RMG (Germany)
- c. Instromet International (Belgium)
- d. Sick Maihak, Germany
- e. FMC, Germany

9.0 MASS FLOW METERS



- a. Daniel Measurement & Control Asia Pacific
- b. Endress + Hauser Instruments International
- c. FMC Measurements Solutions
- d. Heinrichs Messtechnik GMBH
- e. Rheonik MessGerate GMBH

10.0 FIELD INSTRUMENTS (P, DP, F, L, T)

- a. ABB Ltd
- b. Honeywell
- c. Fuji Electric Instruments Co Ltd
- d. Yokogawa
- e. Invensys India Pvt.Ltd

11.0 LEVEL GAUGES/ LEVEL INSTRUMENTS

- a. Bliss Anand
- b. Chemtrols
- c. V-Automat
- d. Levcon
- e. Nivo Controls
- f. Sbeletro Mechanicals
- g. TRAC

12.0 PRESSURE REGULATOR AND SLAM SHUT VALVE

- a. Pietro Fiorentini S.P.A. (Italy)
- b. Emerson
- c. RMG-Regel Messtechnik (Germany)
- d. Mokveld Valves BV (Netherlands)
- e. Schlumberger (USA)
- f. Gortor Controls B V (Netherlands)
- g. Instromet International NV



- h. Nirmal Industrial Controls Pvt Ltd. (up to 6" size only)
- i. ESME Valves Ltd
- j. Kaye & Macdonald Inc.
- k. Nuovo Pignone S.P.A (Italy) (GE Oil Co.)
- l. Richards Industries (Formerly Treloar)
- m. Samson AG Mess-und Regeltechnik
- n. Tormene Gas Technology
- o. Dresser Inc, USA (upto 8" size, 300# class only)

13.0 PRESSURE SAFETY VALVES

- a. Keystone Valves (India) Pvt. Ltd.
- b. Larson & Toubro Ltd.
- c. Lesser GmbH & Co KG
- d. Mekaster Engg Ltd..
- e. Tyco Sanmar Ltd. (New Delhi)
- f. Anderson Greenwood Crosby
- g. BHEL (Trichy)
- h. Curtiss Wright Flow Control Corporation
- i. Dresser Inc.
- j. Fukui Seisakusho Co. Ltd
- k. Nakakita Seisakusho Co Ltd
- l. Nuovo Pignone S.P.A (Italy) (GE Oil co)
- m. Parcol S.P.A
- n. Safety Systems UK Ltd
- o. Tai Milano S.P.A
- p. Weir Valves & Controls France
- q. Bliss Anand Pvt Ltd.

14.0 CONTROL PANEL & ACCESSORIES

- a. Keltron Controls Ltd., Kerala
- b. Elechmec Corporation Ltd., Mumbai
- c. Industrial Controls & Appliances Pvt. Ltd.,
- d. Alstom System Ltd., Noida
- e. Emerson Process Management (I) Pvt. Ltd.
- f. ABB Instruments Ltd., New Delhi
- g. Larsen & Toubro Ltd.



- h. Control & Automation, New Delhi
- i. GE Fanuc Systems Pvt. Ltd., New Delhi
- j. Rockwell Automation (I) Ltd., Ghaziabad
- k. Honeywell Automation Ltd.
- l. Rittal
- m. Pyrotech Elcronics Pvt Ltd.
- n. Positronics Pvt Ltd.
- o. Electronics Corporation of India Ltd.
- p. Phoenix Contact (I) Pvt. Ltd

15.0 JUNCTION BOXES AND CABLES GLANDS

- a. Ex-Protecta
- b. Flameproof Control Gears
- c. Baliga
- d. Flexpro Electricals
- e. RAY Flameproof Pvt Ltd (Flameproof Weatherproof)

16.0 CONTROL AND SIGNAL CABLES

- a. Associated Cables
- b. Brook
- c. Associated Flexibles & Wires (Pvt) Ltd
- d. Universal Cables Ltd, India
- e. Delton Cables Ltd, India
- f. KEI Industries Ltd INDIA
- g. CMI Limited
- h. Cords Cable Industries Ltd, India
- i. Elkay Telelinks (P) Ltd., India
- j. Udey Pyrocables Pvt Ltd, India
- k. Goyolene Fibres (I) Pvt Ltd, India
- l. Netco Cable Industries Pvt Ltd, India
- m. Nicco Corporation Ltd, India
- n. Paramount Communications Ltd, India
- o. Polycab Wires Pvt Ltd, India
- p. Radiant Cables Pvt Ltd, India
- q. Reliance Engineers Ltd., India
- r. Suyog Electricals Ltd, India



- s. Thermo Cables Ltd

17.0 GAS DETECTION SYSTEM

- a. Crowcon Detection Instruments Ltd
- b. Detection Instruments (I) Pvt Ltd
- c. Detector Electronics Corporation
- d. Drager Safety AG & Co. KGAA
- e. General Monitors Ireland Ltd
- f. Mine Safety Appliances Company
- g. MSA – Mines Safety Appliances(India) Ltd
- h. Industrial Scientific Oldham France S.A.
- i. Riken Keiki Co Ltd
- j. Simrad Optronics Icare
- k. Honeywell Analytics
- l. Net Safety Monitoring Inc.
- m. Simtronics SAS
- n. Pollution Protection Systems Mumbai Pvt Ltd

18.0 MOV ACTUATOR:

- a. Rotork- UK, USA & INDIA
- b. Limitorque
- c. Auma- India
- d. Biffi- Italy
- e. DelVal Flow Controls Pvt. Ltd

19.0 PNEUMATIC ACTUATOR (SOLENOID OPERATED ON-OFF TYPE)

- a. Metso Automation
- b. Tyco
- c. Samson Controls
- d. L&T
- e. Emerson
- f. Fisher
- g. Masoneilan Process Control
- h. Instrumentation Limited (IL)-Palghat



- i. Micro Finish
- j. Rotex

20.0 SOLENOID VALVES

- a. Avcon
- b. Festo
- c. DelVal Flow Controls Pvt. Ltd

21.0 ELECTRO – HYDRAULIC ACTUATOR

- a. Avcon Rotork controls (Deutschland Gmbh)
- b. Biffi Italia Srl
- c. Ledeen (Italy)
- d. Virgo Valves and Controls ltd.-India
- e. Limittorque
- f. Reineke
- g. Voith
- h. Bettis
- i. Rotork- UK, USA & INDIA
- j. Rotex
- k. Schuck Group
- l. DelVal Flow Controls Pvt. Ltd

22.0 GAS OVER OIL ACTUATOR

- a. Biffi Italia Srl,
- b. Ledeen(Italy)
- c. Virgo Valves & Control ltd.-India,
- d. Voith,
- e. Bettis,
- f. Rotork-UK, USA, India,
- g. Rotex,
- h. Schuck Group,
- i. Valve Italia.

23.0 OFC

Manufacture/ Procurement, Testing and supply of suitable OFC Joint closures including all necessary accessories of any of the following make:



- a. Raychem
- b. 3M
- c. Siemens
- d. Any other make from the approved vendor list of client with supporting paper.

24.0 FLOW CONTROL VALVES

- a. Fouress Engg. (New Delhi)
- b. Fisher Xomox (New Delhi)
- c. MIL Control Ltd. (Noida)
- d. KOSO India Pvt Ltd
- e. Samson Control (Thane)
- f. Dresser Valves India Pvt Ltd.
- g. Fisher Controls
- h. Valvitalia Italy
- i. CCI Valve technology
- j. Flowserve Pvt Ltd.
- k. Metso Singapore Pvt Ltd.
- l. Instrumentation Ltd Palghat
- m. Dresser Inc. USA

25.0 FLOW COMPUTERS

- a. Emerson
- b. Instromet International (Belgium)
- c. FMC Measurement Solutions (UK)
- d. RMG (Germany)
- e. OMNI Flow Computers Inc.
- f. Thermo Fisher, USA

26.0 INDICATORS & CONTROLLERS

- a. Yokogawa
- b. Eurotherm Chessel
- c. Honeywell
- d. Emerson

27.0 BARRIERS

- a. MTL
- b. STHAL



- c. P&F
- d. Phoenix Contact (I) Pvt. Ltd.

28.0 GAS CHROMATOGRAPH

- a. ABB
- b. Emerson
- c. Instromet International, NV
- d. RMG Regal+Messtechnik GmbH
- e. Yokogawa

29.0 I/P CONVERTERS

- a. ABB
- b. Emerson
- c. IMI Watson Smith Ltd.
- d. Moore Controls Ltd
- e. Shreyas Instruments Pvt Ltd, India
- f. Thermo Brandt Instruments

30.0 SS FITTINGS, INSTRUMENT VALVES & MANIFOLDS**30.1 FOR CNG WORK:**

- a. DK-LOK
- b. Swagelok Co.
- c. Parker
- d. Dawsons Tech Components LLP
- e. ASTEC Valves & Fittings Pvt Ltd
- f. Thomas & Brian Mfg. Pvt. Ltd

30.2 EXCEPT CNG WORK:

- a. Swagelok Co.
- b. Parker
- c. Aura INC.
- d. HOKE
- e. Excelsior Engineering works
- f. Swastik Engineering works India
- g. Comfit and valves pvt ltd
- h. Arya craft and engineering Pvt ltd
- i. DK lok



- j. ASTEC Valves & Fittings Pvt Ltd
- k. Dawsons Tech Components LLP
- l. DelVal Flow Controls Pvt. Ltd**

31.0 SS TUBES

31.1 FOR CNG WORK:

- a. Swagelok Co.
- b. Parker
- c. Sandvik
- d. Jindal Saw
- e. Tubacex Tubes and Pipes Private Ltd
- f. Krystal Global Engineering Ltd.
- g. Maxim Tubes Company Pvt. Ltd.
- h. VENUS PIPES & TUBES LIMITED**

31.2 EXCEPT CNG WORK:

- a. Swagelok Co.
- b. Parker
- c. Sandvik
- d. Heavy metal and tube limited
- e. Nuclear fuel complex India
- f. Scorodite
- g. Ratnamani Metals and Tubes
- h. Jindal Saw

**E. SHOP & FIELD PAINTING****LIST OF RECOMMENDED VENDER/SUPPLIERS OF MAJOR BOUGHT-
OUT ITEMS****1.0 INDIAN VENDORS**

- a. Asian Paints (I) Ltd.
- b. Berger Paints Ltd.
- c. Goodlass Nerlolac Paints Ltd.
- d. Jenson And Nicholson Paint Ltd & chokuGu Jenson & Nicholson Ltd.
- e. Shalimar Paints Ltd.
- f. Sigma Coating, Mumabai
- g. CDC Carboline Ltd.
- h. Premier Products Ltd.
- i. Coromandel Paints & Chemicals Ltd.
- j. Anupam Enterprises
- k. Grand Polycoats
- l. Bombay Paints Ltd.
- m. Vanaprabha Esters & Glycer, Mumbai
- n. Sunil Paints and Varnishes Pvt. Ltd.
- o. Courtaulds Coating & Sealants India (Pvt.) Ltd.
- p. Mark-chem Incorporated, Mumbai (for phosphating chemicals only)
- q. VCM Polyurethane Paint (for polyurethane Paint only)

2.0 FOREIGN VENDORS FOR OVERSEAS PRODUCTS

- a. Sigma Coating, Singapore
- b. Ameron, USA
- c. Kansai Paint, Japan
- d. Hempel Paint, USA
- e. Valspar Corporation, USA
- f. Courtaulds Coating, UK.

**Notes:**

1. Bidder can select equipment of two different makes, selected from this VENDOR LIST and mention the same in the checklist for technical evaluation attached with the tender. The offered bid must include filled datasheet indicating make, model, size, rating of offered instrument/ equipment duly supported by sizing calculation of offered equipment (wherever applicable).
2. Vendors who have already supplied above equipment in other terminals of client, shall also be considered qualified for this tender provided the supplied equipment are commissioned and running successfully and they have not been put on holiday.
3. Equipment / Instruments of any make which is offered by one bidder and acceptable to client shall be accepted for other bidder also. After placement of order, on request of the successful bidder list of other qualified makes for a particular item (for which successful bidder wants to change the vendor) shall be provided.
4. Bidder shall take prior approval of the make / model no of the offered item and it shall be from the list given above. However additional vendors will be considered in exceptional cases, provided they have supplied for similar application to reputed gas transmission/distribution companies, in quantities at least half the numbers being supplied for this tender, and working satisfactorily for minimum 6 months. Documentary evidence substantiating above shall be submitted for taking approval.

**F. FOR PE & LMC WORK (GI/CU)****1.0 MDPE PIPE**

- a. Mukand Poly Products,
- b. Dewa Polymers Pvt. Ltd.

2.0 MDPE FITTINGS & MDPE VALVES

- c. Aliaxis,
- d. George Fischer,
- e. Al-Aziz,
- f. Kimplas,
- g. Banides,
- h. Agru,
- i. Friatech,
- j. Plasson

3.0 GI PIPE

- a. Swastik Pipe Ltd.
- b. Jindal Industries Ltd.
- c. Vishal Pipes Ltd.
- d. Indus Tubes Ltd
- e. Advance steel Tubes Ltd.
- f. Good Luck Tubes Ltd.
- g. Surya Roshni Limited
- h. APL Apollo Tubes Limited
- i. Jindal Pipes Limited
- j. RK Steel Manufacturing Company Private Limited
- k. PSL Tubes Limited

4.0 CASTING GI FITTINGS

- a. Sarin Industries Ltd.
- b. Jupiter Metal Industries Ltd.
- c. Jainsons Industries Ltd.
- d. Jinan Meide Casting Co. Ltd.
- e. Green Malleable Pvt. Ltd.
- f. Chandan Enterprises
- g. Unique Pipe Fitting Co

**5.0 FORGED GI FITTING (FOR HIGH RISE SEGMENT)**

- a. Jainsons Industries
- b. B.M. Meters Pvt. Ltd.

6.0 COPPER TUBES & FITTINGS

- a. Jay Banas Mehta Tubes Limited- Trade Mark "MEXFLOW"
- b. Rajco metal (Tubes & Fittings)
- c. Paras Industries
- d. MERCURE METAL & ALLOYS PVT LTD
- e. Chandan Enterprises (Fittings only)

7.0 BRASS VALVES

- a. Universal srl, Italy
- b. Tiemme Raccorderie Sede, Italy
- c. Enolgas Bonimu s.a.s., Italy
- d. Fratelli Fortis s.r.l, Italy
- e. Giacomo Climbrio, Italy
- f. Parker Hannifin S.P.A., USA
- g. Singapore Valve & Amp; Fittings Pte Limited, Singapore /Bengaluru
- h. Rubinetterie Utensilerie Bonomi (RUB), Italy
- i. Zhegiang Valogin Technology Co. Ltd., China,
- j. Ningbo Zhiqing Industrial Co. Ltd., China,
- k. Zhegiang Dunan Valve Co. Ltd.,
- l. Ningbo Huaping, China.
- m. Chandan Enterprises

8.0 Isolation & Appliance valves

- a. Chandan Enterprises

9.0 BRASS FITTINGS

- a. Chandan Enterprises
- b. Paras Industries Ltd.
- c. Chokhawala Distributors – Brass Adaptor.

10.0 STEEL RE-INFORCED RUBBER HOSE (TYPE-4)

- a. Super Seal Flexible Hose Ltd.



- b. Suraksha Products Pvt. Ltd.
- c. Vansh Industries
- d. T & L Gases

11.0 CORRUGATED FLEXIBLE METAL HOSES (ANACONDA)

- a. KPC Flex Tubes
- b. Vestas Hose Division
- c. Alpha Flexi Tubes
- d. Chandan Enterprises

Note:

1. Vendor may procure material from any of approved vendors listed.
2. For equipment/components other than the above, vendor shall submit past track record for the proposed sub-vendors and obtain written approval from Owner / Consultant before placing order.
3. In case of exigencies like long delivery periods from approved vendors, the contractor shall list down the proposed suppliers/vendors for such items and submit the same for owner review/approval along with necessary documents/PTR.
4. Non-acceptance of a particular proposed vendor due to any reasons whatsoever shall not be a cause of schedule and cost implication. If equipment is sourced from outside India, vendor shall obtain prior approval for make of equipment before placement of order.

Above mentioned vendor list is tentative and further addition/deletion may be done as per discretion of Owner/VCS.



Energising Quality

LIST OF AUTHORIZED THIRD-PARTY INSPECTION AGENCY (TPIA)

DOC. NO.- VCS-C&P-TPIA-001

Rev No : 16

SL. NO.	NAME OF TPIA
1	Det Norske Veritas (DNV)
2	Germanischer Lloyd Industrial Services GmbH
3	Bureau Veritas (India) Pvt. Ltd.
4	Moody International (India) Pvt. Ltd. (Industry Services Division)
5	SGS India Pvt. Ltd.
6	Certification Engineer International Limited (CEIL)
7	TÜV SÜD South Asia Pvt. Ltd.
8	ABS Industrial Verification (India) Pvt. Ltd.
9	Lloyd Register of Industrial Services
10	IRCLASS Systems and Solutions Private Limited
11	Tata Projects Limited
12	International Certification Services Pvt. Ltd.
13	TUV India Pvt. Ltd., Industrial Services Division
14	Intertek India Pvt. Ltd. (Industry Services Division)
15	Quality Austria Central Asia Pvt. Ltd.
16	Edlipse Engineering Global Pvt. Ltd.
17	TUV-Rheinland
18	Hertz Inspection & Services Pvt Ltd
19	Alfred H Knight India Pvt Ltd (Formely known as Dr. Amin Controllers Pvt Ltd)